The German International University for Applied Sciences

Faculty of Business Informatics

Business Intelligence & Analytics [BINF 602]

Dr. Shaimaa Masry

Milestone 3

Submitted by:

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Spring, 2026

Milestone 1: 3
Milestone 2: 27
Milestone 3: 37

Milestone 1:

1. Project Description:

The goal of this project is to design and implement a robust Data Warehouse solution for a retail business environment. Using the Medallion Architecture, we have established a pipeline that ingests raw transactional data into a Bronze Layer for storage and subsequently transforms it into a Silver Layer for data cleansing and standardization. This architecture ensures that data is high-quality, deduplicated, and ready for advanced business intelligence reporting in the final Gold Layer.

2. Description of your Dataset:

The dataset originates from flat CSV files representing an OLTP (Online Transaction Processing) system, including data from CRM and ERP systems. It contains information regarding customers, products, sales (both online and POS), inventory, and logistics.

OLTP Schema Screenshot

	Schema_Name	Table_Name	Rows_Inserted	Created_Date
1	bronze	BRANDS	30	2026-04-24 18:38:51.910
2	bronze	CUSTOMERS	4000	2026-04-24 18:38:51.927
3	bronze	DELIVERIES	1927	2026-04-24 18:38:51.930
4	bronze	DELIVERY_PROVIDERS	8	2026-04-24 18:38:51.940
5	bronze	DEPARTMENTS	12	2026-04-24 18:38:51.947
6	bronze	EMPLOYEES	800	2026-04-24 18:38:51.953
7	bronze	INVENTORY	3600	2026-04-24 18:38:51.960
8	bronze	load_log	19	2026-04-24 18:38:51.860
9	bronze	ONLINE_ORDER_ITEMS	7151	2026-04-24 18:38:51.970
10	bronze	ONLINE_ORDERS	3000	2026-04-24 18:38:51.973
11	bronze	PAYMENTS	3000	2026-04-24 18:38:51.983
12	bronze	POS_TRANSACTIONS	6000	2026-04-24 18:38:51.990
13	bronze	PRODUCT_SUPPLIERS	1214	2026-04-24 18:38:51.993
14	bronze	PRODUCTS	600	2026-04-24 18:38:52.000
15	bronze	PROMOTIONS	120	2026-04-24 18:38:52.003
16	bronze	REGISTERS	218	2026-04-24 18:38:52.010
17	bronze	STORES	64	2026-04-24 18:38:52.017
18	bronze	SUPPLIERS	80	2026-04-24 18:38:52.023
19	bronze	TRANSACTION_ITEMS	14962	2026-04-24 18:38:52.030
20	bronze	WAREHOUSES	8	2026-04-24 18:38:52.033

21	silver	BRANDS	30	2026-04-27 23:28:58.657
22	silver	CUSTOMERS	3988	2026-04-27 23:28:58.667
23	silver	DELIVERIES	1915	2026-04-27 23:28:58.677
24	silver	DELIVERY_PROVIDERS	8	2026-04-27 23:28:58.687
25	silver	DEPARTMENTS	12	2026-04-27 23:28:58.697
26	silver	EMPLOYEES	800	2026-04-27 23:28:58.703
27	silver	etl_log	20	2026-04-27 23:28:58.643
28	silver	INVENTORY	3600	2026-04-27 23:28:58.713
29	silver	ONLINE_ORDER_ITEMS	7121	2026-04-27 23:28:58.720
30	silver	ONLINE_ORDERS	3000	2026-04-27 23:28:58.730
31	silver	PAYMENTS	3000	2026-04-27 23:28:58.737
32	silver	POS_TRANSACTIONS	6000	2026-04-27 23:28:58.743
33	silver	PRODUCT_SUPPLIERS	1214	2026-04-27 23:28:58.750
34	silver	PRODUCTS	600	2026-04-27 23:28:58.760
35	silver	PROMOTIONS	112	2026-04-27 23:28:58.767
36	silver	REGISTERS	218	2026-04-27 23:28:58.777
37	silver	STORES	64	2026-04-27 23:28:58.787
38	silver	SUPPLIERS	80	2026-04-27 23:28:58.797
39	silver	TRANSACTION_ITEMS	14912	2026-04-27 23:28:58.803
40	silver	WAREHOUSES	8	2026-04-27 23:28:58.810

3. Description of the technical work carried out in the Bronze Layer:

The Bronze layer serves as the landing zone where raw data is stored exactly as received from source files.

Environment Setup: A dedicated database named DataWarehouse was created with a bronze schema.

Table Structure: 19 tables were created mirroring the exact structure and column order of the source CSV files. All columns were initially set to NVARCHAR(MAX) to ensure no data loss during ingestion.

Load Method: We implemented a batch processing "Truncate & Insert" strategy.

Stored Procedure: The bronze.load_bronze procedure was developed to automate the loading process. It uses BULK INSERT to move data from the local file system into SQL Server.

Auditing: A log table (bronze.load_log) was created to record the status, start/end times, and the number of rows inserted for every load to ensure auditability.

4. Silver Layer Technical Work

The Silver Layer serves as the "Refinery" of our Data Warehouse. While the Bronze layer contains raw, unprocessed data, the Silver layer transforms it into a reliable **Single Source of Truth** by applying rigorous technical and business rules.

4.1 Silver Layer Specifications

- **Object Type:** Physical Tables.
- **Load Method:** Batch Processing using a "Full Load, Truncate & Insert" strategy.
- **Automation:** Orchestrated through the `silver.load_silver` stored procedure.
- **Data Model:** The operational structure is retained but refined for integration.

4.2 General Transformations (Applied Globally)

Before table-specific logic is applied, several general cleaning steps are executed across the entire dataset to ensure baseline quality:

- **Type Casting:** We transitioned from the `NVARCHAR(MAX)` format used in Bronze to precise SQL types such as `INT`, `DATE`, and `DECIMAL(18,2)`. We utilized `TRY_CAST` to ensure that corrupt data results in a `NULL` rather than a system crash.
- **String Sanitization:** All text fields (e.g., `Brand_Name`, `City`, `SKU`) underwent `TRIM()` processing to remove leading/trailing whitespace that often causes "Join" operations between tables to fail.
- **Case Uniformity:** Identifiers like `Email` were converted to `LOWER()` case to ensure consistency, while status codes were standardized using `UPPER()`.

4.3 Detailed Transformation Summary Table

The following table summarizes the specific cleansing and enrichment logic applied to each entity during the load process:

Table	Transformation Type	Data Quality Resolutions & Derived Logic
CUSTOMERS	Deduplication & Enrichment	Used a CTE with `ROW_NUMBER()` to remove duplicates based on Email. Derived the `Full_Name` column by merging `First_Name` and `Last_Name`.

EMPLOYEES	Standardization	Harmonized varied gender codes (e.g., 'M', 'Male') into a standard 'Male' or 'Female' format using `CASE` logic.
PROMOTIONS	Business Validation	Implemented a strict rule to remove records where `End_Date` was earlier than `Start_Date` or where `Discount_Percent` was missing.
INVENTORY	Imputation	Missing `Stock_Level` values were dynamically filled using the calculated mean stock level across the dataset.
TRANSACTION_ITEMS	Imputation & Calculation	Imputed missing unit prices with the POS mean price and derived the `Total_Line_Amount` (`Quantity * Unit_Price`).
ONLINE_ORDERS	Data Recovery	Recalculated `Order_Total` for records where the value was NULL by summing the related items from the items table.
PAYMENTS	Integrity Sync	Performed a `LEFT JOIN` with `ONLINE_ORDERS` to synchronize and fix mismatched payment amounts with the actual order totals.
DELIVERIES	Integrity Filter	Applied logic to filter out logically impossible records where the `Delivery_Date` occurred before the `Ship_Date`.

4.4 In-Depth Technical Implementation

To provide a deeper understanding of the "Refining" process, here are the four pillars of our Silver layer implementation:

A. Deduplication Strategy

Raw data often contains redundant entries from system glitches. By partitioning our data by unique identifiers (like `Email` in the `CUSTOMERS` table), we ensure that each entity is represented only once. This prevents the overcounting of metrics in the final dashboards.

B. Dynamic Mean Imputation (Handling Missing Data)

"Null" values in financial or inventory columns can break calculations. At the start of our procedure, we calculate the average (mean) for key metrics like `Stock_Level` and `Unit_Price`. If a specific record is missing this data, we fill the gap with the calculated mean, allowing us to maintain a complete dataset for trend analysis without discarding valuable records.

C. Data Enrichment and Performance Optimization

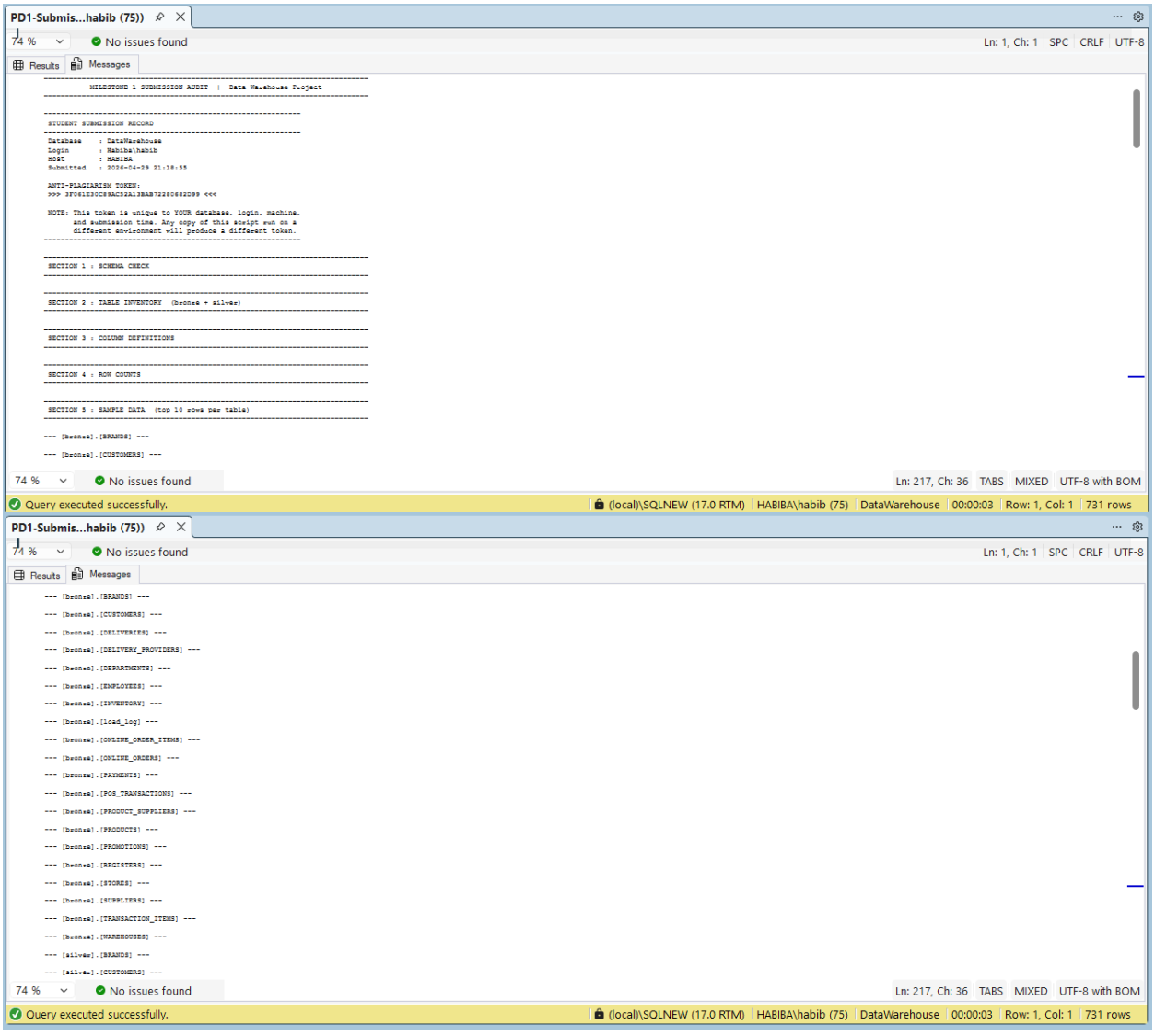
We created new information to make the dataset "Business Ready". For instance, calculating the `Total_Line_Amount` at the Silver layer eliminates the need for Power BI to perform these complex calculations at runtime, significantly improving dashboard responsiveness.

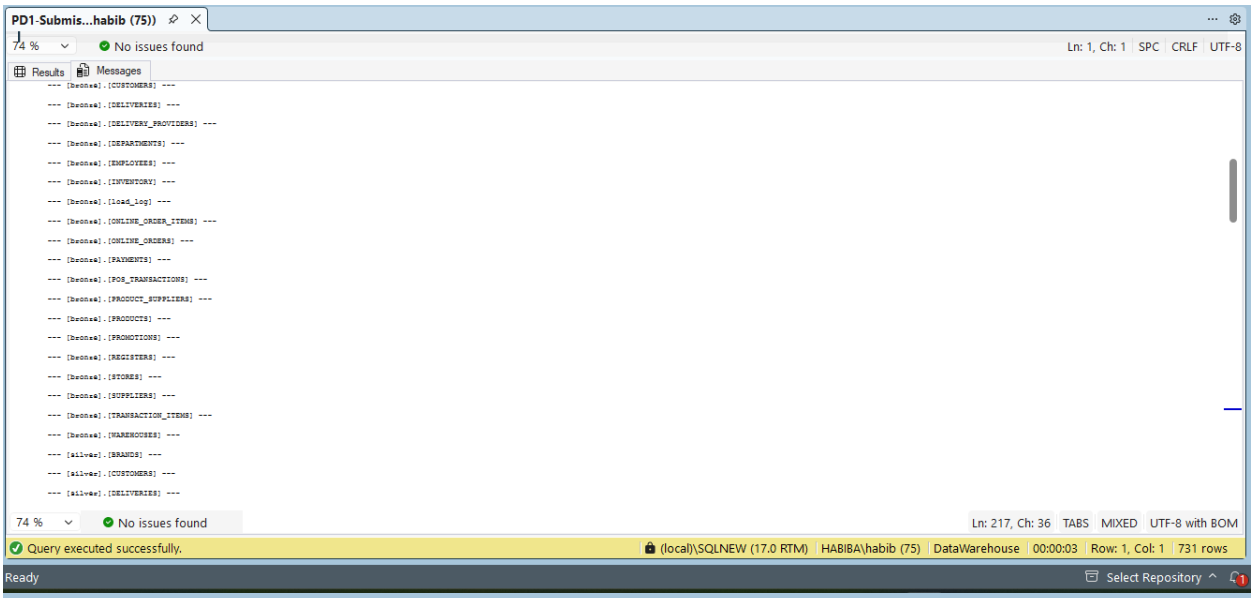
D. Technical Audit and Logging

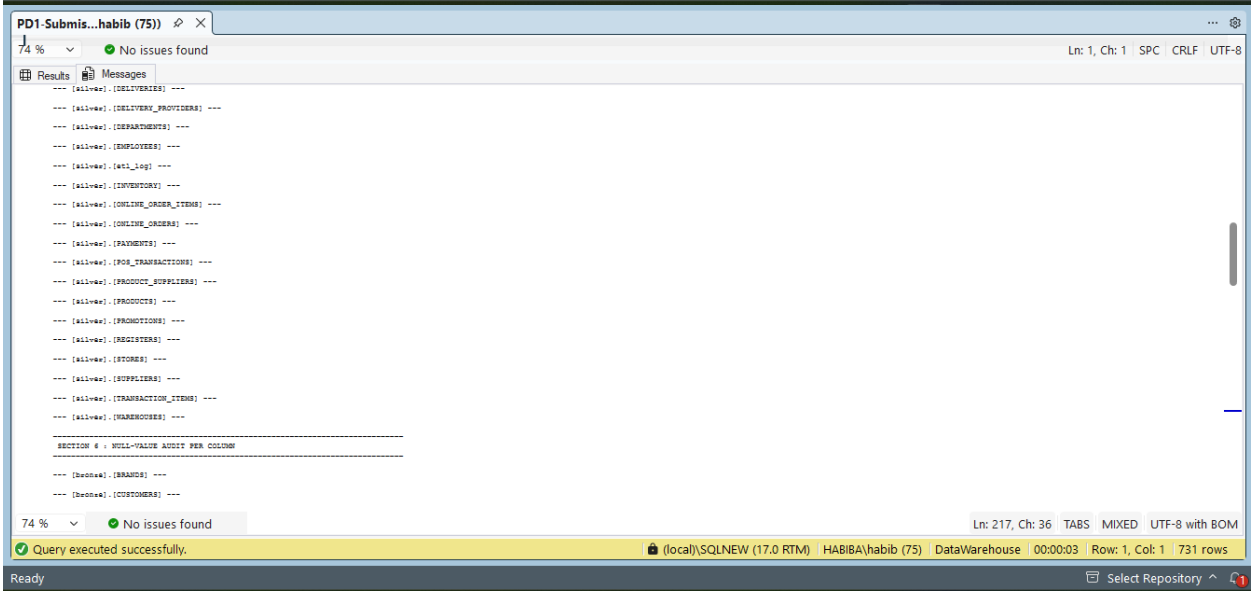
Transparency is maintained through the `silver.etl_log` table. Each step of the transformation logs:

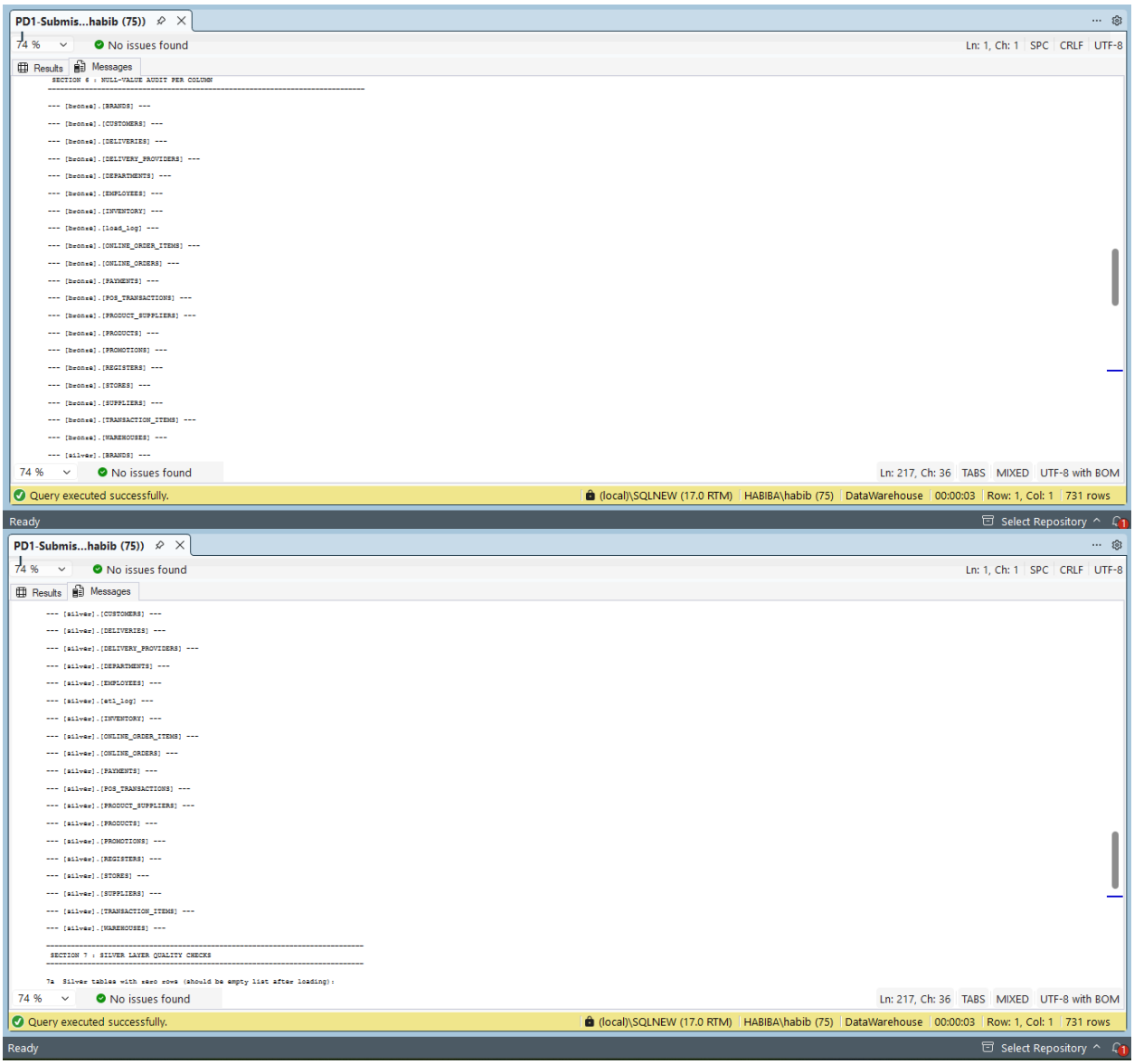
- The specific table and the quality rule applied (e.g., "Removed Date Violations").
- The exact number of rows successfully processed (`@ROWCOUNT`).
- High-precision execution timing (`DATETIME2`) to monitor performance.

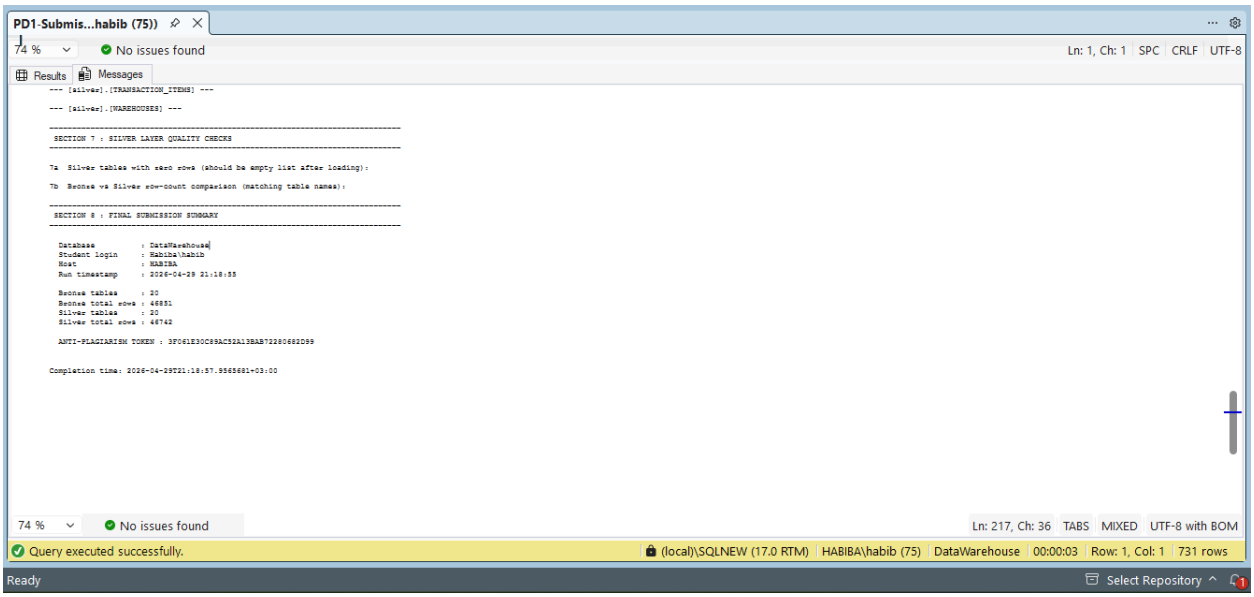
Results Tab:


PD1-Submis...habib (75))
74 %
No issues found
Ln: 1, Ch: 1
SPC
CRLF
UTF-8
Results
Messages
MILESTONE 1 SUBMISSION AUDIT | Data Warehouse Project
STUDENT SUBMISSION RECORD
Database : DataWarehouse
Login : Habiba\habib
Host : HABIBA
Submitted : 2026-04-29 21:18:55
ANTI-PLAGIARISM TOKEN:
>>> 3F561E3CC89AC52A13BAB72280682D99 <<<
NOTE: This token is unique to YOUR database, login, machine,
and submission time. Any copy of this script run on a
different environment will produce a different token.
SECTION 1 : SCHEMA CHECK
SECTION 2 : TABLE INVENTORY (bronze + silver)
SECTION 3 : COLUMN DEFINITIONS
SECTION 4 : ROW COUNTS
SECTION 5 : SAMPLE DATA (top 10 rows per table)
--- [bronze].[BRANDS] ---
--- [bronze].[CUSTOMERS] ---
Ln: 217, Ch: 36
TABS
MIXED
UTF-8 with BOM
Query executed successfully.
(local)\SQLNEW (17.0 RTM)
HABIBA\habib (75)
DataWarehouse
00:00:03
Row: 1, Col: 1
731 rows
--- [bronze].[DELIVERIES] ---
--- [bronze].[DELIVERY_PROVIDERS] ---
--- [bronze].[DEPARTMENTS] ---
--- [bronze].[EMPLOYEES] ---
--- [bronze].[INVENTORY] ---
--- [bronze].[load_log] ---
--- [bronze].[ONLINE_ORDER_ITEMS] ---
--- [bronze].[ONLINE_ORDERS] ---
--- [bronze].[PAYMENTS] ---
--- [bronze].[POS_TRANSACTIONS] ---
--- [bronze].[PRODUCT_SUPPLIERS] ---
--- [bronze].[PRODUCTS] ---
--- [bronze].[PROMOTIONS] ---
--- [bronze].[REGISTERS] ---
--- [bronze].[STORES] ---
--- [bronze].[SUPPLIERS] ---
--- [bronze].[TRANSACTION_ITEMS] ---
--- [bronze].[WAREHOUSES] ---
--- [silver].[BRANDS] ---
--- [silver].[CUSTOMERS] ---


PD1-Submis...habib (75))
74 %
No issues found
Ln: 1, Ch: 1 SPC CRLF UTF-8
Results
Messages
--- [bronze].[CUSTOMERS] ---
--- [bronze].[DELIVERIES] ---
--- [bronze].[DELIVERY_PROVIDERS] ---
--- [bronze].[DEPARTMENTS] ---
--- [bronze].[EMPLOYEES] ---
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--- [bronze].[SUPPLIERS] ---
--- [bronze].[TRANSACTION_ITEMS] ---
--- [bronze].[WAREHOUSES] ---
--- [silver].[BRANDS] ---
--- [silver].[CUSTOMERS] ---
--- [silver].[DELIVERIES] ---
74 %
No issues found
Ln: 217, Ch: 36 TABS MIXED UTF-8 with BOM
Query executed successfully.
(local)\SQLNEW (17.0 RTM) HABIBA\habib (75) DataWarehouse 00:00:03 Row: 1, Col: 1 731 rows
Ready
Select Repository


PD1-Submis...habib (75))
74 %
No issues found
Ln: 1, Ch: 1 SPC CRLF UTF-8
Results
Messages
--- [silver].[DELIVERIES] ---
--- [silver].[DELIVERY_PROVIDERS] ---
--- [silver].[DEPARTMENTS] ---
--- [silver].[EMPLOYEES] ---
--- [silver].[etl_log] ---
--- [silver].[INVENTORY] ---
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--- [silver].[ONLINE_ORDERS] ---
--- [silver].[PAYMENTS] ---
--- [silver].[POS_TRANSACTIONS] ---
--- [silver].[PRODUCT_SUPPLIERS] ---
--- [silver].[PRODUCTS] ---
--- [silver].[PROMOTIONS] ---
--- [silver].[REGISTERS] ---
--- [silver].[STORES] ---
--- [silver].[SUPPLIERS] ---
--- [silver].[TRANSACTION_ITEMS] ---
--- [silver].[WAREHOUSES] ---
SECTION 6 : NULL-VALUE AUDIT PER COLUMN
--- [bronze].[BRANDS] ---
--- [bronze].[CUSTOMERS] ---
74 %
No issues found
Ln: 217, Ch: 36 TABS MIXED UTF-8 with BOM
Query executed successfully.
(local)\SQLNEW (17.0 RTM) HABIBA\habib (75) DataWarehouse 00:00:03 Row: 1, Col: 1 731 rows
Ready
Select Repository


PD1-Submis...habib (75))
74 %
No issues found
Ln: 1, Ch: 1 SPC CRLF UTF-8
Results
Messages
SECTION 6 : NULL-VALUE AUDIT PER COLUMN
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--- [Bronze].[DEPARTMENTS] ---
--- [Bronze].[EMPLOYEES] ---
--- [Bronze].[INVENTORY] ---
--- [Bronze].[load_log] ---
--- [Bronze].[ONLINE_ORDER_ITEMS] ---
--- [Bronze].[ONLINE_ORDERS] ---
--- [Bronze].[PAYMENTS] ---
--- [Bronze].[POS_TRANSACTIONS] ---
--- [Bronze].[PRODUCT_SUPPLIERS] ---
--- [Bronze].[PRODUCTS] ---
--- [Bronze].[PROMOTIONS] ---
--- [Bronze].[REGISTERS] ---
--- [Bronze].[STORES] ---
--- [Bronze].[SUPPLIERS] ---
--- [Bronze].[TRANSACTION_ITEMS] ---
--- [Bronze].[WAREHOUSES] ---
--- [silver].[BRANDS] ---
74 %
No issues found
Ln: 217, Ch: 36 TABS MIXED UTF-8 with BOM
Query executed successfully.
(local)\SQLNEW (17.0 RTM) HABIBA\habib (75) DataWarehouse 00:00:03 Row: 1, Col: 1 731 rows
Ready
Select Repository
PD1-Submis...habib (75))
74 %
No issues found
Ln: 1, Ch: 1 SPC CRLF UTF-8
Results
Messages
--- [silver].[CUSTOMERS] ---
--- [silver].[DELIVERIES] ---
--- [silver].[DELIVERY_PROVIDERS] ---
--- [silver].[DEPARTMENTS] ---
--- [silver].[EMPLOYEES] ---
--- [silver].[etl_log] ---
--- [silver].[INVENTORY] ---
--- [silver].[ONLINE_ORDER_ITEMS] ---
--- [silver].[ONLINE_ORDERS] ---
--- [silver].[PAYMENTS] ---
--- [silver].[POS_TRANSACTIONS] ---
--- [silver].[PRODUCT_SUPPLIERS] ---
--- [silver].[PRODUCTS] ---
--- [silver].[PROMOTIONS] ---
--- [silver].[REGISTERS] ---
--- [silver].[STORES] ---
--- [silver].[SUPPLIERS] ---
--- [silver].[TRANSACTION_ITEMS] ---
--- [silver].[WAREHOUSES] ---
SECTION 7 : SILVER LAYER QUALITY CHECKS
7a Silver tables with zero rows (should be empty list after loading):
74 %
No issues found
Ln: 217, Ch: 36 TABS MIXED UTF-8 with BOM
Query executed successfully.
(local)\SQLNEW (17.0 RTM) HABIBA\habib (75) DataWarehouse 00:00:03 Row: 1, Col: 1 731 rows
Ready
Select Repository




```
================================================================================
        MILESTONE 1 SUBMISSION AUDIT  |  Data Warehouse Project
================================================================================

------------------------------------------------------------
STUDENT SUBMISSION RECORD
------------------------------------------------------------
Database    : DataWarehouse
Login       : Habiba\habib
Host        : HABIBA
Submitted   : 2026-04-29 21:18:55

ANTI-PLAGIARISM TOKEN:
>>> 3F061E30C89AC52A13BAB72280682D99 <<<

NOTE: This token is unique to YOUR database, login, machine,
      and submission time. Any copy of this script run on a
      different environment will produce a different token.
------------------------------------------------------------

================================================================================
SECTION 1 : SCHEMA CHECK
================================================================================

================================================================================
SECTION 2 : TABLE INVENTORY  (bronze + silver)
================================================================================

================================================================================
SECTION 3 : COLUMN DEFINITIONS
================================================================================

================================================================================
SECTION 4 : ROW COUNTS
================================================================================

================================================================================
SECTION 5 : SAMPLE DATA  (top 10 rows per table)
================================================================================

--- [bronze].[BRANDS] ---

--- [bronze].[CUSTOMERS] ---

--- [bronze].[DELIVERIES] ---

--- [bronze].[DELIVERY_PROVIDERS] ---

--- [bronze].[DEPARTMENTS] ---

--- [bronze].[EMPLOYEES] ---

--- [bronze].[INVENTORY] ---

--- [bronze].[load_log] ---

--- [bronze].[ONLINE_ORDER_ITEMS] ---
```

--- [bronze].[ONLINE_ORDERS] ---

--- [bronze].[PAYMENTS] ---

--- [bronze].[POS_TRANSACTIONS] ---

--- [bronze].[PRODUCT_SUPPLIERS] ---

--- [bronze].[PRODUCTS] ---

--- [bronze].[PROMOTIONS] ---

--- [bronze].[REGISTERS] ---

--- [bronze].[STORES] ---

--- [bronze].[SUPPLIERS] ---

--- [bronze].[TRANSACTION_ITEMS] ---

--- [bronze].[WAREHOUSES] ---

--- [silver].[BRANDS] ---

--- [silver].[CUSTOMERS] ---

--- [silver].[DELIVERIES] ---

--- [silver].[DELIVERY_PROVIDERS] ---

--- [silver].[DEPARTMENTS] ---

--- [silver].[EMPLOYEES] ---

--- [silver].[etl_log] ---

--- [silver].[INVENTORY] ---

--- [silver].[ONLINE_ORDER_ITEMS] ---

--- [silver].[ONLINE_ORDERS] ---

--- [silver].[PAYMENTS] ---

--- [silver].[POS_TRANSACTIONS] ---

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--- [silver].[PROMOTIONS] ---

--- [silver].[REGISTERS] ---

--- [silver].[STORES] ---

--- [silver].[SUPPLIERS] ---

--- [silver].[TRANSACTION_ITEMS] ---

--- [silver].[WAREHOUSES] ---

==========
SECTION 6 : NULL-VALUE AUDIT PER COLUMN
==========

--- [bronze].[BRANDS] ---

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--- [bronze].[INVENTORY] ---

--- [bronze].[load_log] ---

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--- [bronze].[ONLINE_ORDERS] ---

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--- [silver].[BRANDS] ---

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--- [silver].[DEPARTMENTS] ---

--- [silver].[EMPLOYEES] ---

--- [silver].[etl_log] ---

--- [silver].[INVENTORY] ---

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--- [silver].[PRODUCTS] ---

--- [silver].[PROMOTIONS] ---

--- [silver].[REGISTERS] ---

--- [silver].[STORES] ---

--- [silver].[SUPPLIERS] ---

--- [silver].[TRANSACTION_ITEMS] ---

--- [silver].[WAREHOUSES] ---

```
================================================================================
SECTION 7 : SILVER LAYER QUALITY CHECKS
================================================================================

7a  Silver tables with zero rows (should be empty list after loading):

7b  Bronze vs Silver row-count comparison (matching table names):

================================================================================
SECTION 8 : FINAL SUBMISSION SUMMARY
================================================================================

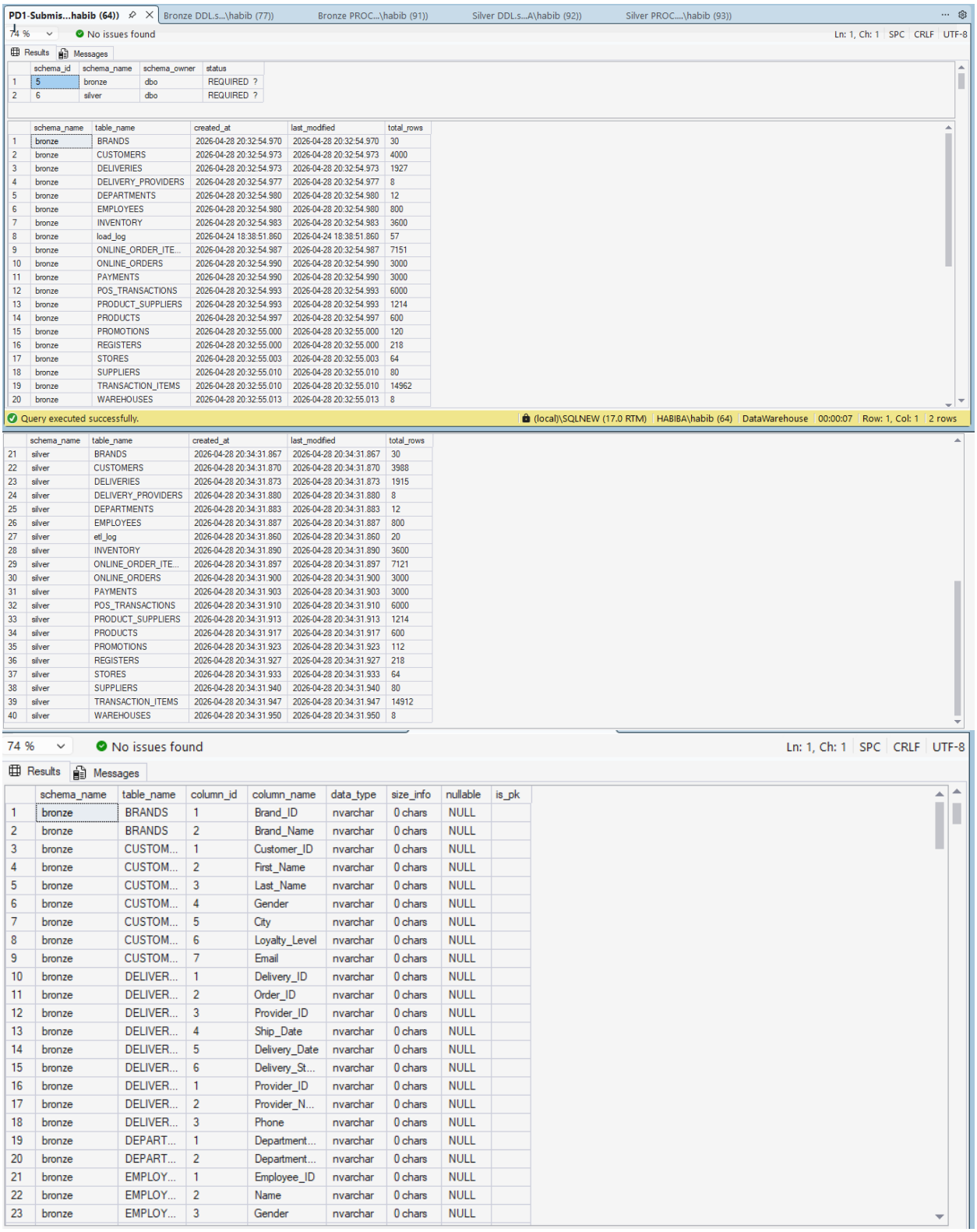
  Database          : DataWarehouse
  Student login     : Habiba\habib
  Host              : HABIBA
  Run timestamp     : 2026-04-29 21:18:55

  Bronze tables     : 20
  Bronze total rows : 46851
  Silver tables     : 20
  Silver total rows : 46742

  ANTI-PLAGIARISM TOKEN : 3F061E30C89AC52A13BAB72280682D99


Completion time: 2026-04-29T21:18:57.9565681+03:00
```

Messages Tab:



PD1-Submis...habib (64)) | Bronze DDL.s...\habib (77)) | Bronze PROC...\habib (91)) | Silver DDL.s...A\habib (92)) | Silver PROC....\habib (93))

74 % | No issues found | Ln: 1, Ch: 1 | SPC | CRLF | UTF-8

Results | Messages

	schema_id	schema_name	schema_owner	status
1	5	bronze	dbo	REQUIRED ?
2	6	silver	dbo	REQUIRED ?

	schema_name	table_name	created_at	last_modified	total_rows
1	bronze	BRANDS	2026-04-28 20:32:54.970	2026-04-28 20:32:54.970	30
2	bronze	CUSTOMERS	2026-04-28 20:32:54.973	2026-04-28 20:32:54.973	4000
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11	bronze	PAYMENTS	2026-04-28 20:32:54.990	2026-04-28 20:32:54.990	3000
12	bronze	POS_TRANSACTIONS	2026-04-28 20:32:54.993	2026-04-28 20:32:54.993	6000
13	bronze	PRODUCT_SUPPLIERS	2026-04-28 20:32:54.993	2026-04-28 20:32:54.993	1214
14	bronze	PRODUCTS	2026-04-28 20:32:54.997	2026-04-28 20:32:54.997	600
15	bronze	PROMOTIONS	2026-04-28 20:32:55.000	2026-04-28 20:32:55.000	120
16	bronze	REGISTERS	2026-04-28 20:32:55.000	2026-04-28 20:32:55.000	218
17	bronze	STORES	2026-04-28 20:32:55.003	2026-04-28 20:32:55.003	64
18	bronze	SUPPLIERS	2026-04-28 20:32:55.010	2026-04-28 20:32:55.010	80
19	bronze	TRANSACTION_ITEMS	2026-04-28 20:32:55.010	2026-04-28 20:32:55.010	14962
20	bronze	WAREHOUSES	2026-04-28 20:32:55.013	2026-04-28 20:32:55.013	8

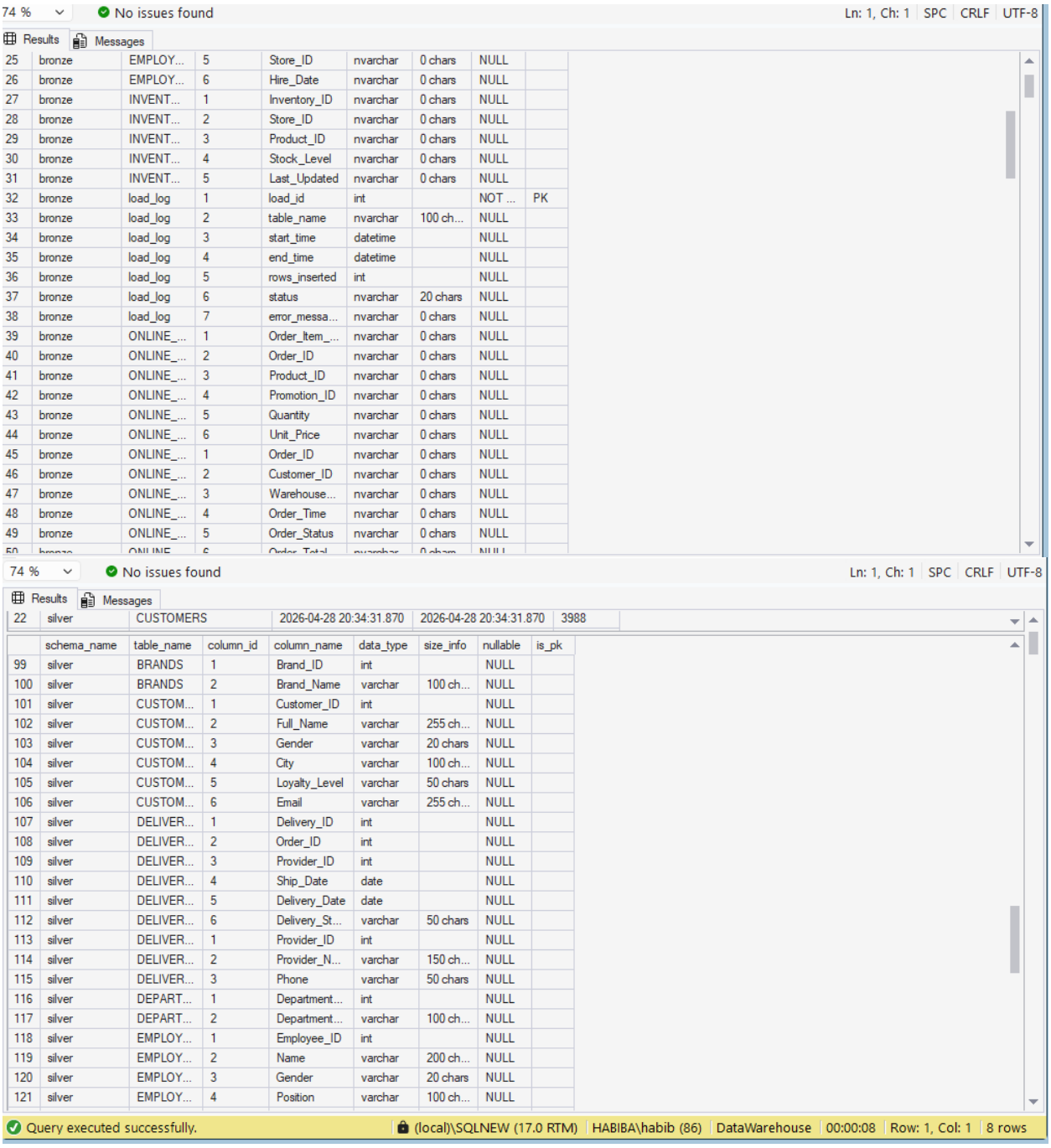
Query executed successfully. | (local)\SQLNEW (17.0 RTM) | HABIBA\habib (64) | DataWarehouse | 00:00:07 | Row: 1, Col: 1 | 2 rows

	schema_name	table_name	created_at	last_modified	total_rows
21	silver	BRANDS	2026-04-28 20:34:31.867	2026-04-28 20:34:31.867	30
22	silver	CUSTOMERS	2026-04-28 20:34:31.870	2026-04-28 20:34:31.870	3988
23	silver	DELIVERIES	2026-04-28 20:34:31.873	2026-04-28 20:34:31.873	1915
24	silver	DELIVERY_PROVIDERS	2026-04-28 20:34:31.880	2026-04-28 20:34:31.880	8
25	silver	DEPARTMENTS	2026-04-28 20:34:31.883	2026-04-28 20:34:31.883	12
26	silver	EMPLOYEES	2026-04-28 20:34:31.887	2026-04-28 20:34:31.887	800
27	silver	etl_log	2026-04-28 20:34:31.860	2026-04-28 20:34:31.860	20
28	silver	INVENTORY	2026-04-28 20:34:31.890	2026-04-28 20:34:31.890	3600
29	silver	ONLINE_ORDER_ITE...	2026-04-28 20:34:31.897	2026-04-28 20:34:31.897	7121
30	silver	ONLINE_ORDERS	2026-04-28 20:34:31.900	2026-04-28 20:34:31.900	3000
31	silver	PAYMENTS	2026-04-28 20:34:31.903	2026-04-28 20:34:31.903	3000
32	silver	POS_TRANSACTIONS	2026-04-28 20:34:31.910	2026-04-28 20:34:31.910	6000
33	silver	PRODUCT_SUPPLIERS	2026-04-28 20:34:31.913	2026-04-28 20:34:31.913	1214
34	silver	PRODUCTS	2026-04-28 20:34:31.917	2026-04-28 20:34:31.917	600
35	silver	PROMOTIONS	2026-04-28 20:34:31.923	2026-04-28 20:34:31.923	112
36	silver	REGISTERS	2026-04-28 20:34:31.927	2026-04-28 20:34:31.927	218
37	silver	STORES	2026-04-28 20:34:31.933	2026-04-28 20:34:31.933	64
38	silver	SUPPLIERS	2026-04-28 20:34:31.940	2026-04-28 20:34:31.940	80
39	silver	TRANSACTION_ITEMS	2026-04-28 20:34:31.947	2026-04-28 20:34:31.947	14912
40	silver	WAREHOUSES	2026-04-28 20:34:31.950	2026-04-28 20:34:31.950	8

74 % | No issues found | Ln: 1, Ch: 1 | SPC | CRLF | UTF-8

Results | Messages

	schema_name	table_name	column_id	column_name	data_type	size_info	nullable	is_pk
1	bronze	BRANDS	1	Brand_ID	nvarchar	0 chars	NULL	
2	bronze	BRANDS	2	Brand_Name	nvarchar	0 chars	NULL	
3	bronze	CUSTOM...	1	Customer_ID	nvarchar	0 chars	NULL	
4	bronze	CUSTOM...	2	First_Name	nvarchar	0 chars	NULL	
5	bronze	CUSTOM...	3	Last_Name	nvarchar	0 chars	NULL	
6	bronze	CUSTOM...	4	Gender	nvarchar	0 chars	NULL	
7	bronze	CUSTOM...	5	City	nvarchar	0 chars	NULL	
8	bronze	CUSTOM...	6	Loyalty_Level	nvarchar	0 chars	NULL	
9	bronze	CUSTOM...	7	Email	nvarchar	0 chars	NULL	
10	bronze	DELIVER...	1	Delivery_ID	nvarchar	0 chars	NULL	
11	bronze	DELIVER...	2	Order_ID	nvarchar	0 chars	NULL	
12	bronze	DELIVER...	3	Provider_ID	nvarchar	0 chars	NULL	
13	bronze	DELIVER...	4	Ship_Date	nvarchar	0 chars	NULL	
14	bronze	DELIVER...	5	Delivery_Date	nvarchar	0 chars	NULL	
15	bronze	DELIVER...	6	Delivery_St...	nvarchar	0 chars	NULL	
16	bronze	DELIVER...	1	Provider_ID	nvarchar	0 chars	NULL	
17	bronze	DELIVER...	2	Provider_N...	nvarchar	0 chars	NULL	
18	bronze	DELIVER...	3	Phone	nvarchar	0 chars	NULL	
19	bronze	DEPART...	1	Department...	nvarchar	0 chars	NULL	
20	bronze	DEPART...	2	Department...	nvarchar	0 chars	NULL	
21	bronze	EMPLOY...	1	Employee_ID	nvarchar	0 chars	NULL	
22	bronze	EMPLOY...	2	Name	nvarchar	0 chars	NULL	
23	bronze	EMPLOY...	3	Gender	nvarchar	0 chars	NULL	



74 % No issues found Ln: 1, Ch: 1 SPC CRLF UTF-8

Results Messages

25	bronze	EMPLOY...	5	Store_ID	nvarchar	0 chars	NULL	
26	bronze	EMPLOY...	6	Hire_Date	nvarchar	0 chars	NULL	
27	bronze	INVENT...	1	Inventory_ID	nvarchar	0 chars	NULL	
28	bronze	INVENT...	2	Store_ID	nvarchar	0 chars	NULL	
29	bronze	INVENT...	3	Product_ID	nvarchar	0 chars	NULL	
30	bronze	INVENT...	4	Stock_Level	nvarchar	0 chars	NULL	
31	bronze	INVENT...	5	Last_Updated	nvarchar	0 chars	NULL	
32	bronze	load_log	1	load_id	int		NOT ...	PK
33	bronze	load_log	2	table_name	nvarchar	100 ch...	NULL	
34	bronze	load_log	3	start_time	datetime		NULL	
35	bronze	load_log	4	end_time	datetime		NULL	
36	bronze	load_log	5	rows_inserted	int		NULL	
37	bronze	load_log	6	status	nvarchar	20 chars	NULL	
38	bronze	load_log	7	error_messa...	nvarchar	0 chars	NULL	
39	bronze	ONLINE_...	1	Order_Item_...	nvarchar	0 chars	NULL	
40	bronze	ONLINE_...	2	Order_ID	nvarchar	0 chars	NULL	
41	bronze	ONLINE_...	3	Product_ID	nvarchar	0 chars	NULL	
42	bronze	ONLINE_...	4	Promotion_ID	nvarchar	0 chars	NULL	
43	bronze	ONLINE_...	5	Quantity	nvarchar	0 chars	NULL	
44	bronze	ONLINE_...	6	Unit_Price	nvarchar	0 chars	NULL	
45	bronze	ONLINE_...	1	Order_ID	nvarchar	0 chars	NULL	
46	bronze	ONLINE_...	2	Customer_ID	nvarchar	0 chars	NULL	
47	bronze	ONLINE_...	3	Warehouse...	nvarchar	0 chars	NULL	
48	bronze	ONLINE_...	4	Order_Time	nvarchar	0 chars	NULL	
49	bronze	ONLINE_...	5	Order_Status	nvarchar	0 chars	NULL	
50	bronze	ONLINE_...	6	Order_Total	nvarchar	0 chars	NULL	

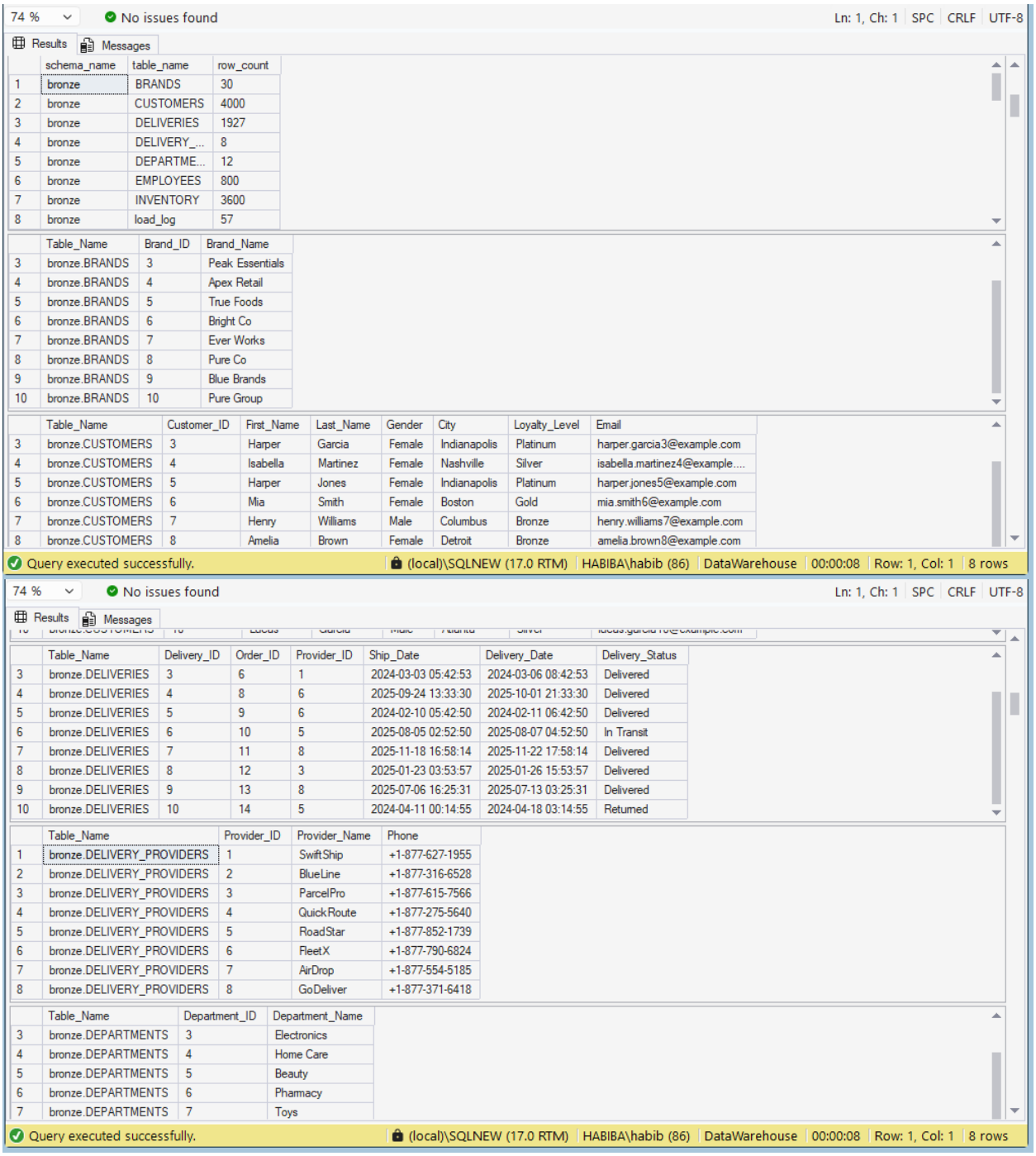
74 % No issues found Ln: 1, Ch: 1 SPC CRLF UTF-8

Results Messages

22	silver	CUSTOMERS	2026-04-28 20:34:31.870	2026-04-28 20:34:31.870	3988

	schema_name	table_name	column_id	column_name	data_type	size_info	nullable	is_pk
99	silver	BRANDS	1	Brand_ID	int		NULL	
100	silver	BRANDS	2	Brand_Name	varchar	100 ch...	NULL	
101	silver	CUSTOM...	1	Customer_ID	int		NULL	
102	silver	CUSTOM...	2	Full_Name	varchar	255 ch...	NULL	
103	silver	CUSTOM...	3	Gender	varchar	20 chars	NULL	
104	silver	CUSTOM...	4	City	varchar	100 ch...	NULL	
105	silver	CUSTOM...	5	Loyalty_Level	varchar	50 chars	NULL	
106	silver	CUSTOM...	6	Email	varchar	255 ch...	NULL	
107	silver	DELIVER...	1	Delivery_ID	int		NULL	
108	silver	DELIVER...	2	Order_ID	int		NULL	
109	silver	DELIVER...	3	Provider_ID	int		NULL	
110	silver	DELIVER...	4	Ship_Date	date		NULL	
111	silver	DELIVER...	5	Delivery_Date	date		NULL	
112	silver	DELIVER...	6	Delivery_St...	varchar	50 chars	NULL	
113	silver	DELIVER...	1	Provider_ID	int		NULL	
114	silver	DELIVER...	2	Provider_N...	varchar	150 ch...	NULL	
115	silver	DELIVER...	3	Phone	varchar	50 chars	NULL	
116	silver	DEPART...	1	Department...	int		NULL	
117	silver	DEPART...	2	Department...	varchar	100 ch...	NULL	
118	silver	EMPLOY...	1	Employee_ID	int		NULL	
119	silver	EMPLOY...	2	Name	varchar	200 ch...	NULL	
120	silver	EMPLOY...	3	Gender	varchar	20 chars	NULL	
121	silver	EMPLOY...	4	Position	varchar	100 ch...	NULL	

Query executed successfully. (local)\SQLNEW (17.0 RTM) HABIBA\habib (86) DataWarehouse 00:00:08 Row: 1, Col: 1 8 rows



74 % No issues found Ln: 1, Ch: 1 SPC CRLF UTF-8

Results Messages

	schema_name	table_name	row_count
1	bronze	BRANDS	30
2	bronze	CUSTOMERS	4000
3	bronze	DELIVERIES	1927
4	bronze	DELIVERY_...	8
5	bronze	DEPARTME...	12
6	bronze	EMPLOYEES	800
7	bronze	INVENTORY	3600
8	bronze	load_log	57

	Table_Name	Brand_ID	Brand_Name
3	bronze.BRANDS	3	Peak Essentials
4	bronze.BRANDS	4	Apex Retail
5	bronze.BRANDS	5	True Foods
6	bronze.BRANDS	6	Bright Co
7	bronze.BRANDS	7	Ever Works
8	bronze.BRANDS	8	Pure Co
9	bronze.BRANDS	9	Blue Brands
10	bronze.BRANDS	10	Pure Group

	Table_Name	Customer_ID	First_Name	Last_Name	Gender	City	Loyalty_Level	Email
3	bronze.CUSTOMERS	3	Harper	Garcia	Female	Indianapolis	Platinum	harper.garcia3@example.com
4	bronze.CUSTOMERS	4	Isabella	Martinez	Female	Nashville	Silver	isabella.martinez4@example....
5	bronze.CUSTOMERS	5	Harper	Jones	Female	Indianapolis	Platinum	harper.jones5@example.com
6	bronze.CUSTOMERS	6	Mia	Smith	Female	Boston	Gold	mia.smith6@example.com
7	bronze.CUSTOMERS	7	Henry	Williams	Male	Columbus	Bronze	henry.williams7@example.com
8	bronze.CUSTOMERS	8	Amelia	Brown	Female	Detroit	Bronze	amelia.brown8@example.com

Query executed successfully. (local)\SQLNEW (17.0 RTM) HABIBA\habib (86) DataWarehouse 00:00:08 Row: 1, Col: 1 8 rows

74 % No issues found Ln: 1, Ch: 1 SPC CRLF UTF-8

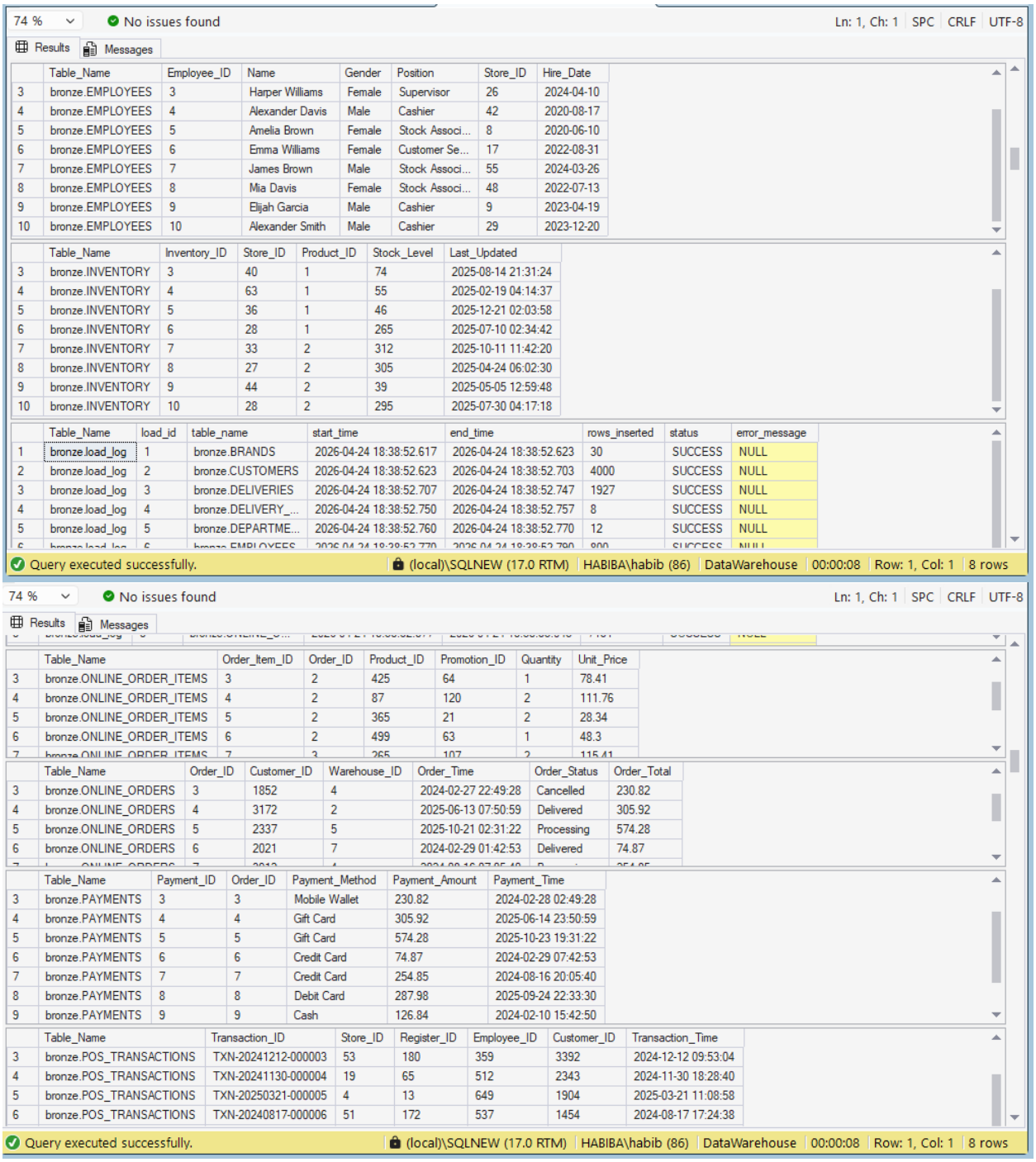
Results Messages

	Table_Name	Delivery_ID	Order_ID	Provider_ID	Ship_Date	Delivery_Date	Delivery_Status
3	bronze.DELIVERIES	3	6	1	2024-03-03 05:42:53	2024-03-06 08:42:53	Delivered
4	bronze.DELIVERIES	4	8	6	2025-09-24 13:33:30	2025-10-01 21:33:30	Delivered
5	bronze.DELIVERIES	5	9	6	2024-02-10 05:42:50	2024-02-11 06:42:50	Delivered
6	bronze.DELIVERIES	6	10	5	2025-08-05 02:52:50	2025-08-07 04:52:50	In Transit
7	bronze.DELIVERIES	7	11	8	2025-11-18 16:58:14	2025-11-22 17:58:14	Delivered
8	bronze.DELIVERIES	8	12	3	2025-01-23 03:53:57	2025-01-26 15:53:57	Delivered
9	bronze.DELIVERIES	9	13	8	2025-07-06 16:25:31	2025-07-13 03:25:31	Delivered
10	bronze.DELIVERIES	10	14	5	2024-04-11 00:14:55	2024-04-18 03:14:55	Returned

	Table_Name	Provider_ID	Provider_Name	Phone
1	bronze.DELIVERY_PROVIDERS	1	SwiftShip	+1-877-627-1955
2	bronze.DELIVERY_PROVIDERS	2	BlueLine	+1-877-316-6528
3	bronze.DELIVERY_PROVIDERS	3	ParcelPro	+1-877-615-7566
4	bronze.DELIVERY_PROVIDERS	4	QuickRoute	+1-877-275-5640
5	bronze.DELIVERY_PROVIDERS	5	RoadStar	+1-877-852-1739
6	bronze.DELIVERY_PROVIDERS	6	FleetX	+1-877-790-6824
7	bronze.DELIVERY_PROVIDERS	7	AirDrop	+1-877-554-5185
8	bronze.DELIVERY_PROVIDERS	8	GoDeliver	+1-877-371-6418

	Table_Name	Department_ID	Department_Name
3	bronze.DEPARTMENTS	3	Electronics
4	bronze.DEPARTMENTS	4	Home Care
5	bronze.DEPARTMENTS	5	Beauty
6	bronze.DEPARTMENTS	6	Pharmacy
7	bronze.DEPARTMENTS	7	Toys

Query executed successfully. (local)\SQLNEW (17.0 RTM) HABIBA\habib (86) DataWarehouse 00:00:08 Row: 1, Col: 1 8 rows



74 % No issues found Ln: 1, Ch: 1 SPC CRLF UTF-8

Results Messages

	Table_Name	Employee_ID	Name	Gender	Position	Store_ID	Hire_Date
3	bronze.EMPLOYEES	3	Harper Williams	Female	Supervisor	26	2024-04-10
4	bronze.EMPLOYEES	4	Alexander Davis	Male	Cashier	42	2020-08-17
5	bronze.EMPLOYEES	5	Amelia Brown	Female	Stock Associ...	8	2020-06-10
6	bronze.EMPLOYEES	6	Emma Williams	Female	Customer Se...	17	2022-08-31
7	bronze.EMPLOYEES	7	James Brown	Male	Stock Associ...	55	2024-03-26
8	bronze.EMPLOYEES	8	Mia Davis	Female	Stock Associ...	48	2022-07-13
9	bronze.EMPLOYEES	9	Elijah Garcia	Male	Cashier	9	2023-04-19
10	bronze.EMPLOYEES	10	Alexander Smith	Male	Cashier	29	2023-12-20

	Table_Name	Inventory_ID	Store_ID	Product_ID	Stock_Level	Last_Updated
3	bronze.INVENTORY	3	40	1	74	2025-08-14 21:31:24
4	bronze.INVENTORY	4	63	1	55	2025-02-19 04:14:37
5	bronze.INVENTORY	5	36	1	46	2025-12-21 02:03:58
6	bronze.INVENTORY	6	28	1	265	2025-07-10 02:34:42
7	bronze.INVENTORY	7	33	2	312	2025-10-11 11:42:20
8	bronze.INVENTORY	8	27	2	305	2025-04-24 06:02:30
9	bronze.INVENTORY	9	44	2	39	2025-05-05 12:59:48
10	bronze.INVENTORY	10	28	2	295	2025-07-30 04:17:18

	Table_Name	load_id	table_name	start_time	end_time	rows_inserted	status	error_message
1	bronze.load_log	1	bronze.BRANDS	2026-04-24 18:38:52.617	2026-04-24 18:38:52.623	30	SUCCESS	NULL
2	bronze.load_log	2	bronze.CUSTOMERS	2026-04-24 18:38:52.623	2026-04-24 18:38:52.703	4000	SUCCESS	NULL
3	bronze.load_log	3	bronze.DELIVERIES	2026-04-24 18:38:52.707	2026-04-24 18:38:52.747	1927	SUCCESS	NULL
4	bronze.load_log	4	bronze.DELIVERY_...	2026-04-24 18:38:52.750	2026-04-24 18:38:52.757	8	SUCCESS	NULL
5	bronze.load_log	5	bronze.DEPARTME...	2026-04-24 18:38:52.760	2026-04-24 18:38:52.770	12	SUCCESS	NULL

Query executed successfully. (local)\SQLNEW (17.0 RTM) HABIBA\habib (86) DataWarehouse 00:00:08 Row: 1, Col: 1 8 rows

74 % No issues found Ln: 1, Ch: 1 SPC CRLF UTF-8

Results Messages

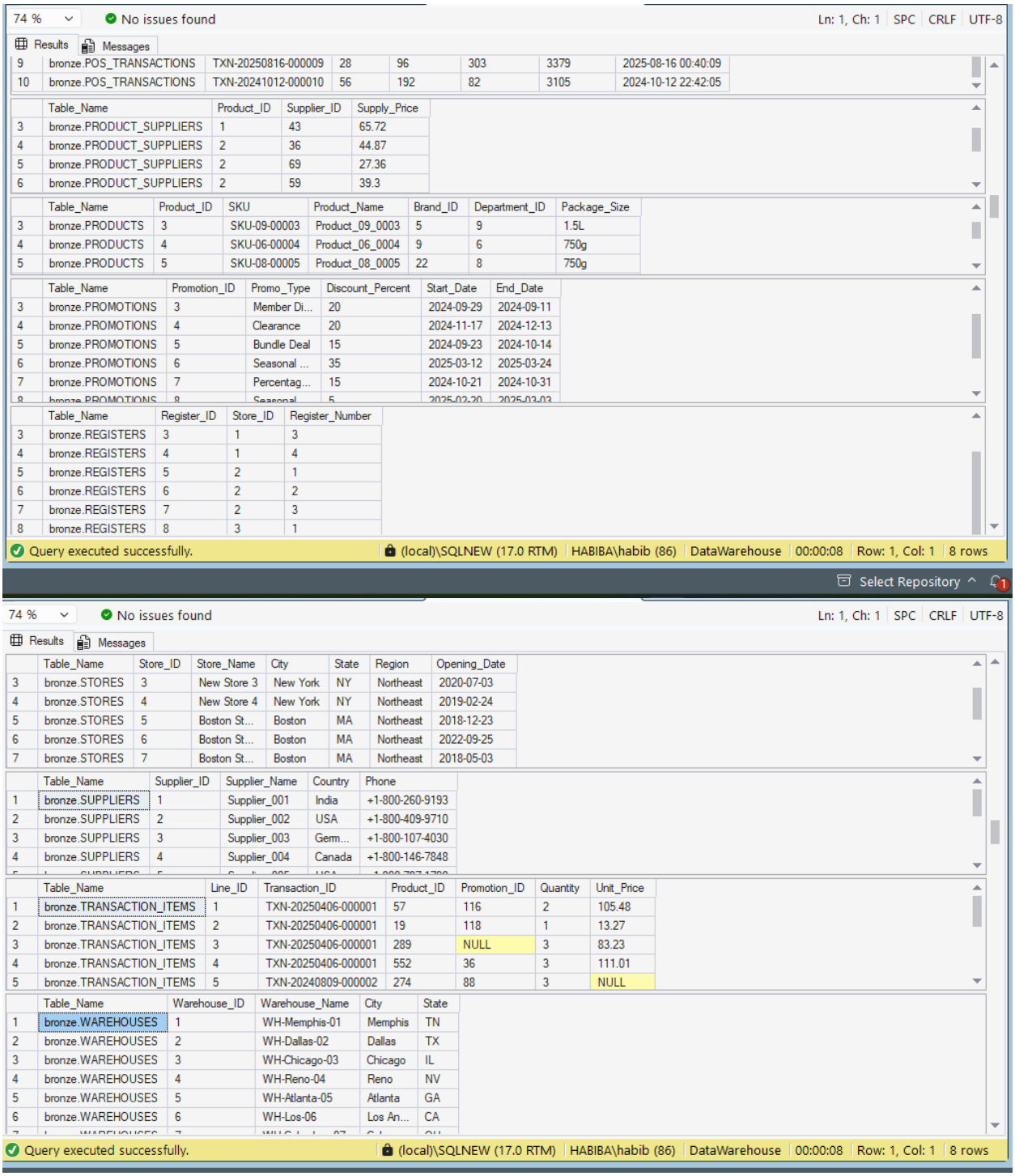
	Table_Name	Order_Item_ID	Order_ID	Product_ID	Promotion_ID	Quantity	Unit_Price
3	bronze.ONLINE_ORDER_ITEMS	3	2	425	64	1	78.41
4	bronze.ONLINE_ORDER_ITEMS	4	2	87	120	2	111.76
5	bronze.ONLINE_ORDER_ITEMS	5	2	365	21	2	28.34
6	bronze.ONLINE_ORDER_ITEMS	6	2	499	63	1	48.3

	Table_Name	Order_ID	Customer_ID	Warehouse_ID	Order_Time	Order_Status	Order_Total
3	bronze.ONLINE_ORDERS	3	1852	4	2024-02-27 22:49:28	Cancelled	230.82
4	bronze.ONLINE_ORDERS	4	3172	2	2025-06-13 07:50:59	Delivered	305.92
5	bronze.ONLINE_ORDERS	5	2337	5	2025-10-21 02:31:22	Processing	574.28
6	bronze.ONLINE_ORDERS	6	2021	7	2024-02-29 01:42:53	Delivered	74.87

	Table_Name	Payment_ID	Order_ID	Payment_Method	Payment_Amount	Payment_Time
3	bronze.PAYMENTS	3	3	Mobile Wallet	230.82	2024-02-28 02:49:28
4	bronze.PAYMENTS	4	4	Gift Card	305.92	2025-06-14 23:50:59
5	bronze.PAYMENTS	5	5	Gift Card	574.28	2025-10-23 19:31:22
6	bronze.PAYMENTS	6	6	Credit Card	74.87	2024-02-29 07:42:53
7	bronze.PAYMENTS	7	7	Credit Card	254.85	2024-08-16 20:05:40
8	bronze.PAYMENTS	8	8	Debit Card	287.98	2025-09-24 22:33:30
9	bronze.PAYMENTS	9	9	Cash	126.84	2024-02-10 15:42:50

	Table_Name	Transaction_ID	Store_ID	Register_ID	Employee_ID	Customer_ID	Transaction_Time
3	bronze.POS_TRANSACTIONS	TXN-20241212-000003	53	180	359	3392	2024-12-12 09:53:04
4	bronze.POS_TRANSACTIONS	TXN-20241130-000004	19	65	512	2343	2024-11-30 18:28:40
5	bronze.POS_TRANSACTIONS	TXN-20250321-000005	4	13	649	1904	2025-03-21 11:08:58
6	bronze.POS_TRANSACTIONS	TXN-20240817-000006	51	172	537	1454	2024-08-17 17:24:38

Query executed successfully. (local)\SQLNEW (17.0 RTM) HABIBA\habib (86) DataWarehouse 00:00:08 Row: 1, Col: 1 8 rows



74 % No issues found Ln: 1, Ch: 1 SPC CRLF UTF-8

Results Messages

9	bronze.POS_TRANSACTIONS	TXN-20250816-000009	28	96	303	3379	2025-08-16 00:40:09
10	bronze.POS_TRANSACTIONS	TXN-20241012-000010	56	192	82	3105	2024-10-12 22:42:05

	Table_Name	Product_ID	Supplier_ID	Supply_Price
3	bronze.PRODUCT_SUPPLIERS	1	43	65.72
4	bronze.PRODUCT_SUPPLIERS	2	36	44.87
5	bronze.PRODUCT_SUPPLIERS	2	69	27.36
6	bronze.PRODUCT_SUPPLIERS	2	59	39.3

	Table_Name	Product_ID	SKU	Product_Name	Brand_ID	Department_ID	Package_Size
3	bronze.PRODUCTS	3	SKU-09-00003	Product_09_0003	5	9	1.5L
4	bronze.PRODUCTS	4	SKU-06-00004	Product_06_0004	9	6	750g
5	bronze.PRODUCTS	5	SKU-08-00005	Product_08_0005	22	8	750g

	Table_Name	Promotion_ID	Promo_Type	Discount_Percent	Start_Date	End_Date
3	bronze.PROMOTIONS	3	Member Di...	20	2024-09-29	2024-09-11
4	bronze.PROMOTIONS	4	Clearance	20	2024-11-17	2024-12-13
5	bronze.PROMOTIONS	5	Bundle Deal	15	2024-09-23	2024-10-14
6	bronze.PROMOTIONS	6	Seasonal ...	35	2025-03-12	2025-03-24
7	bronze.PROMOTIONS	7	Percentag...	15	2024-10-21	2024-10-31
8	bronze.PROMOTIONS	8	Seasonal	5	2025-02-20	2025-03-03

	Table_Name	Register_ID	Store_ID	Register_Number
3	bronze.REGISTERS	3	1	3
4	bronze.REGISTERS	4	1	4
5	bronze.REGISTERS	5	2	1
6	bronze.REGISTERS	6	2	2
7	bronze.REGISTERS	7	2	3
8	bronze.REGISTERS	8	3	1

Query executed successfully. (local)\SQLNEW (17.0 RTM) HABIBA\habib (86) DataWarehouse 00:00:08 Row: 1, Col: 1 8 rows

Select Repository

74 % No issues found Ln: 1, Ch: 1 SPC CRLF UTF-8

Results Messages

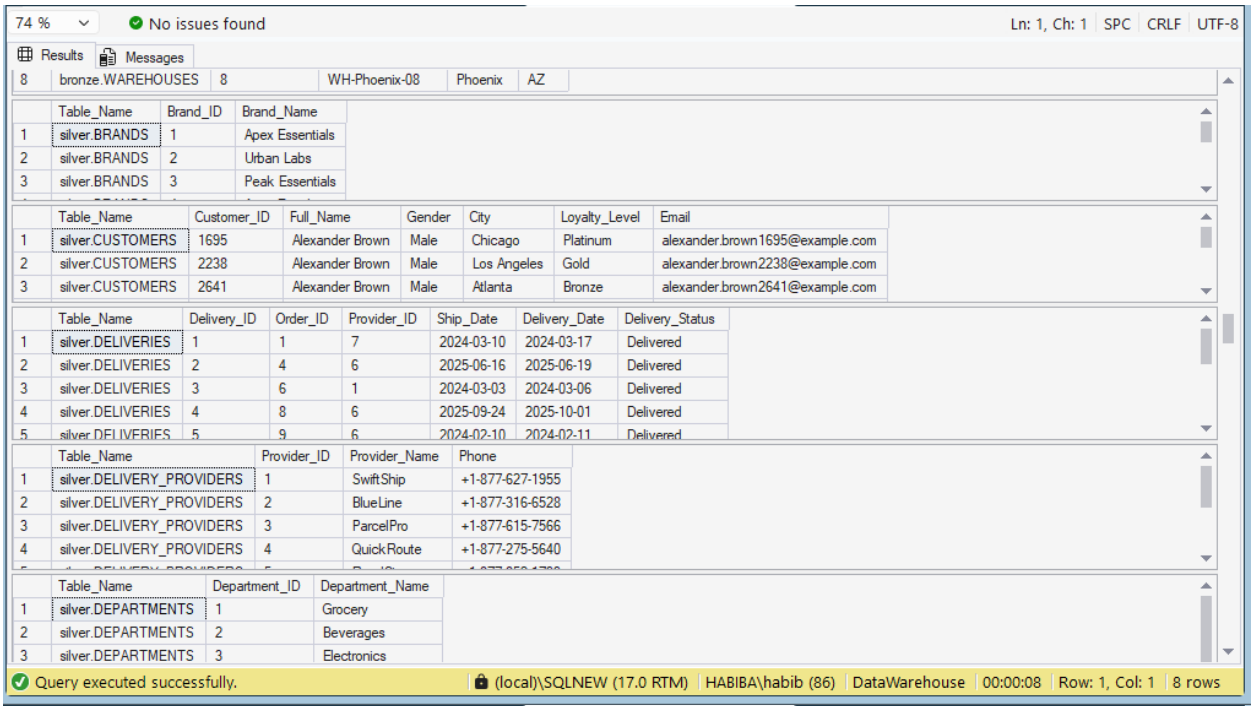
	Table_Name	Store_ID	Store_Name	City	State	Region	Opening_Date
3	bronze.STORES	3	New Store 3	New York	NY	Northeast	2020-07-03
4	bronze.STORES	4	New Store 4	New York	NY	Northeast	2019-02-24
5	bronze.STORES	5	Boston St...	Boston	MA	Northeast	2018-12-23
6	bronze.STORES	6	Boston St...	Boston	MA	Northeast	2022-09-25
7	bronze.STORES	7	Boston St...	Boston	MA	Northeast	2018-05-03

	Table_Name	Supplier_ID	Supplier_Name	Country	Phone
1	bronze.SUPPLIERS	1	Supplier_001	India	+1-800-260-9193
2	bronze.SUPPLIERS	2	Supplier_002	USA	+1-800-409-9710
3	bronze.SUPPLIERS	3	Supplier_003	Germ...	+1-800-107-4030
4	bronze.SUPPLIERS	4	Supplier_004	Canada	+1-800-146-7848

	Table_Name	Line_ID	Transaction_ID	Product_ID	Promotion_ID	Quantity	Unit_Price
1	bronze.TRANSACTION_ITEMS	1	TXN-20250406-000001	57	116	2	105.48
2	bronze.TRANSACTION_ITEMS	2	TXN-20250406-000001	19	118	1	13.27
3	bronze.TRANSACTION_ITEMS	3	TXN-20250406-000001	289	NULL	3	83.23
4	bronze.TRANSACTION_ITEMS	4	TXN-20250406-000001	552	36	3	111.01
5	bronze.TRANSACTION_ITEMS	5	TXN-20240809-000002	274	88	3	NULL

	Table_Name	Warehouse_ID	Warehouse_Name	City	State
1	bronze.WAREHOUSES	1	WH-Memphis-01	Memphis	TN
2	bronze.WAREHOUSES	2	WH-Dallas-02	Dallas	TX
3	bronze.WAREHOUSES	3	WH-Chicago-03	Chicago	IL
4	bronze.WAREHOUSES	4	WH-Reno-04	Reno	NV
5	bronze.WAREHOUSES	5	WH-Atlanta-05	Atlanta	GA
6	bronze.WAREHOUSES	6	WH-Los-06	Los An...	CA

Query executed successfully. (local)\SQLNEW (17.0 RTM) HABIBA\habib (86) DataWarehouse 00:00:08 Row: 1, Col: 1 8 rows



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Results | Messages

8	bronze.WAREHOUSES	8	WH-Phoenix-08	Phoenix	AZ

	Table_Name	Brand_ID	Brand_Name
1	silver.BRANDS	1	Apex Essentials
2	silver.BRANDS	2	Urban Labs
3	silver.BRANDS	3	Peak Essentials

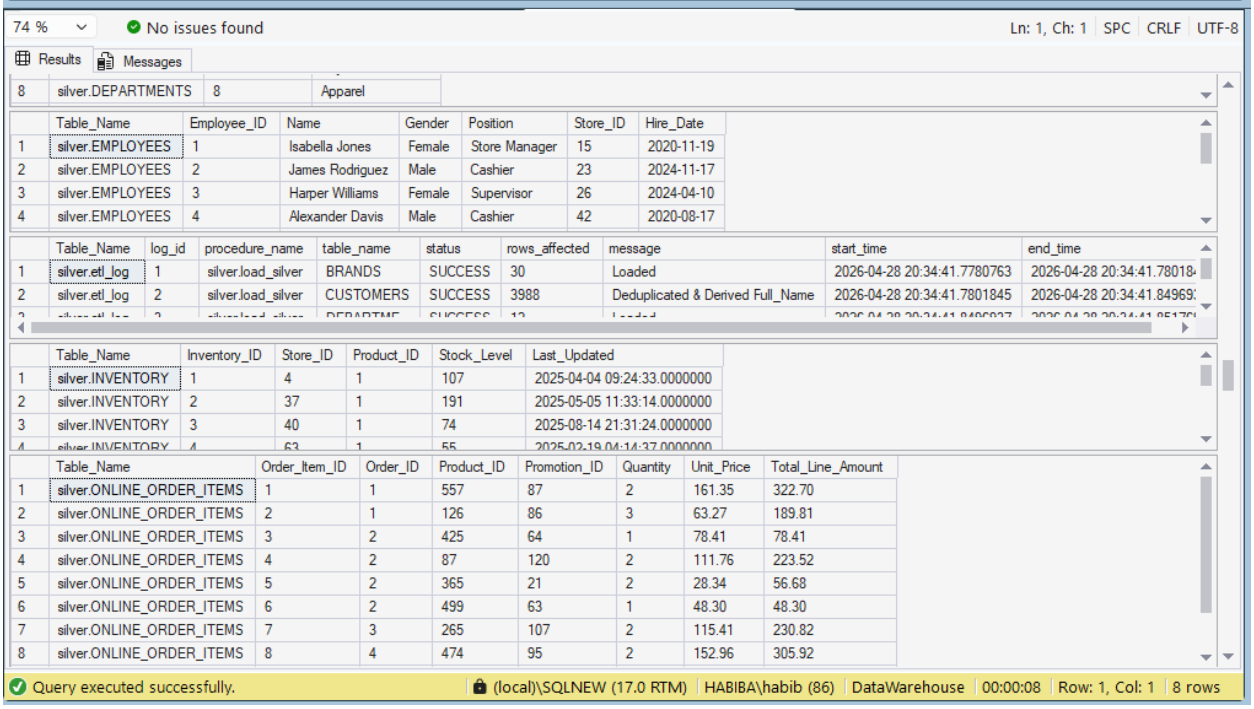
	Table_Name	Customer_ID	Full_Name	Gender	City	Loyalty_Level	Email
1	silver.CUSTOMERS	1695	Alexander Brown	Male	Chicago	Platinum	alexander.brown1695@example.com
2	silver.CUSTOMERS	2238	Alexander Brown	Male	Los Angeles	Gold	alexander.brown2238@example.com
3	silver.CUSTOMERS	2641	Alexander Brown	Male	Atlanta	Bronze	alexander.brown2641@example.com

	Table_Name	Delivery_ID	Order_ID	Provider_ID	Ship_Date	Delivery_Date	Delivery_Status
1	silver.DELIVERIES	1	1	7	2024-03-10	2024-03-17	Delivered
2	silver.DELIVERIES	2	4	6	2025-06-16	2025-06-19	Delivered
3	silver.DELIVERIES	3	6	1	2024-03-03	2024-03-06	Delivered
4	silver.DELIVERIES	4	8	6	2025-09-24	2025-10-01	Delivered
5	silver.DELIVERIES	5	9	6	2024-02-10	2024-02-11	Delivered

	Table_Name	Provider_ID	Provider_Name	Phone
1	silver.DELIVERY_PROVIDERS	1	SwiftShip	+1-877-627-1955
2	silver.DELIVERY_PROVIDERS	2	BlueLine	+1-877-316-6528
3	silver.DELIVERY_PROVIDERS	3	ParcelPro	+1-877-615-7566
4	silver.DELIVERY_PROVIDERS	4	QuickRoute	+1-877-275-5640

	Table_Name	Department_ID	Department_Name
1	silver.DEPARTMENTS	1	Grocery
2	silver.DEPARTMENTS	2	Beverages
3	silver.DEPARTMENTS	3	Electronics

Query executed successfully. | (local)\SQLNEW (17.0 RTM) | HABIBA\habib (86) | DataWarehouse | 00:00:08 | Row: 1, Col: 1 | 8 rows



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Results | Messages

8	silver.DEPARTMENTS	8	Apparel

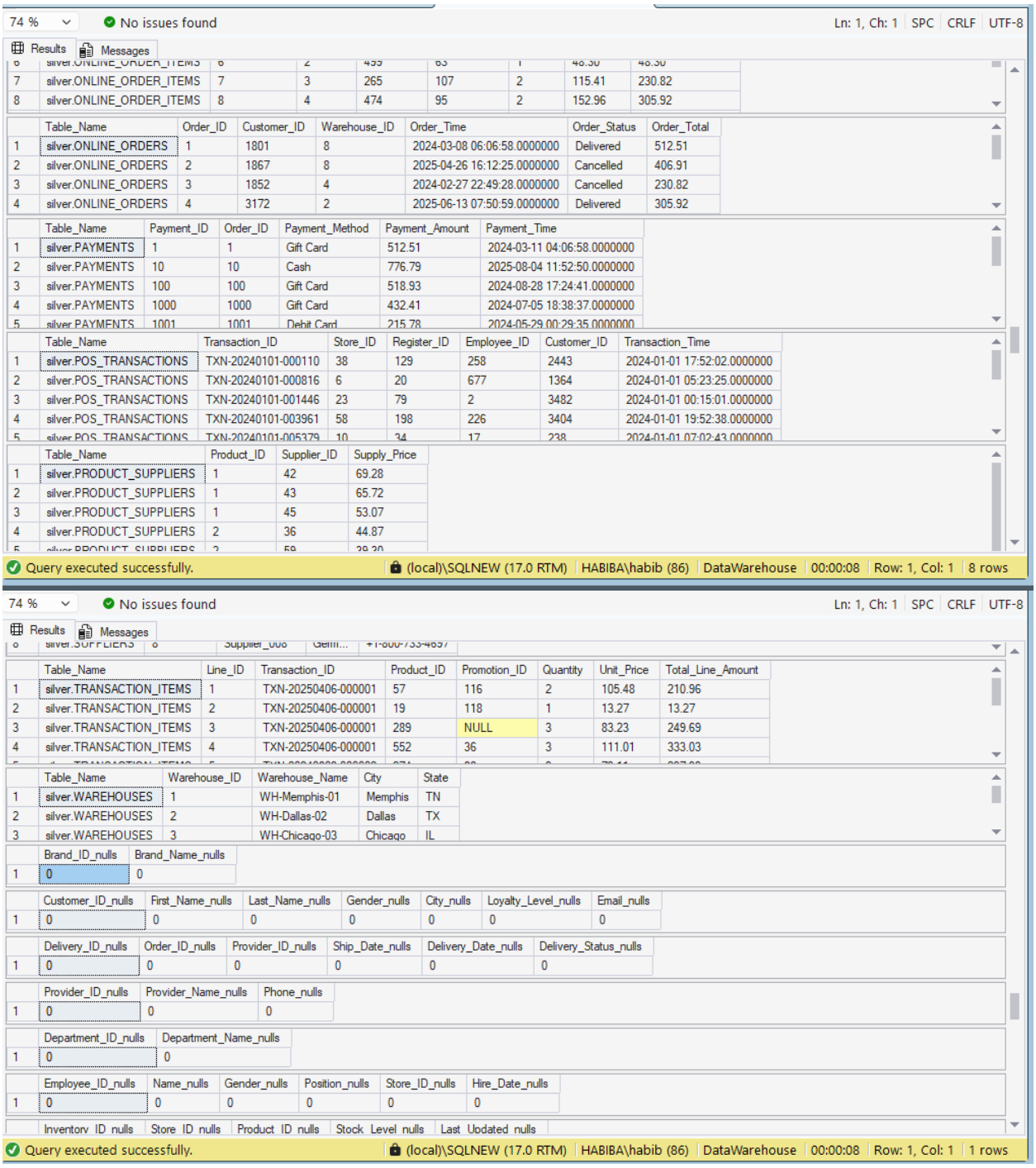
	Table_Name	Employee_ID	Name	Gender	Position	Store_ID	Hire_Date
1	silver.EMPLOYEES	1	Isabella Jones	Female	Store Manager	15	2020-11-19
2	silver.EMPLOYEES	2	James Rodriguez	Male	Cashier	23	2024-11-17
3	silver.EMPLOYEES	3	Harper Williams	Female	Supervisor	26	2024-04-10
4	silver.EMPLOYEES	4	Alexander Davis	Male	Cashier	42	2020-08-17

	Table_Name	log_id	procedure_name	table_name	status	rows_affected	message	start_time	end_time
1	silver.etl_log	1	silver.load_silver	BRANDS	SUCCESS	30	Loaded	2026-04-28 20:34:41.7780763	2026-04-28 20:34:41.78018
2	silver.etl_log	2	silver.load_silver	CUSTOMERS	SUCCESS	3988	Deduplicated & Derived Full_Name	2026-04-28 20:34:41.7801845	2026-04-28 20:34:41.84969

	Table_Name	Inventory_ID	Store_ID	Product_ID	Stock_Level	Last_Updated
1	silver.INVENTORY	1	4	1	107	2025-04-04 09:24:33.0000000
2	silver.INVENTORY	2	37	1	191	2025-05-05 11:33:14.0000000
3	silver.INVENTORY	3	40	1	74	2025-08-14 21:31:24.0000000
4	silver.INVENTORY	4	63	1	55	2025-02-19 04:14:37.0000000

	Table_Name	Order_Item_ID	Order_ID	Product_ID	Promotion_ID	Quantity	Unit_Price	Total_Line_Amount
1	silver.ONLINE_ORDER_ITEMS	1	1	557	87	2	161.35	322.70
2	silver.ONLINE_ORDER_ITEMS	2	1	126	86	3	63.27	189.81
3	silver.ONLINE_ORDER_ITEMS	3	2	425	64	1	78.41	78.41
4	silver.ONLINE_ORDER_ITEMS	4	2	87	120	2	111.76	223.52
5	silver.ONLINE_ORDER_ITEMS	5	2	365	21	2	28.34	56.68
6	silver.ONLINE_ORDER_ITEMS	6	2	499	63	1	48.30	48.30
7	silver.ONLINE_ORDER_ITEMS	7	3	265	107	2	115.41	230.82
8	silver.ONLINE_ORDER_ITEMS	8	4	474	95	2	152.96	305.92

Query executed successfully. | (local)\SQLNEW (17.0 RTM) | HABIBA\habib (86) | DataWarehouse | 00:00:08 | Row: 1, Col: 1 | 8 rows



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Results Messages

7	silver.ONLINE_ORDER_ITEMS	7	3	265	107	2	115.41	230.82
8	silver.ONLINE_ORDER_ITEMS	8	4	474	95	2	152.96	305.92

	Table_Name	Order_ID	Customer_ID	Warehouse_ID	Order_Time	Order_Status	Order_Total
1	silver.ONLINE_ORDERS	1	1801	8	2024-03-08 06:06:58.0000000	Delivered	512.51
2	silver.ONLINE_ORDERS	2	1867	8	2025-04-26 16:12:25.0000000	Cancelled	406.91
3	silver.ONLINE_ORDERS	3	1852	4	2024-02-27 22:49:28.0000000	Cancelled	230.82
4	silver.ONLINE_ORDERS	4	3172	2	2025-06-13 07:50:59.0000000	Delivered	305.92

	Table_Name	Payment_ID	Order_ID	Payment_Method	Payment_Amount	Payment_Time
1	silver.PAYMENTS	1	1	Gift Card	512.51	2024-03-11 04:06:58.0000000
2	silver.PAYMENTS	10	10	Cash	776.79	2025-08-04 11:52:50.0000000
3	silver.PAYMENTS	100	100	Gift Card	518.93	2024-08-28 17:24:41.0000000
4	silver.PAYMENTS	1000	1000	Gift Card	432.41	2024-07-05 18:38:37.0000000
5	silver.PAYMENTS	1001	1001	Debit Card	215.78	2024-05-29 00:29:35.0000000

	Table_Name	Transaction_ID	Store_ID	Register_ID	Employee_ID	Customer_ID	Transaction_Time
1	silver.POS_TRANSACTIONS	TXN-20240101-000110	38	129	258	2443	2024-01-01 17:52:02.0000000
2	silver.POS_TRANSACTIONS	TXN-20240101-000816	6	20	677	1364	2024-01-01 05:23:25.0000000
3	silver.POS_TRANSACTIONS	TXN-20240101-001446	23	79	2	3482	2024-01-01 00:15:01.0000000
4	silver.POS_TRANSACTIONS	TXN-20240101-003961	58	198	226	3404	2024-01-01 19:52:38.0000000
5	silver.POS_TRANSACTIONS	TXN-20240101-005379	10	34	17	238	2024-01-01 07:02:43.0000000

	Table_Name	Product_ID	Supplier_ID	Supply_Price
1	silver.PRODUCT_SUPPLIERS	1	42	69.28
2	silver.PRODUCT_SUPPLIERS	1	43	65.72
3	silver.PRODUCT_SUPPLIERS	1	45	53.07
4	silver.PRODUCT_SUPPLIERS	2	36	44.87
5	silver.PRODUCT_SUPPLIERS	2	59	29.20

Query executed successfully. (local)\SQLNEW (17.0 RTM) HABIBA\habib (86) DataWarehouse 00:00:08 Row: 1, Col: 1 8 rows

74 % No issues found Ln: 1, Ch: 1 SPC CRLF UTF-8

Results Messages

	Table_Name	Line_ID	Transaction_ID	Product_ID	Promotion_ID	Quantity	Unit_Price	Total_Line_Amount
1	silver.TRANSACTION_ITEMS	1	TXN-20250406-000001	57	116	2	105.48	210.96
2	silver.TRANSACTION_ITEMS	2	TXN-20250406-000001	19	118	1	13.27	13.27
3	silver.TRANSACTION_ITEMS	3	TXN-20250406-000001	289	NULL	3	83.23	249.69
4	silver.TRANSACTION_ITEMS	4	TXN-20250406-000001	552	36	3	111.01	333.03

	Table_Name	Warehouse_ID	Warehouse_Name	City	State
1	silver.WAREHOUSES	1	WH-Memphis-01	Memphis	TN
2	silver.WAREHOUSES	2	WH-Dallas-02	Dallas	TX
3	silver.WAREHOUSES	3	WH-Chicago-03	Chicago	IL

	Brand_ID_nulls	Brand_Name_nulls
1	0	0

	Customer_ID_nulls	First_Name_nulls	Last_Name_nulls	Gender_nulls	City_nulls	Loyalty_Level_nulls	Email_nulls
1	0	0	0	0	0	0	0

	Delivery_ID_nulls	Order_ID_nulls	Provider_ID_nulls	Ship_Date_nulls	Delivery_Date_nulls	Delivery_Status_nulls
1	0	0	0	0	0	0

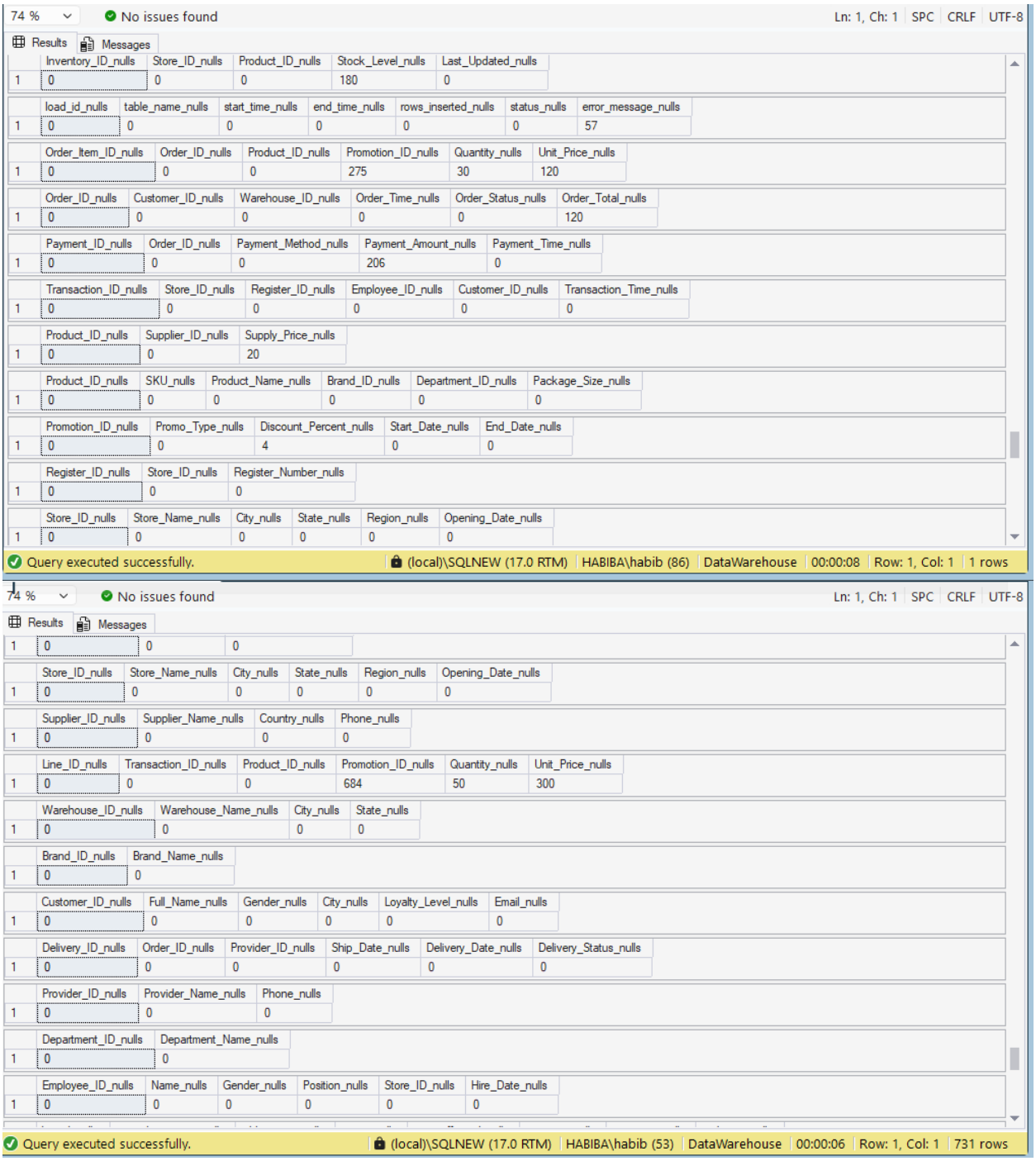
	Provider_ID_nulls	Provider_Name_nulls	Phone_nulls
1	0	0	0

	Department_ID_nulls	Department_Name_nulls
1	0	0

	Employee_ID_nulls	Name_nulls	Gender_nulls	Position_nulls	Store_ID_nulls	Hire_Date_nulls
1	0	0	0	0	0	0

Inventory_ID_nulls	Store_ID_nulls	Product_ID_nulls	Stock_Level_nulls	Last_Updated_nulls

Query executed successfully. (local)\SQLNEW (17.0 RTM) HABIBA\habib (86) DataWarehouse 00:00:08 Row: 1, Col: 1 1 rows



74 % No issues found Ln: 1, Ch: 1 SPC CRLF UTF-8

Results Messages

	Inventory_ID_nulls	Store_ID_nulls	Product_ID_nulls	Stock_Level_nulls	Last_Updated_nulls
1	0	0	0	180	0

	load_id_nulls	table_name_nulls	start_time_nulls	end_time_nulls	rows_inserted_nulls	status_nulls	error_message_nulls
1	0	0	0	0	0	0	57

	Order_Item_ID_nulls	Order_ID_nulls	Product_ID_nulls	Promotion_ID_nulls	Quantity_nulls	Unit_Price_nulls
1	0	0	0	275	30	120

	Order_ID_nulls	Customer_ID_nulls	Warehouse_ID_nulls	Order_Time_nulls	Order_Status_nulls	Order_Total_nulls
1	0	0	0	0	0	120

	Payment_ID_nulls	Order_ID_nulls	Payment_Method_nulls	Payment_Amount_nulls	Payment_Time_nulls
1	0	0	0	206	0

	Transaction_ID_nulls	Store_ID_nulls	Register_ID_nulls	Employee_ID_nulls	Customer_ID_nulls	Transaction_Time_nulls
1	0	0	0	0	0	0

	Product_ID_nulls	Supplier_ID_nulls	Supply_Price_nulls
1	0	0	20

	Product_ID_nulls	SKU_nulls	Product_Name_nulls	Brand_ID_nulls	Department_ID_nulls	Package_Size_nulls
1	0	0	0	0	0	0

	Promotion_ID_nulls	Promo_Type_nulls	Discount_Percent_nulls	Start_Date_nulls	End_Date_nulls
1	0	0	4	0	0

	Register_ID_nulls	Store_ID_nulls	Register_Number_nulls
1	0	0	0

	Store_ID_nulls	Store_Name_nulls	City_nulls	State_nulls	Region_nulls	Opening_Date_nulls
1	0	0	0	0	0	0

Query executed successfully. (local)\SQLNEW (17.0 RTM) HABIBA\habib (86) DataWarehouse 00:00:08 Row: 1, Col: 1 1 rows

74 % No issues found Ln: 1, Ch: 1 SPC CRLF UTF-8

Results Messages

1	0	0	0

	Store_ID_nulls	Store_Name_nulls	City_nulls	State_nulls	Region_nulls	Opening_Date_nulls
1	0	0	0	0	0	0

	Supplier_ID_nulls	Supplier_Name_nulls	Country_nulls	Phone_nulls
1	0	0	0	0

	Line_ID_nulls	Transaction_ID_nulls	Product_ID_nulls	Promotion_ID_nulls	Quantity_nulls	Unit_Price_nulls
1	0	0	0	684	50	300

	Warehouse_ID_nulls	Warehouse_Name_nulls	City_nulls	State_nulls
1	0	0	0	0

	Brand_ID_nulls	Brand_Name_nulls
1	0	0

	Customer_ID_nulls	Full_Name_nulls	Gender_nulls	City_nulls	Loyalty_Level_nulls	Email_nulls
1	0	0	0	0	0	0

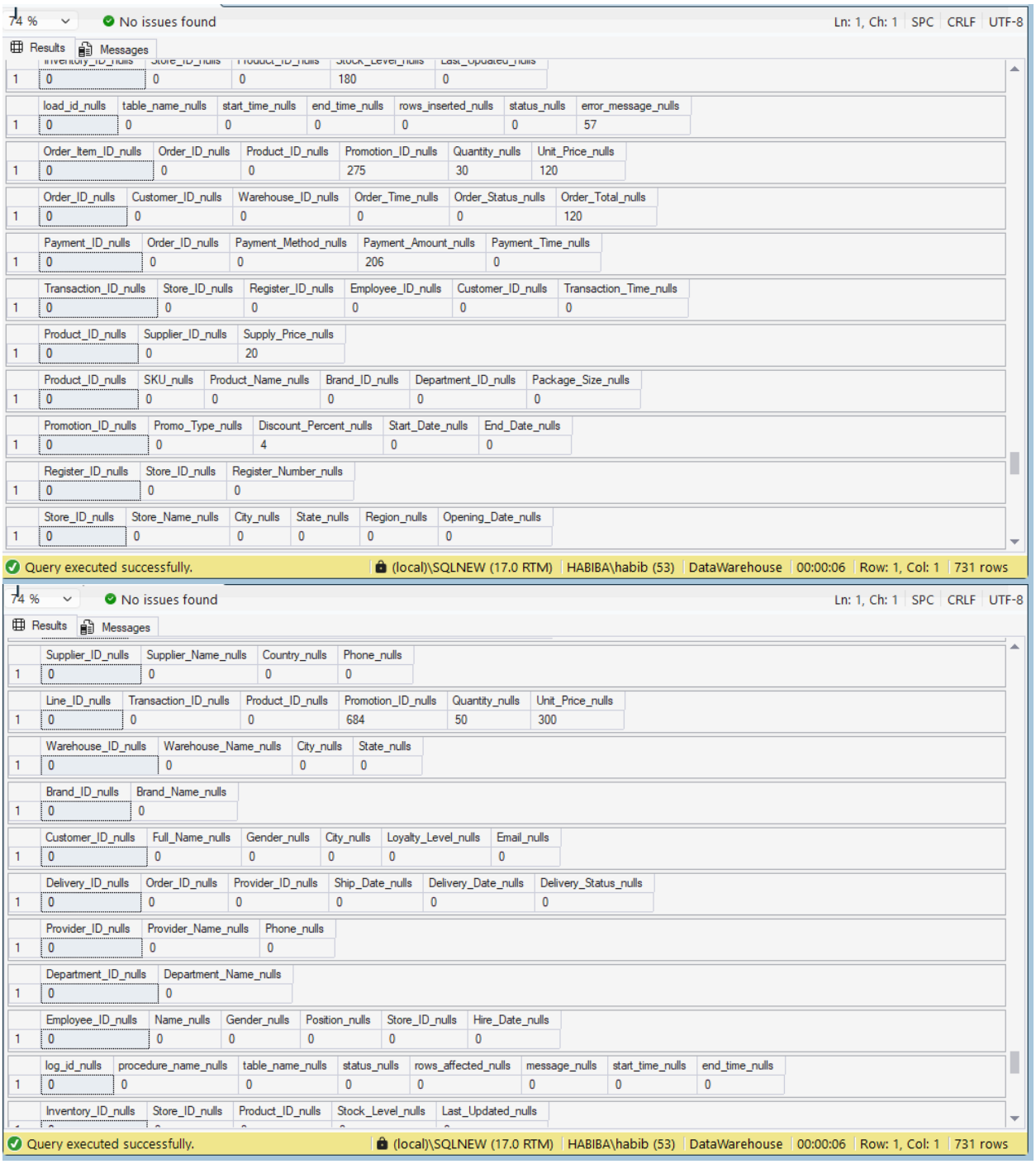
	Delivery_ID_nulls	Order_ID_nulls	Provider_ID_nulls	Ship_Date_nulls	Delivery_Date_nulls	Delivery_Status_nulls
1	0	0	0	0	0	0

	Provider_ID_nulls	Provider_Name_nulls	Phone_nulls
1	0	0	0

	Department_ID_nulls	Department_Name_nulls
1	0	0

	Employee_ID_nulls	Name_nulls	Gender_nulls	Position_nulls	Store_ID_nulls	Hire_Date_nulls
1	0	0	0	0	0	0

Query executed successfully. (local)\SQLNEW (17.0 RTM) HABIBA\habib (53) DataWarehouse 00:00:06 Row: 1, Col: 1 731 rows



74 % No issues found Ln: 1, Ch: 1 SPC CRLF UTF-8

Results Messages

	Inventory_ID_nulls	Store_ID_nulls	Product_ID_nulls	Stock_Level_nulls	Last_Updated_nulls
1	0	0	0	180	0

	load_id_nulls	table_name_nulls	start_time_nulls	end_time_nulls	rows_inserted_nulls	status_nulls	error_message_nulls
1	0	0	0	0	0	0	57

	Order_Item_ID_nulls	Order_ID_nulls	Product_ID_nulls	Promotion_ID_nulls	Quantity_nulls	Unit_Price_nulls
1	0	0	0	275	30	120

	Order_ID_nulls	Customer_ID_nulls	Warehouse_ID_nulls	Order_Time_nulls	Order_Status_nulls	Order_Total_nulls
1	0	0	0	0	0	120

	Payment_ID_nulls	Order_ID_nulls	Payment_Method_nulls	Payment_Amount_nulls	Payment_Time_nulls
1	0	0	0	206	0

	Transaction_ID_nulls	Store_ID_nulls	Register_ID_nulls	Employee_ID_nulls	Customer_ID_nulls	Transaction_Time_nulls
1	0	0	0	0	0	0

	Product_ID_nulls	Supplier_ID_nulls	Supply_Price_nulls
1	0	0	20

	Product_ID_nulls	SKU_nulls	Product_Name_nulls	Brand_ID_nulls	Department_ID_nulls	Package_Size_nulls
1	0	0	0	0	0	0

	Promotion_ID_nulls	Promo_Type_nulls	Discount_Percent_nulls	Start_Date_nulls	End_Date_nulls
1	0	0	4	0	0

	Register_ID_nulls	Store_ID_nulls	Register_Number_nulls
1	0	0	0

	Store_ID_nulls	Store_Name_nulls	City_nulls	State_nulls	Region_nulls	Opening_Date_nulls
1	0	0	0	0	0	0

Query executed successfully. (local)\SQLNEW (17.0 RTM) HABIBA\habib (53) DataWarehouse 00:00:06 Row: 1, Col: 1 731 rows

74 % No issues found Ln: 1, Ch: 1 SPC CRLF UTF-8

Results Messages

	Supplier_ID_nulls	Supplier_Name_nulls	Country_nulls	Phone_nulls
1	0	0	0	0

	Line_ID_nulls	Transaction_ID_nulls	Product_ID_nulls	Promotion_ID_nulls	Quantity_nulls	Unit_Price_nulls
1	0	0	0	684	50	300

	Warehouse_ID_nulls	Warehouse_Name_nulls	City_nulls	State_nulls
1	0	0	0	0

	Brand_ID_nulls	Brand_Name_nulls
1	0	0

	Customer_ID_nulls	Full_Name_nulls	Gender_nulls	City_nulls	Loyalty_Level_nulls	Email_nulls
1	0	0	0	0	0	0

	Delivery_ID_nulls	Order_ID_nulls	Provider_ID_nulls	Ship_Date_nulls	Delivery_Date_nulls	Delivery_Status_nulls
1	0	0	0	0	0	0

	Provider_ID_nulls	Provider_Name_nulls	Phone_nulls
1	0	0	0

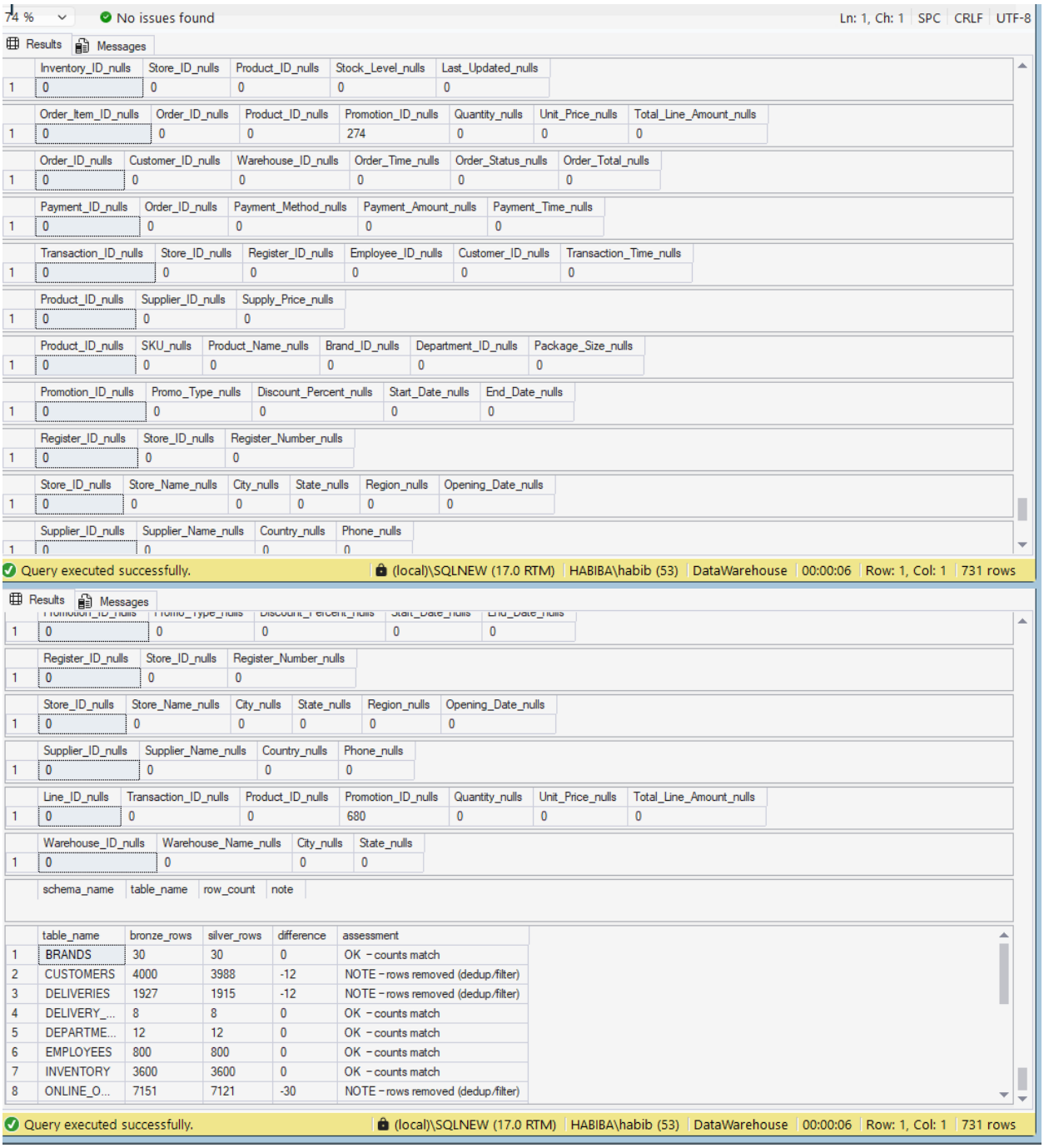
	Department_ID_nulls	Department_Name_nulls
1	0	0

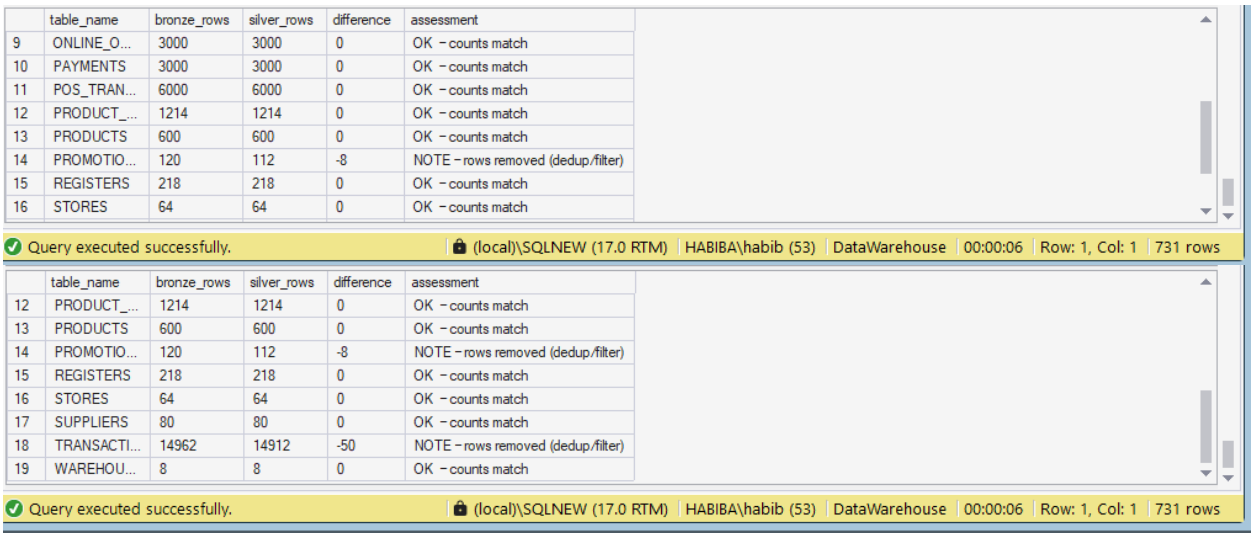
	Employee_ID_nulls	Name_nulls	Gender_nulls	Position_nulls	Store_ID_nulls	Hire_Date_nulls
1	0	0	0	0	0	0

	log_id_nulls	procedure_name_nulls	table_name_nulls	status_nulls	rows_affected_nulls	message_nulls	start_time_nulls	end_time_nulls
1	0	0	0	0	0	0	0	0

	Inventory_ID_nulls	Store_ID_nulls	Product_ID_nulls	Stock_Level_nulls	Last_Updated_nulls

Query executed successfully. (local)\SQLNEW (17.0 RTM) HABIBA\habib (53) DataWarehouse 00:00:06 Row: 1, Col: 1 731 rows


74 %
No issues found
Ln: 1, Ch: 1
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UTF-8
Results
Messages
Inventory_ID_nulls | Store_ID_nulls | Product_ID_nulls | Stock_Level_nulls | Last_Updated_nulls
1 | 0 | 0 | 0 | 0 | 0
Order_Item_ID_nulls | Order_ID_nulls | Product_ID_nulls | Promotion_ID_nulls | Quantity_nulls | Unit_Price_nulls | Total_Line_Amount_nulls
1 | 0 | 0 | 0 | 274 | 0 | 0 | 0
Order_ID_nulls | Customer_ID_nulls | Warehouse_ID_nulls | Order_Time_nulls | Order_Status_nulls | Order_Total_nulls
1 | 0 | 0 | 0 | 0 | 0 | 0
Payment_ID_nulls | Order_ID_nulls | Payment_Method_nulls | Payment_Amount_nulls | Payment_Time_nulls
1 | 0 | 0 | 0 | 0 | 0
Transaction_ID_nulls | Store_ID_nulls | Register_ID_nulls | Employee_ID_nulls | Customer_ID_nulls | Transaction_Time_nulls
1 | 0 | 0 | 0 | 0 | 0 | 0
Product_ID_nulls | Supplier_ID_nulls | Supply_Price_nulls
1 | 0 | 0 | 0
Product_ID_nulls | SKU_nulls | Product_Name_nulls | Brand_ID_nulls | Department_ID_nulls | Package_Size_nulls
1 | 0 | 0 | 0 | 0 | 0 | 0
Promotion_ID_nulls | Promo_Type_nulls | Discount_Percent_nulls | Start_Date_nulls | End_Date_nulls
1 | 0 | 0 | 0 | 0 | 0
Register_ID_nulls | Store_ID_nulls | Register_Number_nulls
1 | 0 | 0 | 0
Store_ID_nulls | Store_Name_nulls | City_nulls | State_nulls | Region_nulls | Opening_Date_nulls
1 | 0 | 0 | 0 | 0 | 0 | 0
Supplier_ID_nulls | Supplier_Name_nulls | Country_nulls | Phone_nulls
1 | 0 | 0 | 0 | 0
Query executed successfully.
(local)\SQLNEW (17.0 RTM)
HABIBA\habib (53)
DataWarehouse
00:00:06
Row: 1, Col: 1
731 rows
Results
Messages
Register_ID_nulls | Store_ID_nulls | Register_Number_nulls
1 | 0 | 0 | 0
Store_ID_nulls | Store_Name_nulls | City_nulls | State_nulls | Region_nulls | Opening_Date_nulls
1 | 0 | 0 | 0 | 0 | 0 | 0
Supplier_ID_nulls | Supplier_Name_nulls | Country_nulls | Phone_nulls
1 | 0 | 0 | 0 | 0
Line_ID_nulls | Transaction_ID_nulls | Product_ID_nulls | Promotion_ID_nulls | Quantity_nulls | Unit_Price_nulls | Total_Line_Amount_nulls
1 | 0 | 0 | 0 | 680 | 0 | 0 | 0
Warehouse_ID_nulls | Warehouse_Name_nulls | City_nulls | State_nulls
1 | 0 | 0 | 0 | 0
schema_name | table_name | row_count | note
table_name | bronze_rows | silver_rows | difference | assessment
1 | BRANDS | 30 | 30 | 0 | OK – counts match
2 | CUSTOMERS | 4000 | 3988 | -12 | NOTE – rows removed (dedup/filter)
3 | DELIVERIES | 1927 | 1915 | -12 | NOTE – rows removed (dedup/filter)
4 | DELIVERY_... | 8 | 8 | 0 | OK – counts match
5 | DEPARTME... | 12 | 12 | 0 | OK – counts match
6 | EMPLOYEES | 800 | 800 | 0 | OK – counts match
7 | INVENTORY | 3600 | 3600 | 0 | OK – counts match
8 | ONLINE_O... | 7151 | 7121 | -30 | NOTE – rows removed (dedup/filter)
Query executed successfully.
(local)\SQLNEW (17.0 RTM)
HABIBA\habib (53)
DataWarehouse
00:00:06
Row: 1, Col: 1
731 rows



	table_name	bronze_rows	silver_rows	difference	assessment
9	ONLINE_O...	3000	3000	0	OK – counts match
10	PAYMENTS	3000	3000	0	OK – counts match
11	POS_TRAN...	6000	6000	0	OK – counts match
12	PRODUCT_...	1214	1214	0	OK – counts match
13	PRODUCTS	600	600	0	OK – counts match
14	PROMOTIO...	120	112	-8	NOTE – rows removed (dedup/filter)
15	REGISTERS	218	218	0	OK – counts match
16	STORES	64	64	0	OK – counts match

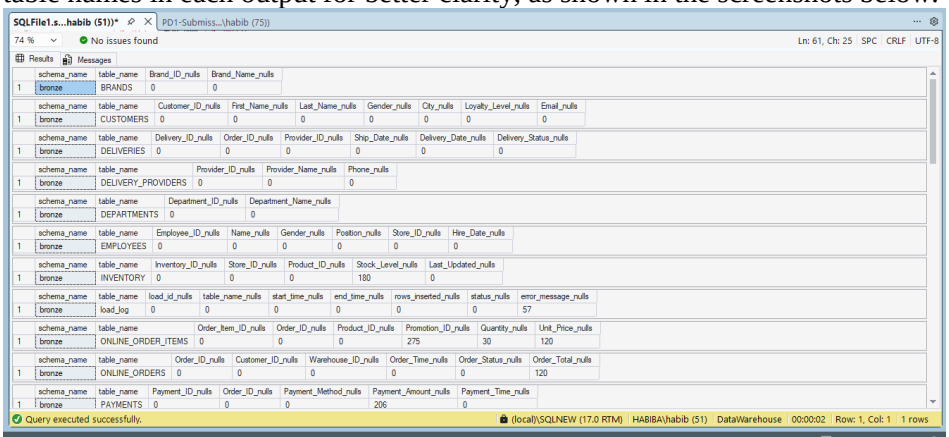
Query executed successfully. (local)\SQLNEW (17.0 RTM) HABIBA\habib (53) DataWarehouse 00:00:06 Row: 1, Col: 1 731 rows

	table_name	bronze_rows	silver_rows	difference	assessment
12	PRODUCT_...	1214	1214	0	OK – counts match
13	PRODUCTS	600	600	0	OK – counts match
14	PROMOTIO...	120	112	-8	NOTE – rows removed (dedup/filter)
15	REGISTERS	218	218	0	OK – counts match
16	STORES	64	64	0	OK – counts match
17	SUPPLIERS	80	80	0	OK – counts match
18	TRANSACTI...	14962	14912	-50	NOTE – rows removed (dedup/filter)
19	WAREHOU...	8	8	0	OK – counts match

Query executed successfully. (local)\SQLNEW (17.0 RTM) HABIBA\habib (53) DataWarehouse 00:00:06 Row: 1, Col: 1 731 rows

Since the original script lacked a column to distinguish between the **Bronze** and **Silver** layers in the results(in SECTION 6 : NULL-VALUE AUDIT PER COLUMN), it made it difficult to identify exactly which schema the

null counts belonged to; therefore, I have generated updated code that explicitly includes the schema and table names in each output for better clarity, as shown in the screenshots below.


SQLFile1.s...habib (51))* | PD1-Submiss...\habib (75))

74 % | No issues found | Ln: 61, Ch: 25 | SPC | CRLF | UTF-8

Results | Messages

	schema_name	table_name	Brand_ID_nulls	Brand_Name_nulls
1	bronze	BRANDS	0	0

	schema_name	table_name	Customer_ID_nulls	First_Name_nulls	Last_Name_nulls	Gender_nulls	City_nulls	Loyalty_Level_nulls	Email_nulls
1	bronze	CUSTOMERS	0	0	0	0	0	0	0

	schema_name	table_name	Delivery_ID_nulls	Order_ID_nulls	Provider_ID_nulls	Ship_Date_nulls	Delivery_Date_nulls	Delivery_Status_nulls
1	bronze	DELIVERIES	0	0	0	0	0	0

	schema_name	table_name	Provider_ID_nulls	Provider_Name_nulls	Phone_nulls
1	bronze	DELIVERY_PROVIDERS	0	0	0

	schema_name	table_name	Department_ID_nulls	Department_Name_nulls
1	bronze	DEPARTMENTS	0	0

	schema_name	table_name	Employee_ID_nulls	Name_nulls	Gender_nulls	Position_nulls	Store_ID_nulls	Hire_Date_nulls
1	bronze	EMPLOYEES	0	0	0	0	0	0

	schema_name	table_name	Inventory_ID_nulls	Store_ID_nulls	Product_ID_nulls	Stock_Level_nulls	Last_Updated_nulls
1	bronze	INVENTORY	0	0	0	180	0

	schema_name	table_name	load_id_nulls	table_name_nulls	start_time_nulls	end_time_nulls	rows_inserted_nulls	status_nulls	error_message_nulls
1	bronze	load_log	0	0	0	0	0	0	57

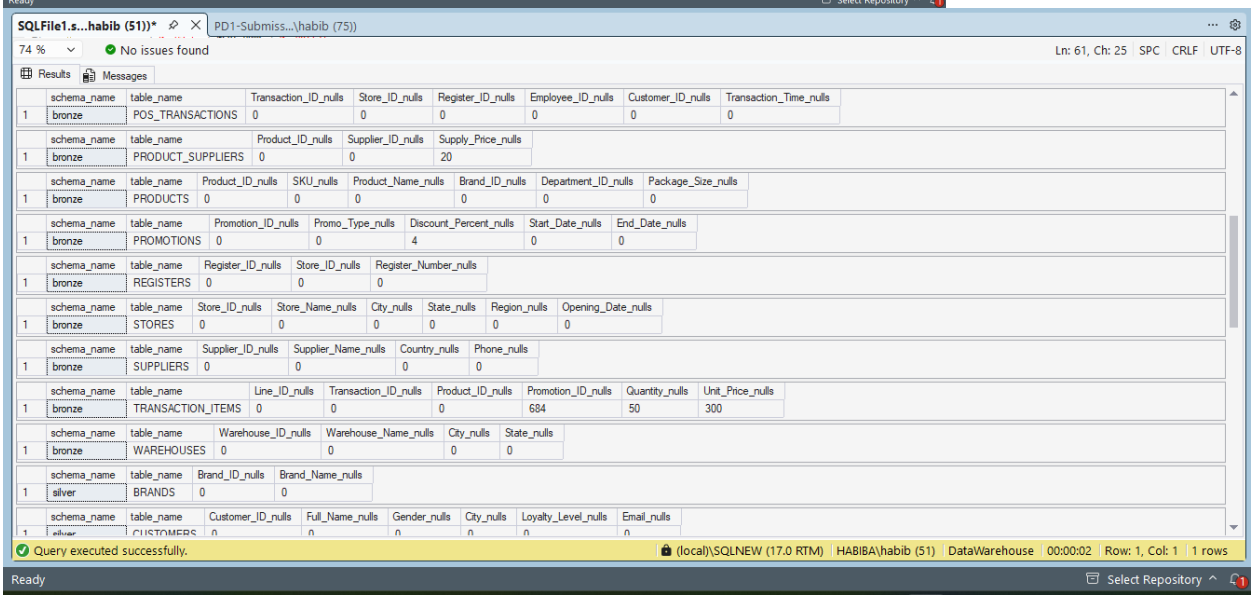
	schema_name	table_name	Order_Item_ID_nulls	Order_ID_nulls	Product_ID_nulls	Promotion_ID_nulls	Quantity_nulls	Unit_Price_nulls
1	bronze	ONLINE_ORDER_ITEMS	0	0	0	275	30	120

	schema_name	table_name	Order_ID_nulls	Customer_ID_nulls	Warehouse_ID_nulls	Order_Time_nulls	Order_Status_nulls	Order_Total_nulls
1	bronze	ONLINE_ORDERS	0	0	0	0	0	120

	schema_name	table_name	Payment_ID_nulls	Order_ID_nulls	Payment_Method_nulls	Payment_Amount_nulls	Payment_Time_nulls
1	bronze	PAYMENTS	0	0	0	206	0

Query executed successfully. | (local)\SQLNEW (17.0 RTM) | HABIBA\habib (51) | DataWarehouse | 00:00:02 | Row: 1, Col: 1 | 1 rows

Ready | Select Repository


SQLFile1.s...habib (51))* | PD1-Submiss...\habib (75))

74 % | No issues found | Ln: 61, Ch: 25 | SPC | CRLF | UTF-8

Results | Messages

	schema_name	table_name	Transaction_ID_nulls	Store_ID_nulls	Register_ID_nulls	Employee_ID_nulls	Customer_ID_nulls	Transaction_Time_nulls
1	bronze	POS_TRANSACTIONS	0	0	0	0	0	0

	schema_name	table_name	Product_ID_nulls	Supplier_ID_nulls	Supply_Price_nulls
1	bronze	PRODUCT_SUPPLIERS	0	0	20

	schema_name	table_name	Product_ID_nulls	SKU_nulls	Product_Name_nulls	Brand_ID_nulls	Department_ID_nulls	Package_Size_nulls
1	bronze	PRODUCTS	0	0	0	0	0	0

	schema_name	table_name	Promotion_ID_nulls	Promo_Type_nulls	Discount_Percent_nulls	Start_Date_nulls	End_Date_nulls
1	bronze	PROMOTIONS	0	0	4	0	0

	schema_name	table_name	Register_ID_nulls	Store_ID_nulls	Register_Number_nulls
1	bronze	REGISTERS	0	0	0

	schema_name	table_name	Store_ID_nulls	Store_Name_nulls	City_nulls	State_nulls	Region_nulls	Opening_Date_nulls
1	bronze	STORES	0	0	0	0	0	0

	schema_name	table_name	Supplier_ID_nulls	Supplier_Name_nulls	Country_nulls	Phone_nulls
1	bronze	SUPPLIERS	0	0	0	0

	schema_name	table_name	Line_ID_nulls	Transaction_ID_nulls	Product_ID_nulls	Promotion_ID_nulls	Quantity_nulls	Unit_Price_nulls
1	bronze	TRANSACTION_ITEMS	0	0	0	684	50	300

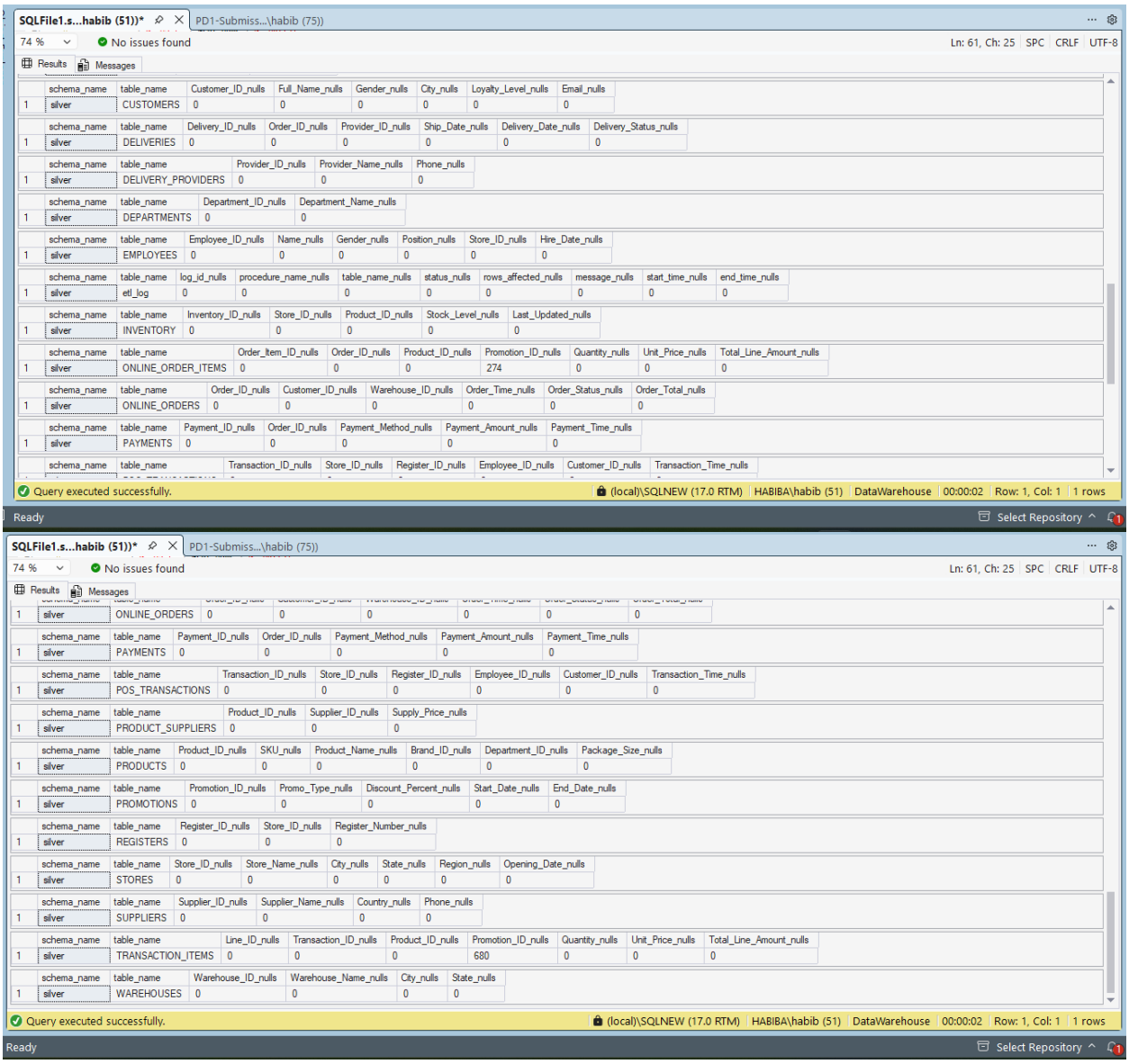
	schema_name	table_name	Warehouse_ID_nulls	Warehouse_Name_nulls	City_nulls	State_nulls
1	bronze	WAREHOUSES	0	0	0	0

	schema_name	table_name	Brand_ID_nulls	Brand_Name_nulls
1	silver	BRANDS	0	0

	schema_name	table_name	Customer_ID_nulls	Full_Name_nulls	Gender_nulls	City_nulls	Loyalty_Level_nulls	Email_nulls
1	silver	CUSTOMERS	0	0	0	0	0	0

Query executed successfully. | (local)\SQLNEW (17.0 RTM) | HABIBA\habib (51) | DataWarehouse | 00:00:02 | Row: 1, Col: 1 | 1 rows

Ready | Select Repository


SQLFile1.s...habib (51))*
PD1-Submiss...\habib (75))
74 %
No issues found
Ln: 61, Ch: 25 SPC CRLF UTF-8
Results
Messages
schema_name table_name Customer_ID_nulls Full_Name_nulls Gender_nulls City_nulls Loyalty_Level_nulls Email_nulls
1 silver CUSTOMERS 0 0 0 0 0 0
schema_name table_name Delivery_ID_nulls Order_ID_nulls Provider_ID_nulls Ship_Date_nulls Delivery_Date_nulls Delivery_Status_nulls
1 silver DELIVERIES 0 0 0 0 0 0
schema_name table_name Provider_ID_nulls Provider_Name_nulls Phone_nulls
1 silver DELIVERY_PROVIDERS 0 0 0
schema_name table_name Department_ID_nulls Department_Name_nulls
1 silver DEPARTMENTS 0 0
schema_name table_name Employee_ID_nulls Name_nulls Gender_nulls Position_nulls Store_ID_nulls Hire_Date_nulls
1 silver EMPLOYEES 0 0 0 0 0 0
schema_name table_name log_id_nulls procedure_name_nulls table_name_nulls status_nulls rows_affected_nulls message_nulls start_time_nulls end_time_nulls
1 silver etl_log 0 0 0 0 0 0 0 0
schema_name table_name Inventory_ID_nulls Store_ID_nulls Product_ID_nulls Stock_Level_nulls Last_Updated_nulls
1 silver INVENTORY 0 0 0 0 0
schema_name table_name Order_Item_ID_nulls Order_ID_nulls Product_ID_nulls Promotion_ID_nulls Quantity_nulls Unit_Price_nulls Total_Line_Amount_nulls
1 silver ONLINE_ORDER_ITEMS 0 0 0 274 0 0 0
schema_name table_name Order_ID_nulls Customer_ID_nulls Warehouse_ID_nulls Order_Time_nulls Order_Status_nulls Order_Total_nulls
1 silver ONLINE_ORDERS 0 0 0 0 0 0
schema_name table_name Payment_ID_nulls Order_ID_nulls Payment_Method_nulls Payment_Amount_nulls Payment_Time_nulls
1 silver PAYMENTS 0 0 0 0 0
schema_name table_name Transaction_ID_nulls Store_ID_nulls Register_ID_nulls Employee_ID_nulls Customer_ID_nulls Transaction_Time_nulls
Query executed successfully.
(local)\SQLNEW (17.0 RTM) HABIBA\habib (51) DataWarehouse 00:00:02 Row: 1, Col: 1 1 rows
Ready
Select Repository

SQLFile1.s...habib (51))*
PD1-Submiss...\habib (75))
74 %
No issues found
Ln: 61, Ch: 25 SPC CRLF UTF-8
Results
Messages
1 silver ONLINE_ORDERS 0 0 0 0 0
schema_name table_name Payment_ID_nulls Order_ID_nulls Payment_Method_nulls Payment_Amount_nulls Payment_Time_nulls
1 silver PAYMENTS 0 0 0 0 0
schema_name table_name Transaction_ID_nulls Store_ID_nulls Register_ID_nulls Employee_ID_nulls Customer_ID_nulls Transaction_Time_nulls
1 silver POS_TRANSACTIONS 0 0 0 0 0 0
schema_name table_name Product_ID_nulls Supplier_ID_nulls Supply_Price_nulls
1 silver PRODUCT_SUPPLIERS 0 0 0
schema_name table_name Product_ID_nulls SKU_nulls Product_Name_nulls Brand_ID_nulls Department_ID_nulls Package_Size_nulls
1 silver PRODUCTS 0 0 0 0 0 0
schema_name table_name Promotion_ID_nulls Promo_Type_nulls Discount_Percent_nulls Start_Date_nulls End_Date_nulls
1 silver PROMOTIONS 0 0 0 0 0
schema_name table_name Register_ID_nulls Store_ID_nulls Register_Number_nulls
1 silver REGISTERS 0 0 0
schema_name table_name Store_ID_nulls Store_Name_nulls City_nulls State_nulls Region_nulls Opening_Date_nulls
1 silver STORES 0 0 0 0 0 0
schema_name table_name Supplier_ID_nulls Supplier_Name_nulls Country_nulls Phone_nulls
1 silver SUPPLIERS 0 0 0 0
schema_name table_name Line_ID_nulls Transaction_ID_nulls Product_ID_nulls Promotion_ID_nulls Quantity_nulls Unit_Price_nulls Total_Line_Amount_nulls
1 silver TRANSACTION_ITEMS 0 0 0 680 0 0 0
schema_name table_name Warehouse_ID_nulls Warehouse_Name_nulls City_nulls State_nulls
1 silver WAREHOUSES 0 0 0 0
Query executed successfully.
(local)\SQLNEW (17.0 RTM) HABIBA\habib (51) DataWarehouse 00:00:02 Row: 1, Col: 1 1 rows
Ready
Select Repository

Milestone 2:

1.Description of the technical work carried out in the Gold Layer Layer

The Gold Layer was implemented to provide business ready analytical data structured using star schema design, as required by the project guidelines

A dedicated schema named gold was created to store all dimension and fact tables.

Star Schema Design

Two star schemas were implemented representing:

- Store Sales business process
- Online Sales business process

These schemas together form a **fact constellation** through shared dimensions.

Dimension Tables

Dimension tables were created using cleaned and enriched data from the Silver Layer.
Each dimension includes a surrogate key generated using IDENTITY(1,1).

Shared dimensions include:

- Date
- Customer
- Product
- Promotion

Additional dimensions:

- Store, Employee (store sales)
- Warehouse, Order Status, Payment Method (online sales)

Fact Tables

Two fact tables were created:

- fact_store_sales
- fact_online_sales

Each fact table:

- Stores transactional measures (quantity, unit price, sales amount)
- References dimensions using surrogate keys (foreign keys)

Granularity

- fact_store_sales: One row per transaction item
- fact_online_sales: One row per online order item

No aggregation was applied; data is stored at detailed transactional level.

Data Loading Process

The Gold Layer is populated using a stored procedure (gold.load_gold)

The procedure performs:

- Full refresh using DELETE operations
- Loading dimension tables first
- Then loading fact tables

Data Integration and Business Logic

Data was integrated from multiple Silver tables using joins.

Additional business logic was applied, including:

- Mapping order status to descriptive meanings
- Categorizing payment methods into providers
- Combining product, brand, and department information

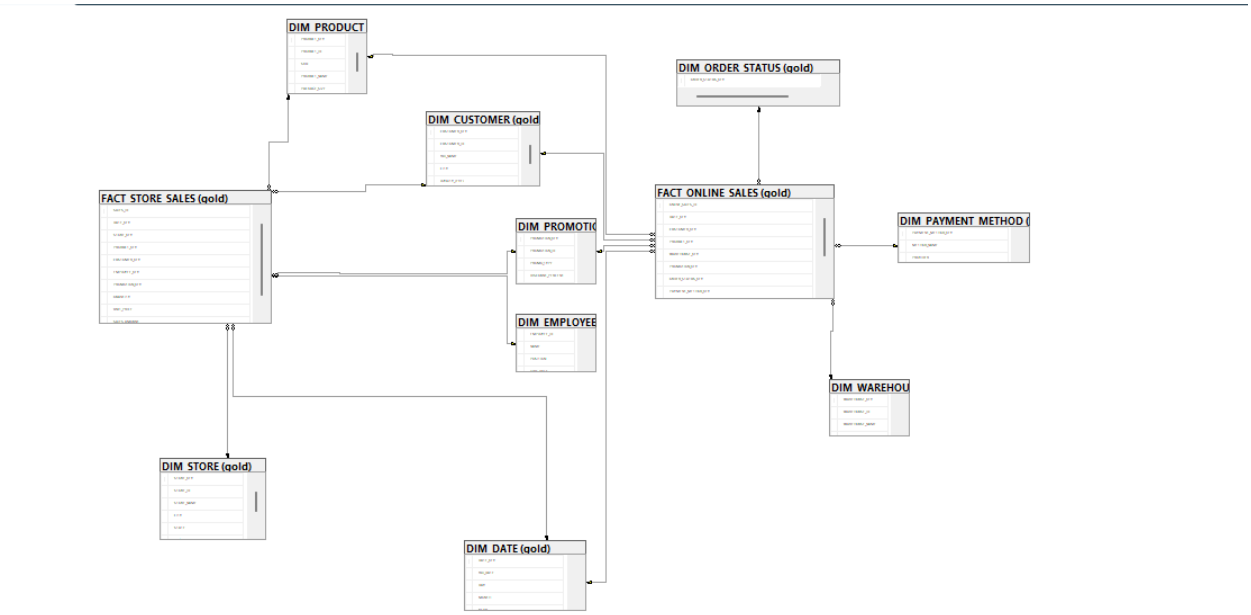
Referential Integrity

All fact tables reference dimension tables using surrogate keys, ensuring consistency and accurate analytical relationships.

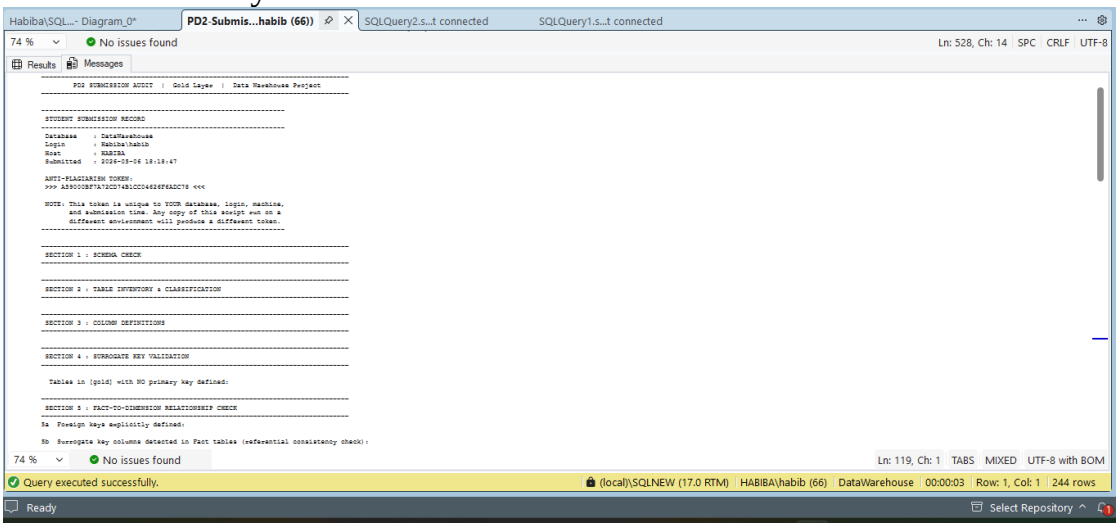
Optional Relationships

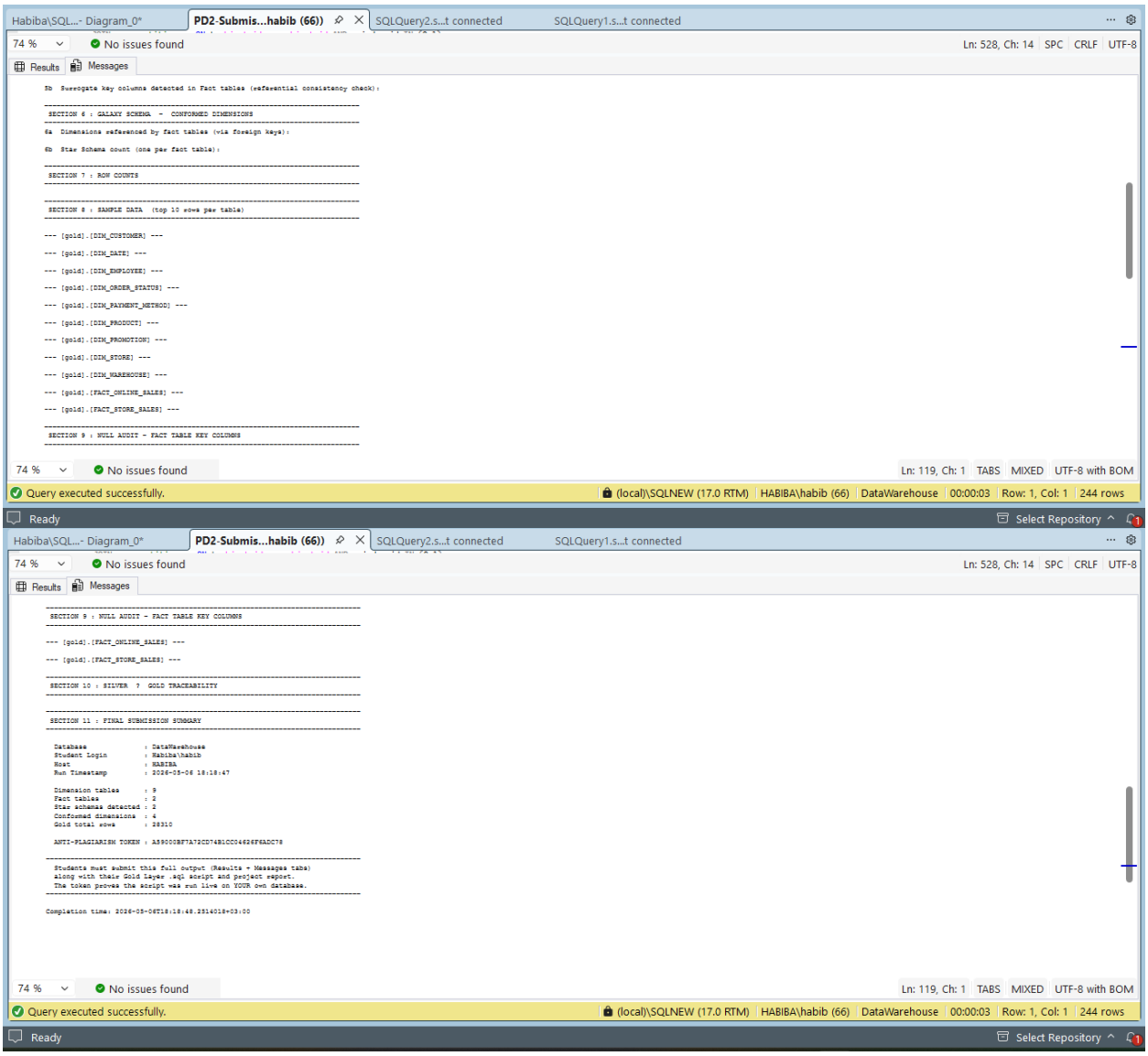
Some relationships (promotions, payment methods) were implemented using LEFT JOINs to handle missing or optional data.

2. A screenshot of your Data Warehouse Schema from SQL Server



3.Gold Layer Technical Audit:






```
================================================================================
      PD2 SUBMISSION AUDIT | Gold Layer | Data Warehouse Project
================================================================================

------------------------------------------------------------
STUDENT SUBMISSION RECORD
------------------------------------------------------------
Database    : DataWarehouse
Login       : Habiba\habib
Host        : HABIBA
Submitted   : 2026-05-06 18:18:47

ANTI-PLAGIARISM TOKEN:
>>> A59000BF7A72CD74B1CC04626F6ADC78 <<<

NOTE: This token is unique to YOUR database, login, machine,
      and submission time. Any copy of this script run on a
      different environment will produce a different token.
------------------------------------------------------------

================================================================================
SECTION 1 : SCHEMA CHECK
================================================================================

================================================================================
SECTION 2 : TABLE INVENTORY & CLASSIFICATION
================================================================================
```

```
================================================================================
SECTION 3 : COLUMN DEFINITIONS
================================================================================

================================================================================
SECTION 4 : SURROGATE KEY VALIDATION
================================================================================

  Tables in [gold] with NO primary key defined:

================================================================================
SECTION 5 : FACT-TO-DIMENSION RELATIONSHIP CHECK
================================================================================
5a  Foreign keys explicitly defined:

5b  Surrogate key columns detected in Fact tables (referential consistency check):

================================================================================
SECTION 6 : GALAXY SCHEMA  –  CONFORMED DIMENSIONS
================================================================================
6a  Dimensions referenced by fact tables (via foreign keys):

6b  Star Schema count (one per fact table):

================================================================================
SECTION 7 : ROW COUNTS
================================================================================

================================================================================
SECTION 8 : SAMPLE DATA  (top 10 rows per table)
================================================================================

--- [gold].[DIM_CUSTOMER] ---

--- [gold].[DIM_DATE] ---

--- [gold].[DIM_EMPLOYEE] ---

--- [gold].[DIM_ORDER_STATUS] ---

--- [gold].[DIM_PAYMENT_METHOD] ---

--- [gold].[DIM_PRODUCT] ---

--- [gold].[DIM_PROMOTION] ---

--- [gold].[DIM_STORE] ---

--- [gold].[DIM_WAREHOUSE] ---

--- [gold].[FACT_ONLINE_SALES] ---

--- [gold].[FACT_STORE_SALES] ---

================================================================================
SECTION 9 : NULL AUDIT – FACT TABLE KEY COLUMNS
================================================================================

--- [gold].[FACT_ONLINE_SALES] ---

--- [gold].[FACT_STORE_SALES] ---

================================================================================
SECTION 10 : SILVER  ?  GOLD TRACEABILITY
================================================================================

================================================================================
SECTION 11 : FINAL SUBMISSION SUMMARY
================================================================================

  Database             : DataWarehouse
  Student Login        : Habiba\habib
  Host                 : HABIBA
  Run Timestamp        : 2026-05-06 18:18:47

  Dimension tables     : 9
  Fact tables          : 2
  Star schemas detected : 2
  Conformed dimensions  : 4
```

Gold total rows : 28310

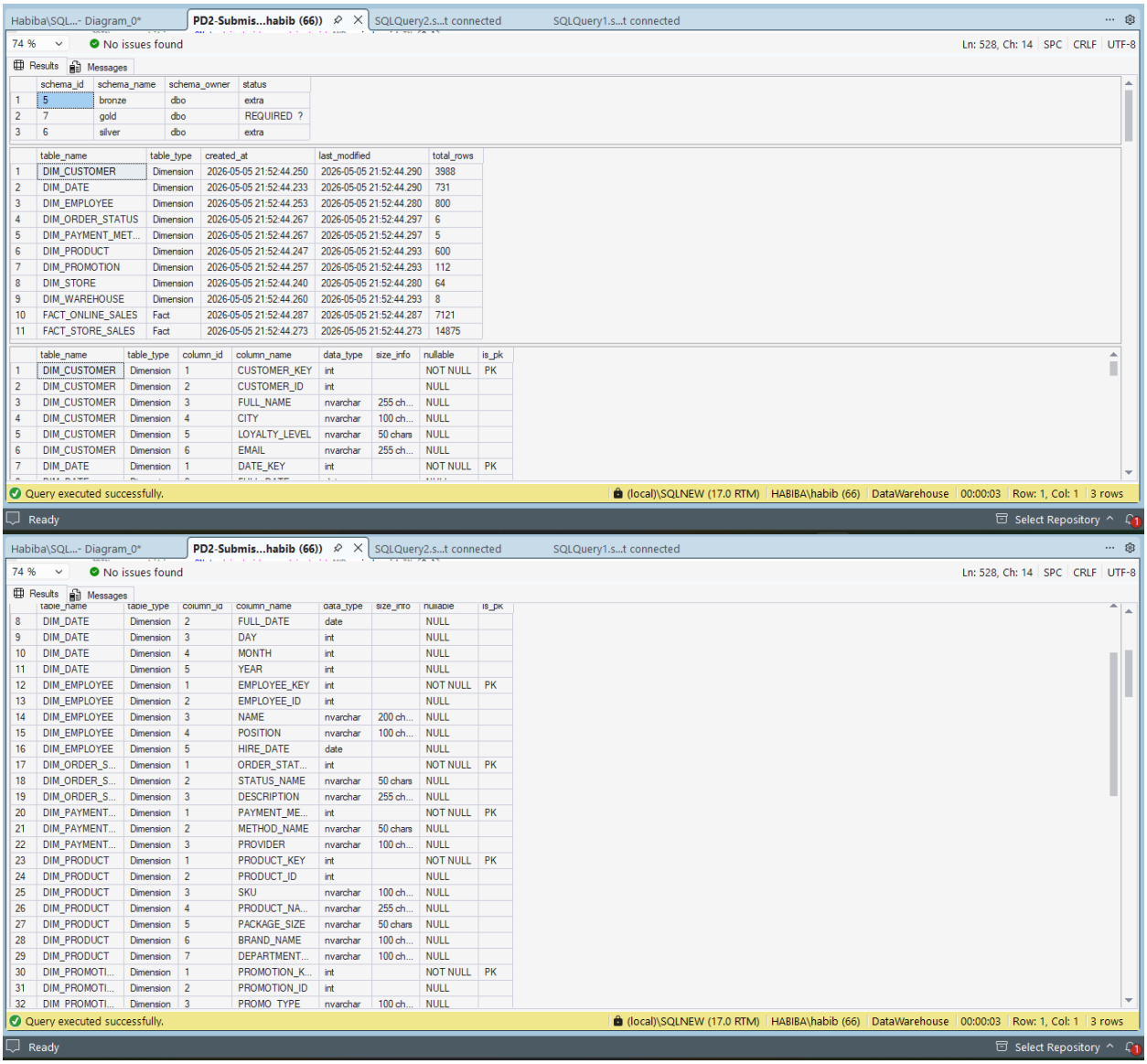
ANTI-PLAGIARISM TOKEN : A59000BF7A72CD74B1CC04626F6ADC78

==========

Students must submit this full output (Results + Messages tabs)
along with their Gold Layer .sql script and project report.
The token proves the script was run live on YOUR own database.

==========

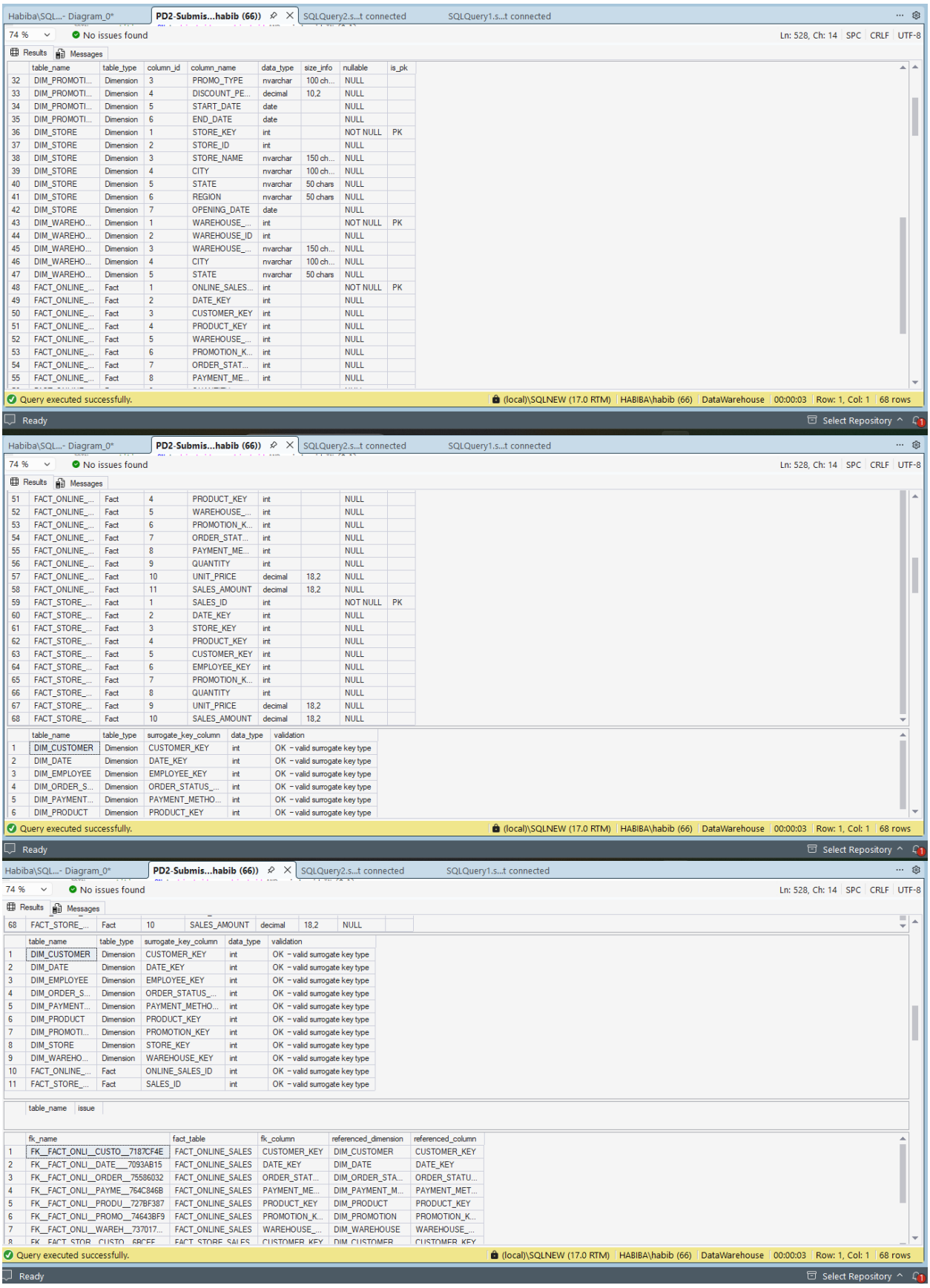
Completion time: 2026-05-06T18:18:48.2514018+03:00



	schema_id	schema_name	schema_owner	status
1	5	bronze	dbo	extra
2	7	gold	dbo	REQUIRED ?
3	6	silver	dbo	extra

	table_name	table_type	created_at	last_modified	total_rows
1	DIM_CUSTOMER	Dimension	2026-05-05 21:52:44.250	2026-05-05 21:52:44.290	3988
2	DIM_DATE	Dimension	2026-05-05 21:52:44.233	2026-05-05 21:52:44.290	731
3	DIM_EMPLOYEE	Dimension	2026-05-05 21:52:44.253	2026-05-05 21:52:44.280	800
4	DIM_ORDER_STATUS	Dimension	2026-05-05 21:52:44.267	2026-05-05 21:52:44.297	6
5	DIM_PAYMENT_MET...	Dimension	2026-05-05 21:52:44.267	2026-05-05 21:52:44.297	5
6	DIM_PRODUCT	Dimension	2026-05-05 21:52:44.247	2026-05-05 21:52:44.293	600
7	DIM_PROMOTION	Dimension	2026-05-05 21:52:44.257	2026-05-05 21:52:44.293	112
8	DIM_STORE	Dimension	2026-05-05 21:52:44.240	2026-05-05 21:52:44.280	64
9	DIM_WAREHOUSE	Dimension	2026-05-05 21:52:44.260	2026-05-05 21:52:44.293	8
10	FACT_ONLINE_SALES	Fact	2026-05-05 21:52:44.287	2026-05-05 21:52:44.287	7121
11	FACT_STORE_SALES	Fact	2026-05-05 21:52:44.273	2026-05-05 21:52:44.273	14875

	table_name	table_type	column_id	column_name	data_type	size_info	nullable	is_pk
1	DIM_CUSTOMER	Dimension	1	CUSTOMER_KEY	int		NOT NULL	PK
2	DIM_CUSTOMER	Dimension	2	CUSTOMER_ID	int		NULL	
3	DIM_CUSTOMER	Dimension	3	FULL_NAME	nvarchar	255 ch...	NULL	
4	DIM_CUSTOMER	Dimension	4	CITY	nvarchar	100 ch...	NULL	
5	DIM_CUSTOMER	Dimension	5	LOYALTY_LEVEL	nvarchar	50 chars	NULL	
6	DIM_CUSTOMER	Dimension	6	EMAIL	nvarchar	255 ch...	NULL	
7	DIM_DATE	Dimension	1	DATE_KEY	int		NOT NULL	PK
8	DIM_DATE	Dimension	2	FULL_DATE	date		NULL	
9	DIM_DATE	Dimension	3	DAY	int		NULL	
10	DIM_DATE	Dimension	4	MONTH	int		NULL	
11	DIM_DATE	Dimension	5	YEAR	int		NULL	
12	DIM_EMPLOYEE	Dimension	1	EMPLOYEE_KEY	int		NOT NULL	PK
13	DIM_EMPLOYEE	Dimension	2	EMPLOYEE_ID	int		NULL	
14	DIM_EMPLOYEE	Dimension	3	NAME	nvarchar	200 ch...	NULL	
15	DIM_EMPLOYEE	Dimension	4	POSITION	nvarchar	100 ch...	NULL	
16	DIM_EMPLOYEE	Dimension	5	HIRE_DATE	date		NULL	
17	DIM_ORDER_S...	Dimension	1	ORDER_STAT...	int		NOT NULL	PK
18	DIM_ORDER_S...	Dimension	2	STATUS_NAME	nvarchar	50 chars	NULL	
19	DIM_ORDER_S...	Dimension	3	DESCRIPTION	nvarchar	255 ch...	NULL	
20	DIM_PAYMENT...	Dimension	1	PAYMENT_ME...	int		NOT NULL	PK
21	DIM_PAYMENT...	Dimension	2	METHOD_NAME	nvarchar	50 chars	NULL	
22	DIM_PAYMENT...	Dimension	3	PROVIDER	nvarchar	100 ch...	NULL	
23	DIM_PRODUCT	Dimension	1	PRODUCT_KEY	int		NOT NULL	PK
24	DIM_PRODUCT	Dimension	2	PRODUCT_ID	int		NULL	
25	DIM_PRODUCT	Dimension	3	SKU	nvarchar	100 ch...	NULL	
26	DIM_PRODUCT	Dimension	4	PRODUCT_NA...	nvarchar	255 ch...	NULL	
27	DIM_PRODUCT	Dimension	5	PACKAGE_SIZE	nvarchar	50 chars	NULL	
28	DIM_PRODUCT	Dimension	6	BRAND_NAME	nvarchar	100 ch...	NULL	
29	DIM_PRODUCT	Dimension	7	DEPARTMENT...	nvarchar	100 ch...	NULL	
30	DIM_PROMOTI...	Dimension	1	PROMOTION_K...	int		NOT NULL	PK
31	DIM_PROMOTI...	Dimension	2	PROMOTION_ID	int		NULL	
32	DIM_PROMOTI...	Dimension	3	PROMO_TYPE	nvarchar	100 ch...	NULL	



	table_name	table_type	column_id	column_name	data_type	size_info	nullable	is_pk
32	DIM_PROMOTI...	Dimension	3	PROMO_TYPE	nvarchar	100 ch...	NULL	
33	DIM_PROMOTI...	Dimension	4	DISCOUNT_PE...	decimal	10,2	NULL	
34	DIM_PROMOTI...	Dimension	5	START_DATE	date		NULL	
35	DIM_PROMOTI...	Dimension	6	END_DATE	date		NULL	
36	DIM_STORE	Dimension	1	STORE_KEY	int		NOT NULL	PK
37	DIM_STORE	Dimension	2	STORE_ID	int		NULL	
38	DIM_STORE	Dimension	3	STORE_NAME	nvarchar	150 ch...	NULL	
39	DIM_STORE	Dimension	4	CITY	nvarchar	100 ch...	NULL	
40	DIM_STORE	Dimension	5	STATE	nvarchar	50 chars	NULL	
41	DIM_STORE	Dimension	6	REGION	nvarchar	50 chars	NULL	
42	DIM_STORE	Dimension	7	OPENING_DATE	date		NULL	
43	DIM_WAREHO...	Dimension	1	WAREHOUSE_...	int		NOT NULL	PK
44	DIM_WAREHO...	Dimension	2	WAREHOUSE_ID	int		NULL	
45	DIM_WAREHO...	Dimension	3	WAREHOUSE_...	nvarchar	150 ch...	NULL	
46	DIM_WAREHO...	Dimension	4	CITY	nvarchar	100 ch...	NULL	
47	DIM_WAREHO...	Dimension	5	STATE	nvarchar	50 chars	NULL	
48	FACT_ONLINE_...	Fact	1	ONLINE_SALES...	int		NOT NULL	PK
49	FACT_ONLINE_...	Fact	2	DATE_KEY	int		NULL	
50	FACT_ONLINE_...	Fact	3	CUSTOMER_KEY	int		NULL	
51	FACT_ONLINE_...	Fact	4	PRODUCT_KEY	int		NULL	
52	FACT_ONLINE_...	Fact	5	WAREHOUSE_...	int		NULL	
53	FACT_ONLINE_...	Fact	6	PROMOTION_K...	int		NULL	
54	FACT_ONLINE_...	Fact	7	ORDER_STAT...	int		NULL	
55	FACT_ONLINE_...	Fact	8	PAYMENT_ME...	int		NULL	

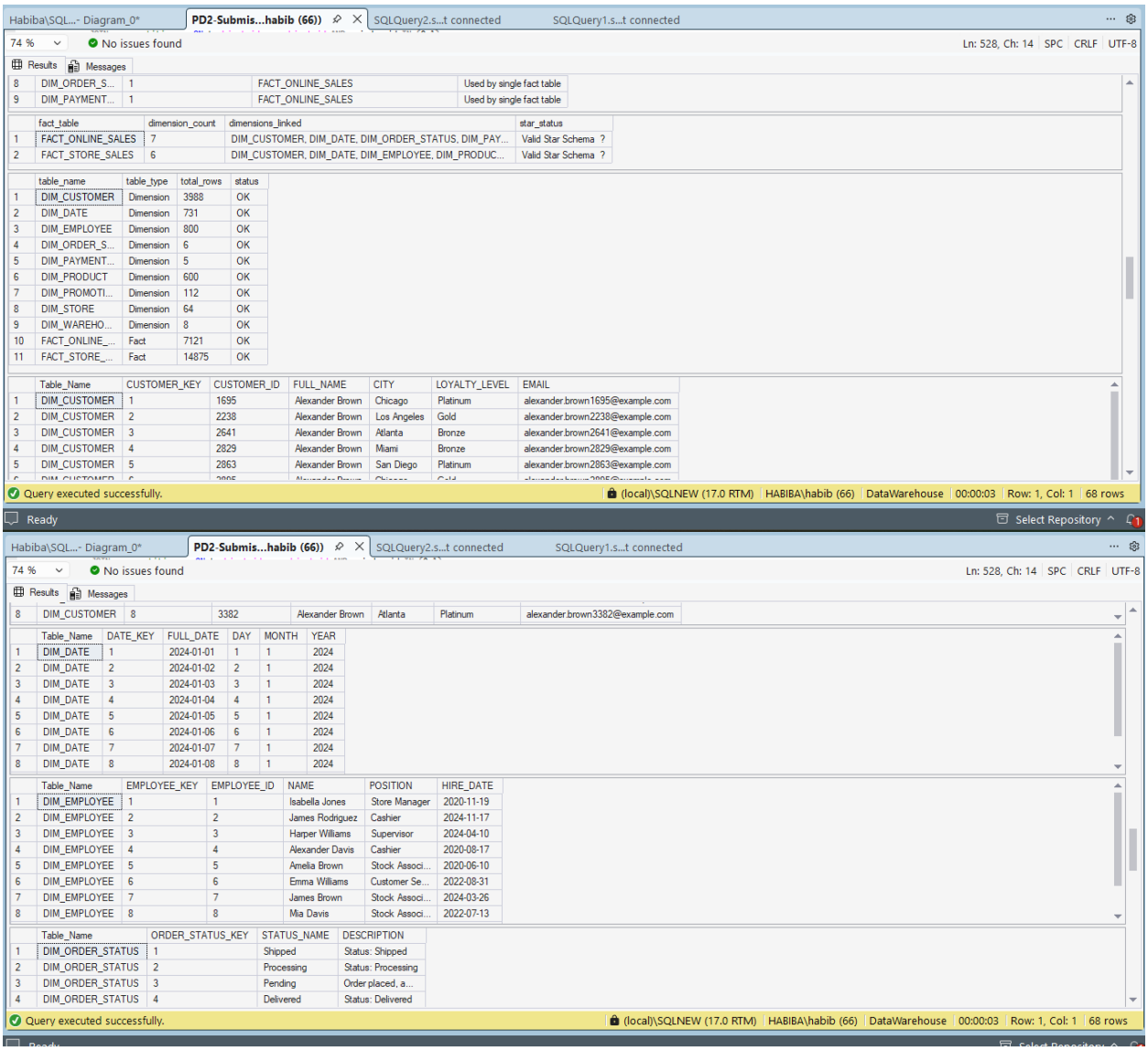
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51	FACT_ONLINE_...	Fact	4	PRODUCT_KEY	int		NULL	
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53	FACT_ONLINE_...	Fact	6	PROMOTION_K...	int		NULL	
54	FACT_ONLINE_...	Fact	7	ORDER_STAT...	int		NULL	
55	FACT_ONLINE_...	Fact	8	PAYMENT_ME...	int		NULL	
56	FACT_ONLINE_...	Fact	9	QUANTITY	int		NULL	
57	FACT_ONLINE_...	Fact	10	UNIT_PRICE	decimal	18,2	NULL	
58	FACT_ONLINE_...	Fact	11	SALES_AMOUNT	decimal	18,2	NULL	
59	FACT_STORE_...	Fact	1	SALES_ID	int		NOT NULL	PK
60	FACT_STORE_...	Fact	2	DATE_KEY	int		NULL	
61	FACT_STORE_...	Fact	3	STORE_KEY	int		NULL	
62	FACT_STORE_...	Fact	4	PRODUCT_KEY	int		NULL	
63	FACT_STORE_...	Fact	5	CUSTOMER_KEY	int		NULL	
64	FACT_STORE_...	Fact	6	EMPLOYEE_KEY	int		NULL	
65	FACT_STORE_...	Fact	7	PROMOTION_K...	int		NULL	
66	FACT_STORE_...	Fact	8	QUANTITY	int		NULL	
67	FACT_STORE_...	Fact	9	UNIT_PRICE	decimal	18,2	NULL	
68	FACT_STORE_...	Fact	10	SALES_AMOUNT	decimal	18,2	NULL	

	table_name	table_type	surrogate_key_column	data_type	validation
1	DIM_CUSTOMER	Dimension	CUSTOMER_KEY	int	OK - valid surrogate key type
2	DIM_DATE	Dimension	DATE_KEY	int	OK - valid surrogate key type
3	DIM_EMPLOYEE	Dimension	EMPLOYEE_KEY	int	OK - valid surrogate key type
4	DIM_ORDER_S...	Dimension	ORDER_STATUS_...	int	OK - valid surrogate key type
5	DIM_PAYMENT...	Dimension	PAYMENT_METHO...	int	OK - valid surrogate key type
6	DIM_PRODUCT	Dimension	PRODUCT_KEY	int	OK - valid surrogate key type

	table_name	table_type	surrogate_key_column	data_type	validation
1	DIM_CUSTOMER	Dimension	CUSTOMER_KEY	int	OK - valid surrogate key type
2	DIM_DATE	Dimension	DATE_KEY	int	OK - valid surrogate key type
3	DIM_EMPLOYEE	Dimension	EMPLOYEE_KEY	int	OK - valid surrogate key type
4	DIM_ORDER_S...	Dimension	ORDER_STATUS_...	int	OK - valid surrogate key type
5	DIM_PAYMENT...	Dimension	PAYMENT_METHO...	int	OK - valid surrogate key type
6	DIM_PRODUCT	Dimension	PRODUCT_KEY	int	OK - valid surrogate key type
7	DIM_PROMOTI...	Dimension	PROMOTION_KEY	int	OK - valid surrogate key type
8	DIM_STORE	Dimension	STORE_KEY	int	OK - valid surrogate key type
9	DIM_WAREHO...	Dimension	WAREHOUSE_KEY	int	OK - valid surrogate key type
10	FACT_ONLINE_...	Fact	ONLINE_SALES_ID	int	OK - valid surrogate key type
11	FACT_STORE_...	Fact	SALES_ID	int	OK - valid surrogate key type

	table_name	issue

	fk_name	fact_table	fk_column	referenced_dimension	referenced_column
1	FK__FACT_ONLI__CUSTO__7187CF4E	FACT_ONLINE_SALES	CUSTOMER_KEY	DIM_CUSTOMER	CUSTOMER_KEY
2	FK__FACT_ONLI__DATE___7093AB15	FACT_ONLINE_SALES	DATE_KEY	DIM_DATE	DATE_KEY
3	FK__FACT_ONLI__ORDER__75586032	FACT_ONLINE_SALES	ORDER_STAT...	DIM_ORDER_STA...	ORDER_STATU...
4	FK__FACT_ONLI__PAYME__764C846B	FACT_ONLINE_SALES	PAYMENT_ME...	DIM_PAYMENT_M...	PAYMENT_MET...
5	FK__FACT_ONLI__PRODU__727BF387	FACT_ONLINE_SALES	PRODUCT_KEY	DIM_PRODUCT	PRODUCT_KEY
6	FK__FACT_ONLI__PROMO__74643BF9	FACT_ONLINE_SALES	PROMOTION_K...	DIM_PROMOTION	PROMOTION_K...
7	FK__FACT_ONLI__WAREH__737017...	FACT_ONLINE_SALES	WAREHOUSE_...	DIM_WAREHOUSE	WAREHOUSE_...
8	FK__FACT_STOR__CUSTO__6BCEE...	FACT_STORE_SALES	CUSTOMER_KEY	DIM_CUSTOMER	CUSTOMER_KEY



8	DIM_ORDER_S...	1	FACT_ONLINE_SALES	Used by single fact table
9	DIM_PAYMENT...	1	FACT_ONLINE_SALES	Used by single fact table

	fact_table	dimension_count	dimensions_linked	star_status
1	FACT_ONLINE_SALES	7	DIM_CUSTOMER, DIM_DATE, DIM_ORDER_STATUS, DIM_PAY...	Valid Star Schema ?
2	FACT_STORE_SALES	6	DIM_CUSTOMER, DIM_DATE, DIM_EMPLOYEE, DIM_PRODUC...	Valid Star Schema ?

	table_name	table_type	total_rows	status
1	DIM_CUSTOMER	Dimension	3988	OK
2	DIM_DATE	Dimension	731	OK
3	DIM_EMPLOYEE	Dimension	800	OK
4	DIM_ORDER_S...	Dimension	6	OK
5	DIM_PAYMENT...	Dimension	5	OK
6	DIM_PRODUCT	Dimension	600	OK
7	DIM_PROMOTI...	Dimension	112	OK
8	DIM_STORE	Dimension	64	OK
9	DIM_WAREHO...	Dimension	8	OK
10	FACT_ONLINE_...	Fact	7121	OK
11	FACT_STORE_...	Fact	14875	OK

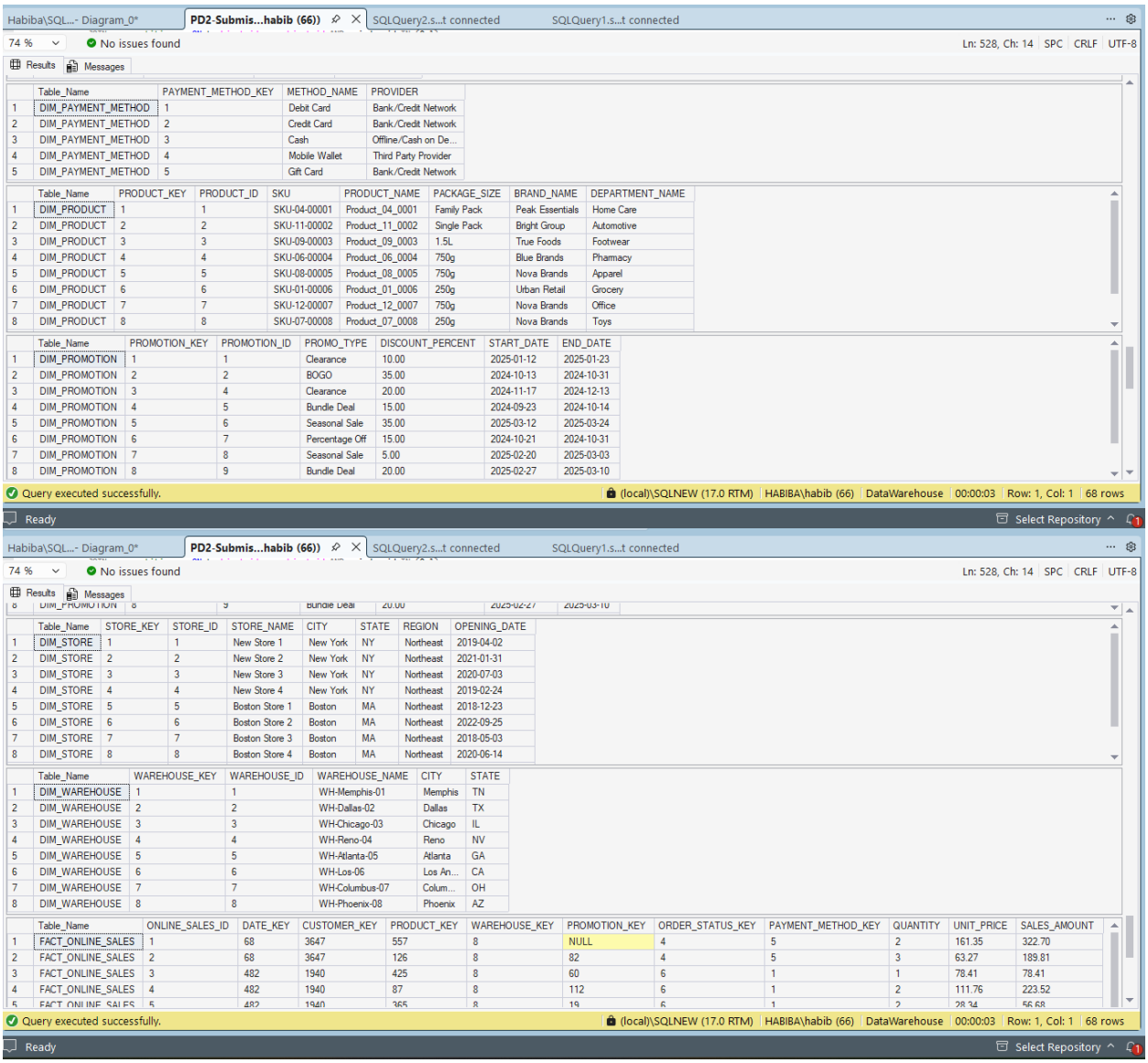
	Table_Name	CUSTOMER_KEY	CUSTOMER_ID	FULL_NAME	CITY	LOYALTY_LEVEL	EMAIL
1	DIM_CUSTOMER	1	1695	Alexander Brown	Chicago	Platinum	alexander.brown1695@example.com
2	DIM_CUSTOMER	2	2238	Alexander Brown	Los Angeles	Gold	alexander.brown2238@example.com
3	DIM_CUSTOMER	3	2641	Alexander Brown	Atlanta	Bronze	alexander.brown2641@example.com
4	DIM_CUSTOMER	4	2829	Alexander Brown	Miami	Bronze	alexander.brown2829@example.com
5	DIM_CUSTOMER	5	2863	Alexander Brown	San Diego	Platinum	alexander.brown2863@example.com

8	DIM_CUSTOMER	8	3382	Alexander Brown	Atlanta	Platinum	alexander.brown3382@example.com

	Table_Name	DATE_KEY	FULL_DATE	DAY	MONTH	YEAR
1	DIM_DATE	1	2024-01-01	1	1	2024
2	DIM_DATE	2	2024-01-02	2	1	2024
3	DIM_DATE	3	2024-01-03	3	1	2024
4	DIM_DATE	4	2024-01-04	4	1	2024
5	DIM_DATE	5	2024-01-05	5	1	2024
6	DIM_DATE	6	2024-01-06	6	1	2024
7	DIM_DATE	7	2024-01-07	7	1	2024
8	DIM_DATE	8	2024-01-08	8	1	2024

	Table_Name	EMPLOYEE_KEY	EMPLOYEE_ID	NAME	POSITION	HIRE_DATE
1	DIM_EMPLOYEE	1	1	Isabella Jones	Store Manager	2020-11-19
2	DIM_EMPLOYEE	2	2	James Rodriguez	Cashier	2024-11-17
3	DIM_EMPLOYEE	3	3	Harper Williams	Supervisor	2024-04-10
4	DIM_EMPLOYEE	4	4	Alexander Davis	Cashier	2020-08-17
5	DIM_EMPLOYEE	5	5	Amelia Brown	Stock Associ...	2020-06-10
6	DIM_EMPLOYEE	6	6	Emma Williams	Customer Se...	2022-08-31
7	DIM_EMPLOYEE	7	7	James Brown	Stock Associ...	2024-03-26
8	DIM_EMPLOYEE	8	8	Mia Davis	Stock Associ...	2022-07-13

	Table_Name	ORDER_STATUS_KEY	STATUS_NAME	DESCRIPTION
1	DIM_ORDER_STATUS	1	Shipped	Status: Shipped
2	DIM_ORDER_STATUS	2	Processing	Status: Processing
3	DIM_ORDER_STATUS	3	Pending	Order placed, a...
4	DIM_ORDER_STATUS	4	Delivered	Status: Delivered



	Table_Name	PAYMENT_METHOD_KEY	METHOD_NAME	PROVIDER
1	DIM_PAYMENT_METHOD	1	Debit Card	Bank/Credit Network
2	DIM_PAYMENT_METHOD	2	Credit Card	Bank/Credit Network
3	DIM_PAYMENT_METHOD	3	Cash	Offline/Cash on De...
4	DIM_PAYMENT_METHOD	4	Mobile Wallet	Third Party Provider
5	DIM_PAYMENT_METHOD	5	Gift Card	Bank/Credit Network

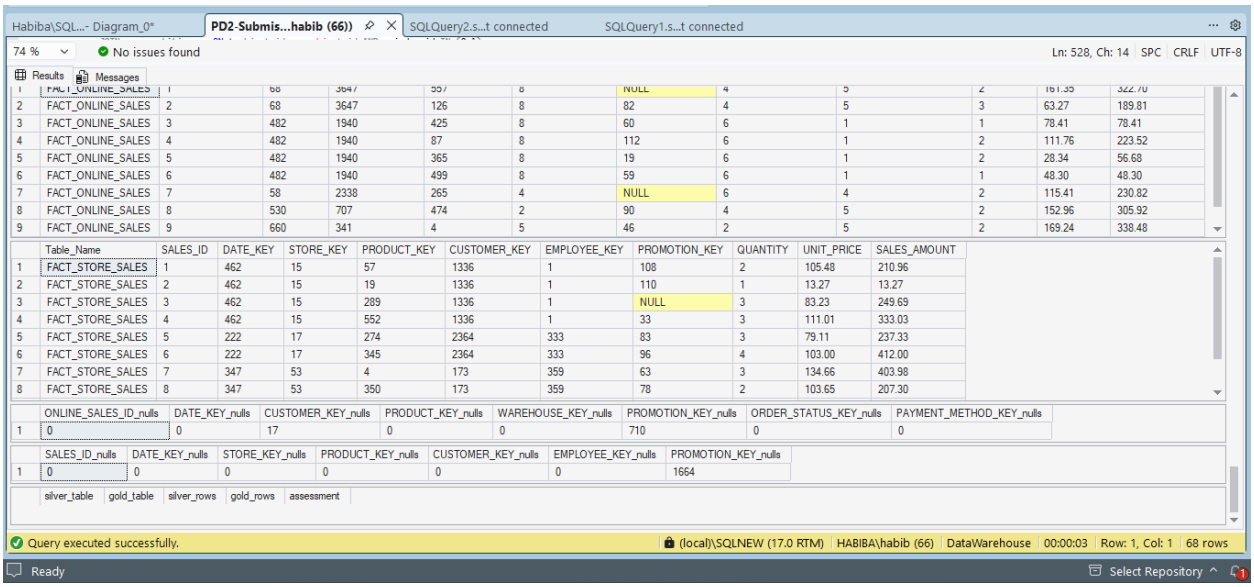
	Table_Name	PRODUCT_KEY	PRODUCT_ID	SKU	PRODUCT_NAME	PACKAGE_SIZE	BRAND_NAME	DEPARTMENT_NAME
1	DIM_PRODUCT	1	1	SKU-04-00001	Product_04_0001	Family Pack	Peak Essentials	Home Care
2	DIM_PRODUCT	2	2	SKU-11-00002	Product_11_0002	Single Pack	Bright Group	Automotive
3	DIM_PRODUCT	3	3	SKU-09-00003	Product_09_0003	1.5L	True Foods	Footwear
4	DIM_PRODUCT	4	4	SKU-06-00004	Product_06_0004	750g	Blue Brands	Pharmacy
5	DIM_PRODUCT	5	5	SKU-08-00005	Product_08_0005	750g	Nova Brands	Apparel
6	DIM_PRODUCT	6	6	SKU-01-00006	Product_01_0006	250g	Urban Retail	Grocery
7	DIM_PRODUCT	7	7	SKU-12-00007	Product_12_0007	750g	Nova Brands	Office
8	DIM_PRODUCT	8	8	SKU-07-00008	Product_07_0008	250g	Nova Brands	Toys

	Table_Name	PROMOTION_KEY	PROMOTION_ID	PROMO_TYPE	DISCOUNT_PERCENT	START_DATE	END_DATE
1	DIM_PROMOTION	1	1	Clearance	10.00	2025-01-12	2025-01-23
2	DIM_PROMOTION	2	2	BOGO	35.00	2024-10-13	2024-10-31
3	DIM_PROMOTION	3	4	Clearance	20.00	2024-11-17	2024-12-13
4	DIM_PROMOTION	4	5	Bundle Deal	15.00	2024-09-23	2024-10-14
5	DIM_PROMOTION	5	6	Seasonal Sale	35.00	2025-03-12	2025-03-24
6	DIM_PROMOTION	6	7	Percentage Off	15.00	2024-10-21	2024-10-31
7	DIM_PROMOTION	7	8	Seasonal Sale	5.00	2025-02-20	2025-03-03
8	DIM_PROMOTION	8	9	Bundle Deal	20.00	2025-02-27	2025-03-10

	Table_Name	STORE_KEY	STORE_ID	STORE_NAME	CITY	STATE	REGION	OPENING_DATE
1	DIM_STORE	1	1	New Store 1	New York	NY	Northeast	2019-04-02
2	DIM_STORE	2	2	New Store 2	New York	NY	Northeast	2021-01-31
3	DIM_STORE	3	3	New Store 3	New York	NY	Northeast	2020-07-03
4	DIM_STORE	4	4	New Store 4	New York	NY	Northeast	2019-02-24
5	DIM_STORE	5	5	Boston Store 1	Boston	MA	Northeast	2018-12-23
6	DIM_STORE	6	6	Boston Store 2	Boston	MA	Northeast	2022-09-25
7	DIM_STORE	7	7	Boston Store 3	Boston	MA	Northeast	2018-05-03
8	DIM_STORE	8	8	Boston Store 4	Boston	MA	Northeast	2020-06-14

	Table_Name	WAREHOUSE_KEY	WAREHOUSE_ID	WAREHOUSE_NAME	CITY	STATE
1	DIM_WAREHOUSE	1	1	WH-Memphis-01	Memphis	TN
2	DIM_WAREHOUSE	2	2	WH-Dallas-02	Dallas	TX
3	DIM_WAREHOUSE	3	3	WH-Chicago-03	Chicago	IL
4	DIM_WAREHOUSE	4	4	WH-Reno-04	Reno	NV
5	DIM_WAREHOUSE	5	5	WH-Atlanta-05	Atlanta	GA
6	DIM_WAREHOUSE	6	6	WH-Los-06	Los An...	CA
7	DIM_WAREHOUSE	7	7	WH-Columbus-07	Colum...	OH
8	DIM_WAREHOUSE	8	8	WH-Phoenix-08	Phoenix	AZ

	Table_Name	ONLINE_SALES_ID	DATE_KEY	CUSTOMER_KEY	PRODUCT_KEY	WAREHOUSE_KEY	PROMOTION_KEY	ORDER_STATUS_KEY	PAYMENT_METHOD_KEY	QUANTITY	UNIT_PRICE	SALES_AMOUNT
1	FACT_ONLINE_SALES	1	68	3647	557	8	NULL	4	5	2	161.35	322.70
2	FACT_ONLINE_SALES	2	68	3647	126	8	82	4	5	3	63.27	189.81
3	FACT_ONLINE_SALES	3	482	1940	425	8	60	6	1	1	78.41	78.41
4	FACT_ONLINE_SALES	4	482	1940	87	8	112	6	1	2	111.76	223.52



1	FACT_ONLINE_SALES	1	68	3647	557	8	NULL	4	5	2	161.35	322.70
2	FACT_ONLINE_SALES	2	68	3647	126	8	82	4	5	3	63.27	189.81
3	FACT_ONLINE_SALES	3	482	1940	425	8	60	6	1	1	78.41	78.41
4	FACT_ONLINE_SALES	4	482	1940	87	8	112	6	1	2	111.76	223.52
5	FACT_ONLINE_SALES	5	482	1940	365	8	19	6	1	2	28.34	56.68
6	FACT_ONLINE_SALES	6	482	1940	499	8	59	6	1	1	48.30	48.30
7	FACT_ONLINE_SALES	7	58	2338	265	4	NULL	6	4	2	115.41	230.82
8	FACT_ONLINE_SALES	8	530	707	474	2	90	4	5	2	152.96	305.92
9	FACT_ONLINE_SALES	9	660	341	4	5	46	2	5	2	169.24	338.48

	Table_Name	SALES_ID	DATE_KEY	STORE_KEY	PRODUCT_KEY	CUSTOMER_KEY	EMPLOYEE_KEY	PROMOTION_KEY	QUANTITY	UNIT_PRICE	SALES_AMOUNT
1	FACT_STORE_SALES	1	462	15	57	1336	1	108	2	105.48	210.96
2	FACT_STORE_SALES	2	462	15	19	1336	1	110	1	13.27	13.27
3	FACT_STORE_SALES	3	462	15	289	1336	1	NULL	3	83.23	249.69
4	FACT_STORE_SALES	4	462	15	552	1336	1	33	3	111.01	333.03
5	FACT_STORE_SALES	5	222	17	274	2364	333	83	3	79.11	237.33
6	FACT_STORE_SALES	6	222	17	345	2364	333	96	4	103.00	412.00
7	FACT_STORE_SALES	7	347	53	4	173	359	63	3	134.66	403.98
8	FACT_STORE_SALES	8	347	53	350	173	359	78	2	103.65	207.30

	ONLINE_SALES_ID_nulls	DATE_KEY_nulls	CUSTOMER_KEY_nulls	PRODUCT_KEY_nulls	WAREHOUSE_KEY_nulls	PROMOTION_KEY_nulls	ORDER_STATUS_KEY_nulls	PAYMENT_METHOD_KEY_nulls
1	0	0	17	0	0	710	0	0

	SALES_ID_nulls	DATE_KEY_nulls	STORE_KEY_nulls	PRODUCT_KEY_nulls	CUSTOMER_KEY_nulls	EMPLOYEE_KEY_nulls	PROMOTION_KEY_nulls
1	0	0	0	0	0	0	1664

	silver_table	gold_table	silver_rows	gold_rows	assessment

Milestone 3:

In this milestone, we developed a complete Business Intelligence solution designed to transform raw omnichannel data into actionable insights. We began by importing server data and structuring it into two distinct star schemas to effectively manage both physical and digital sales environments. Next, we engineered custom DAX KPIs and established strategic targets to accurately benchmark our performance, ultimately giving managers a unified, easily digestible view of the business.

Our final reporting suite provides end-to-end visibility across four specialized dashboards. It begins with an Executive Overview of combined omnichannel revenue and growth, followed by a Product & Promotion analysis evaluating cross-channel marketing ROI. The suite then drills into channel-specific execution, tracking physical Store Operations productivity and digital Online Fulfillment logistics.

1. User Personas

To design an effective dashboard, we must tailor the views to the people consuming the data.

- **Persona A: The Executive (VP of Sales / C-Suite)**
    - *Goal:* Monitor overall company health, high-level omnichannel performance, and year-over-year (YoY) growth.
- **Persona B: The Omnichannel Retail Manager**
    - *Goal:* Understand cross-channel behavior, product popularity, and how marketing/promotions drive sales across both physical and digital storefronts.
- **Persona C: The Regional Store Manager**
    - *Goal:* Track physical store performance, employee productivity, and localized sales trends.
- **Persona D: The E-Commerce & Fulfillment Manager**
    - *Goal:* Monitor online order volume, warehouse fulfillment efficiency, payment method trends, and order completion rates.

2. Business Questions

Executive & Omnichannel Manager Questions

1. How does the current period's overall revenue performance compare dynamically to the exact same period last year (YoY Growth Metrics via Time Intelligence)?
2. What is the trend of Online sales over time (daily, monthly, quarterly)?
3. What is the total combined revenue and quantity across the company?
4. How do Online Vs. Store sales compare, and how is it split by month?

In-Store Operations Questions

5. Which physical stores generate the highest revenue?
6. Who are the top-performing employees by total sales amount?
7. How do physical store sales vary by geographic region and state?
8. What is the total transaction value per weekdays?

Product and Promotion Insight Questions

9. What is the Store, and Online Revenue across the company, and how is it split by Department?
10. Which corporate marketing promotions yield the highest return on investment (ROI)?
11. How do discounts (Avg Discount %) impact overall sales volume?
12. What are our best selling Product, Brand, Category and promotion?

Online & Fulfillment Operations Questions

13. Which warehouses fulfill the highest volume of online orders?
14. What is the status distribution across the fulfillment pipeline?
15. Which payment providers generate the highest total online revenue?
16. How does online transaction volume distribute across customer loyalty tiers?
17. Are we meeting our operational targets for Online Sales, AOV, and Cancellation Rate?

3. Comprehensive Measure & KPI Dictionary

Measure / KPI Name	DAX Logic / Source Data	Business Purpose (What it shows)	Executive Value (Why we use it)
Total Store / Online Revenue	SUM(Fact[Sales_Amount])	Aggregates the raw transactional revenue into readable totals for each specific channel.	It is the ultimate baseline of company health. Shows if physical stores and the website are generating enough cash flow.
Total Combined Revenue	[Total Store Revenue] + [Total Online Revenue]	Merges both fact tables to show the absolute gross income of the entire omnichannel company.	The definitive metric of overall corporate scale, market share, and enterprise growth.
Total Combined Quantity	SUM(Fact_Store[Quantity]) + SUM(Fact_Online[Quantity])	Shows the actual physical volume of goods moving out of stores and warehouses.	High revenue can hide problems if you just raised prices. Volume ensures consumer demand is actually still there.
YoY Revenue	DIVIDE([Current Sales] - [Combined Revenue PY], [Combined Revenue PY], 0)	Measures year-over-year momentum using Time Intelligence functions	Shows the actual trajectory of

Growth %		(SAMEPERIODLASTYEAR).	the business, proving to stakeholders if recent strategies are working.
Online vs. Store Split %	DIVIDE([Total Online Revenue], [Total Combined Revenue], 0)	Calculates the exact percentage share each channel contributes to the grand total.	Shows digital adoption rates. Executives need this to know where to invest their budget next year (real estate vs. web dev).
Avg Order Value (Online AOV) / Avg Unit Price	DIVIDE([Total Sales], [Total Transactions], 0)	Calculates the average customer "basket size" per checkout.	Shows upselling success. If AOV goes up, cashiers and website algorithms are successfully convincing customers to buy more per visit.
Total Transactions / Orders	COUNTROWS(FactTable)	Counts the volume of customer interactions and checkouts, rather than just the money made.	Measures foot traffic in-store and conversion traffic online, independent of pricing strategies.

Avg Discount %	AVERAGE(Dim_Promotion[Discount_Percent])	Calculates the average markdown percentage given away during sales events.	Tracks promotional efficiency. Crucial for ensuring marketing campaigns aren't destroying overall profit margins.
Top Promotion	String Measure (e.g., TOPN / FIRSTNONBLANK)	Dynamically identifies the exact marketing campaign that drove the most combined revenue.	Guides future marketing budgets toward the most effective discount strategies to maximize ROI.
Best Selling Product / Brand / Category	String Measure (e.g., TOPN / FIRSTNONBLANK)	Dynamically displays the name of the highest-grossing SKU or product grouping.	Drives inventory purchasing decisions, supply chain focus, and homepage product placement.
Cancellation Rate %	DIVIDE([Cancelled Orders], [Total Orders], 0)	Calculates the percentage of digital orders that fail to reach the customer.	High cancellations destroy profit margins due to wasted labor. Shows the health of the logistics pipeline.

Warehouse Revenue / Total Qty Shipped	CALCULATE(SUM(Fact_Online[Quantity]), USERELATIONSHIP(...))	Tracks exact fulfillment volume and revenue generated by specific distribution centers.	Identifies logistical bottlenecks and dictates capacity planning for the supply chain network.
Employee Rank	RANKX(ALL(Dim_Employe), [Total Store Revenue])	Ranks physical store staff by their individual sales performance.	Identifies top talent for promotion and identifies underperformers requiring additional sales training.
Top Performing Store	String Measure (e.g., TOPN / FIRSTNONBLANK)	Identifies the physical brick-and-mortar location generating the most revenue.	Highlights successful regional management and proves which real estate investments are paying off.

4.Target Measures & Strategic Goals Dictionary

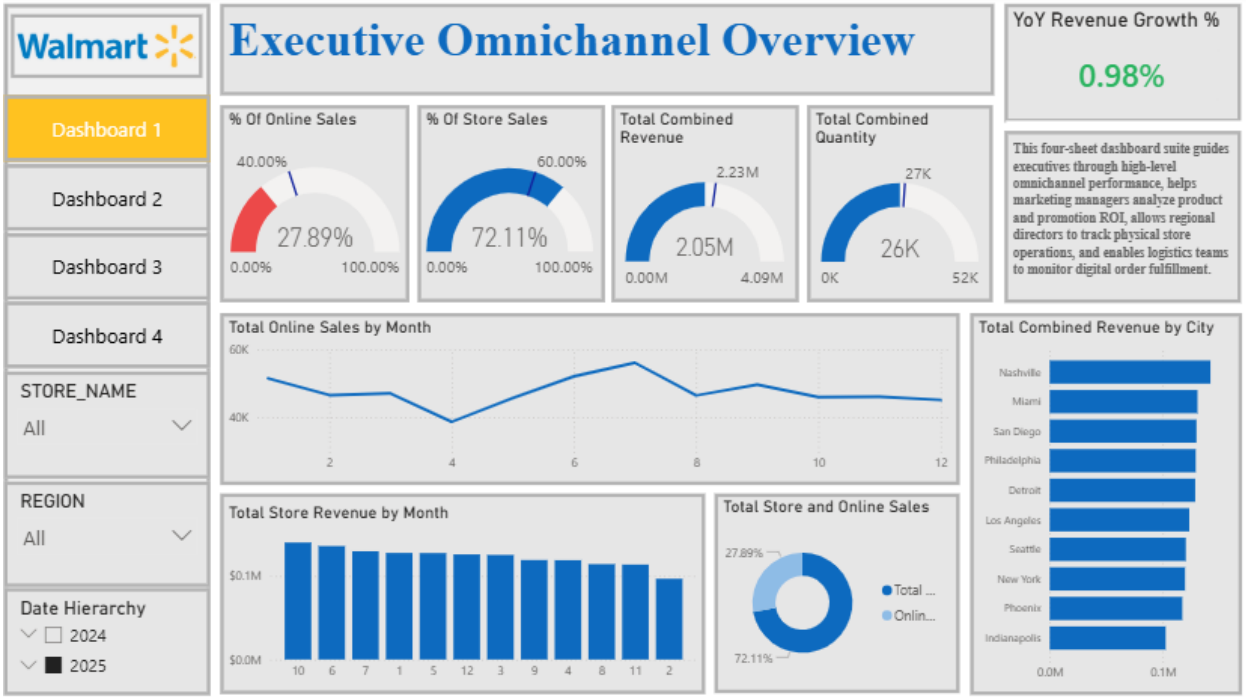
Based on what did we set these targets?

Target Measure Name	Target Value / DAX Formula	Business Rationale (Why this target?)
Target Store Revenue	[Last Year Store Sales] * 1.10	**Historical Baseline:** In retail, budgets are built on last year's performance. Adding a 10% multiplier represents expected strategic growth.
Target Online Sales	[Last Year Online Sales] * 1.15	**Digital Growth:** E-commerce typically scales faster than physical retail. A 15% growth target pushes the digital marketing team to expand market reach.
Target Combined Revenue	[Last Year Combined Sales] * 1.10	**Corporate Target:** The overarching goal for the entire business, ensuring omnichannel efforts sum up to a minimum 10% gross increase.
Target Combined Quantity	[Last Year Quantity] * 1.05	**Volume Goal:** Aiming for a 5% increase in physical units moved to ensure product demand is growing alongside revenue (and not just driven by inflation/price hikes).
Target Store Transactions	[Last Year Transactions] * 1.05	**Foot Traffic Goal:** Ensuring physical stores are successfully bringing 5% more individual shoppers through the doors compared to last year.
Target YoY Growth	0.10 (10%)	**Shareholder Expectation:** A flat percentage dictated by the board of directors to ensure the company is outpacing inflation and market competitors.
Target Online Sales %	0.40 (40%)	**Strategic Shift:** E-commerce usually has lower overhead. Pushing online adoption to 40% of total revenue increases overall company profit margins.
Target Store Revenue %	0.60 (60%)	**Physical Foundation:** The complement to the online target, ensuring the core brick-and-mortar footprint remains the dominant, stable revenue driver.
Target Online AOV	175	**Strategic Threshold:** A flat goal set by marketing. Setting the goal higher than the current average forces the web team to build better "Frequently Bought Together" prompts.
Target Avg Unit Price	85	**Upselling Target:** A static goal used on the Store Operations dashboard to encourage floor staff and cashiers to cross-sell premium products.

Target Avg Discount %	0.15 (15%)	**Margin Protection:** A flat ceiling limit to prevent the marketing department from giving away too much profit during promotional events.
Target Cancellation Rate	0.05 (5%)	**Operational Tolerance:** In logistics, a 0% cancellation rate is impossible. 5% is the maximum acceptable "failure threshold" before the company loses money on wasted shipping.

6. Dashboard Pages & Business Insights

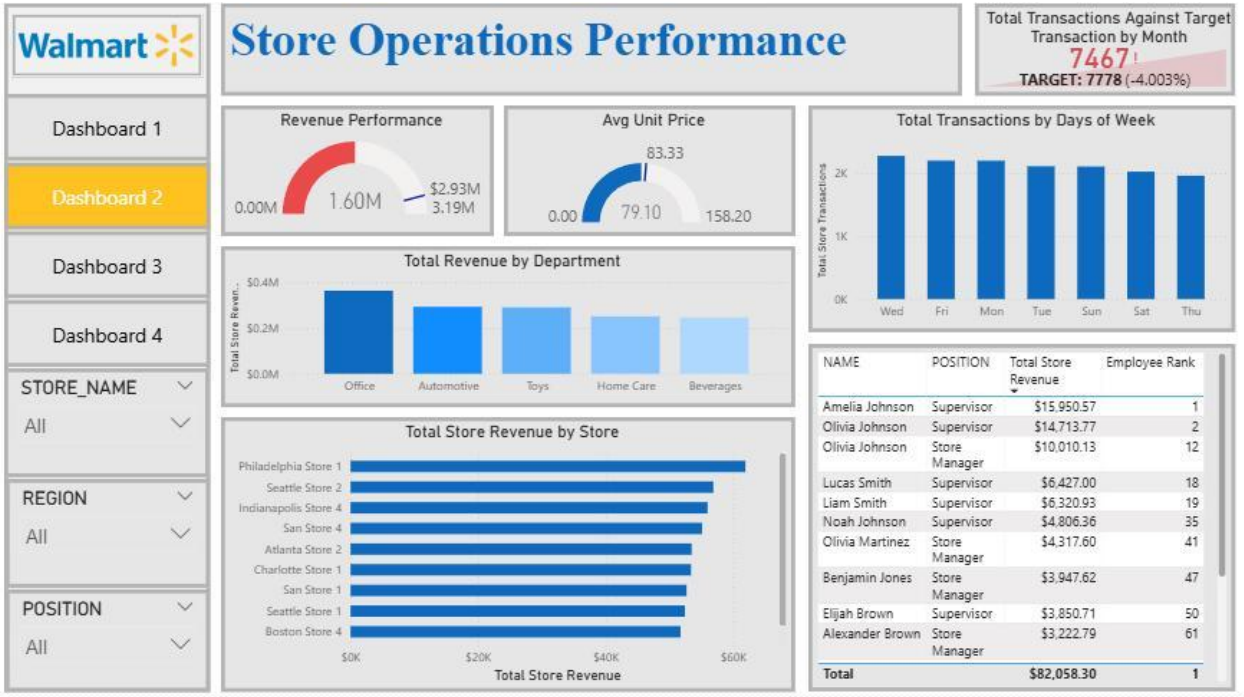
6.1 Executive Omnichannel Overview



Executive Omnichannel Overview: Provides a high-level summary combining online and physical sales KPIs to monitor overall business growth, total volume, and strategic channel performance against historical targets. **Target Users:** C-Suite Executives, VP of Sales, and Omnichannel Retail Managers.

Visual Used	Business Question Answered
KPI Gauges (4)	Are we meeting our strategic targets for % of Online Sales, % of Store Sales, Total Combined Revenue, and Total Combined Quantity?
KPI Card	How does current overall sales performance compare to the same period last year (YoY Revenue Growth %)?
Line Chart	What is the monthly trend and peak period for online e-commerce sales?
Column Chart	How does physical store revenue fluctuate on a month-to-month basis?
Donut Chart	What is the exact percentage split of revenue coming from physical stores versus digital channels?
Horizontal Bar Chart	Which specific cities generate the highest combined cross-channel revenue?

6.2 Store Operations Performance

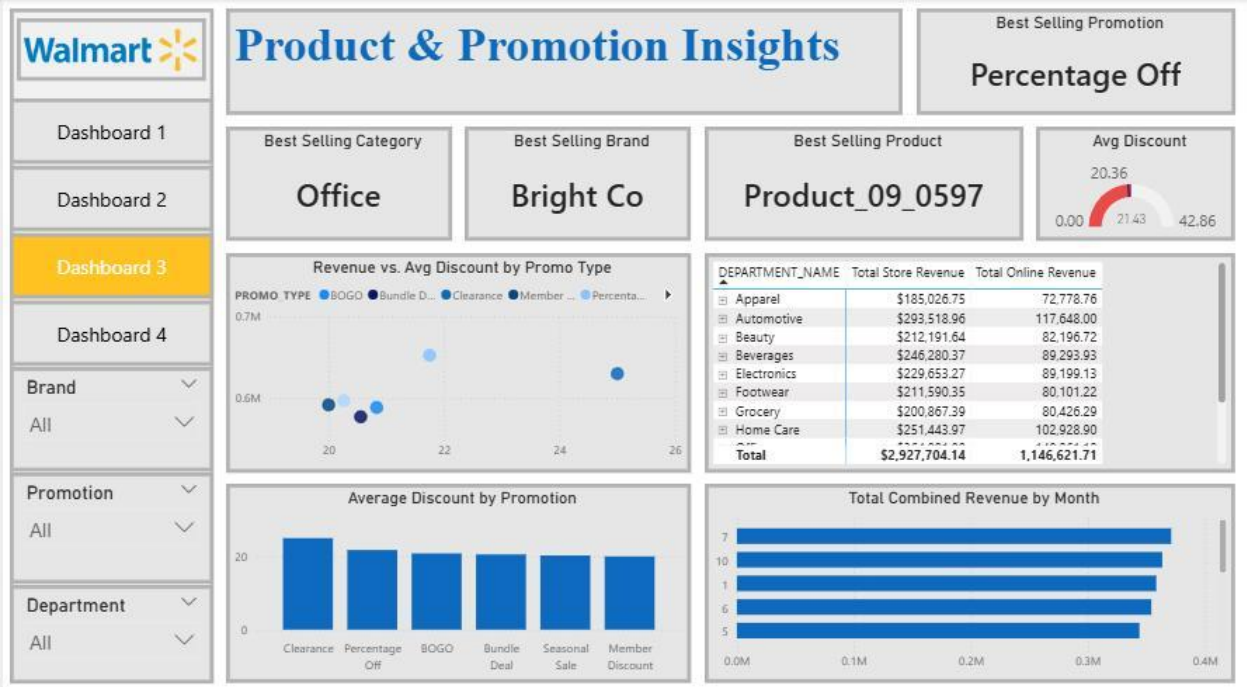


Store Operations Performance: Tracks the physical retail side of the business, focusing strictly on localized store revenue, foot traffic (transactions), departmental sales, and individual employee productivity.

Target Users: Regional Directors and Local Store Managers.

Visual Used	Business Question Answered
KPI Card & Gauges (3)	Are we hitting our localized targets for Total Transactions and Revenue, and what is the Average Unit Price sold in-store?
Column Chart (Departments)	Which specific product departments (e.g., Office, Automotive, Toys) generate the most revenue in stores?
Column Chart (Days of Week)	Which days of the week drive the highest physical foot traffic and total transaction volume?
Horizontal Bar Chart	Which specific physical store locations generate the absolute highest total revenue?
Table Visual	Who are the top-performing employees by total store revenue, and how do they rank against their peers?

6.3 Product & Promotion Insights

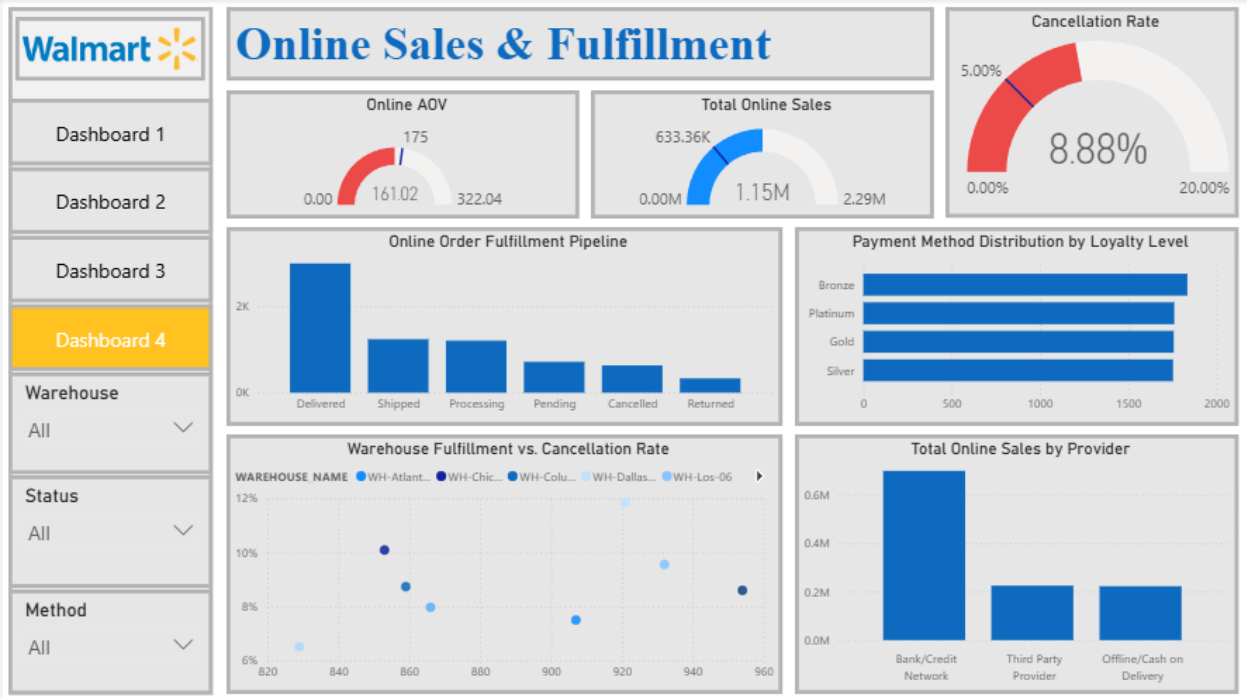


Product & Promotion Insights: Analyzes product popularity, identifies top-performing categories and brands, and measures the financial return and discount depth of various marketing promotions across all sales channels.

Target Users: Omnichannel Retail Managers and Marketing Teams.

Visual Used	**Business Question Answered**
KPI Cards & Gauge (5)	What are our absolute best-selling promotions, categories, brands, and products, and are we staying within our target Average Discount %?
Scatter Plot	How does the average discount percentage correlate with total revenue across different promotion types?
Matrix (Table)	How does total revenue directly compare between physical stores and online channels for each product department?
Column Chart	Which specific marketing promotions (e.g., Clearance, Percentage Off, BOGO) drive the deepest average discounts?
Horizontal Bar Chart	Which specific months historically generate the highest total combined revenue

6.4 Online Sales & Fulfillment



Purpose: Monitors digital operations, specifically tracking e-commerce revenue, warehouse fulfillment efficiency, cancellation rates, and customer payment behaviors.
Target Users: E-Commerce Managers, Logistics & Fulfillment Managers.

Visual Used	Business Question Answered
KPI Gauges (3)	Are we hitting our targets for Total Online Sales, Average Order Value (AOV), and keeping our Cancellation Rate under the threshold?
Column Chart (Pipeline)	What is the distribution and drop-off of online order statuses (from delivered to returned)?
Scatter Plot	How does warehouse fulfillment volume (Total Orders) correlate with their specific Cancellation Rate %?
Bar Chart	How do online order volumes distribute across different customer Loyalty Levels (Bronze, Silver, Gold, Platinum)?
Column Chart	What is the most used payment provider that our customers use?